

# **MMseqs2: The Missing Manual**

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# 1. MMseqs2 Documentation Overview

MMseqs2 is designed for a difficult constraint set: very large sequence collections, sensitivity targets close to established alignment tools, and runtime envelopes that still fit production workflows. The project solves this by treating performance as a systems property instead of a single-kernel optimization problem. Workflows, core compute modules, storage format, and split policy are all part of one execution model.

This manual follows that same model. It starts from the big picture, then moves to algorithmic and systems acceleration details, then to functional modules and command-level references. The goal is that a reader can move from a task question (“what should I run?”) to a design question (“why does this run behave this way?”) without switching mental frameworks.

## 1.1. What This Manual Assumes

The documentation assumes that you will run MMseqs2 as a cascade of stages, not as isolated commands. In practice, prefiltering and indexing choices set the candidate volume and memory behavior, alignment settings control the quality-cost boundary, and result-handling choices decide how downstream interpretation is constrained. If one stage is misconfigured, later stages may still produce outputs, but those outputs may no longer be comparable across runs.

For that reason, this manual repeatedly emphasizes contract stability: database type compatibility, sidecar availability, mode consistency, and reproducible parameter sets. These are not housekeeping details; they are part of algorithmic correctness at scale.

## 1.2. Reading Strategy

Start with the conceptual chapters before diving into command pages:

1. Read Performance Foundations for algorithmic acceleration, storage/indexing behavior, split policy, and parallel execution tradeoffs.
2. Read System Map to understand the layered API model and command cascade.
3. Use Functional Modules Manual for task-oriented command discovery.
4. Use Command Reference Index and Dependency Map for exact command topology and CLI details.

If you already know your task and only need command selection, you can jump directly to Functional Modules Manual. If you are diagnosing runtime or output drift, read the conceptual chapters first; they explain the cross-stage couplings that usually cause expensive surprises.

## 1.3. Why the Cascade Matters

A typical MMseqs2 workflow performs database preparation, candidate generation, alignment or rescoring on surviving pairs, and downstream transformation such as clustering, taxonomy assignment, or export. This order is the primary scaling mechanism. Expensive operations are delayed until fast filters shrink the search space.

The practical consequence is that late-stage tuning often has less impact than early-stage tuning. Raising sensitivity or adding richer alignment outputs can be useful, but only after index strategy,

load behavior, and split policy are stable. In large production runs, infrastructure misalignment can dominate wall time long before algorithmic thresholds are the true bottleneck.

## 1.4. Source-of-Truth Policy

Narrative chapters explain design and usage strategy. Command snapshots in `mmseqs_help_output` define concrete CLI behavior for this repository state. Source code under `MMseqs2/src/` and workflow scripts under `MMseqs2/data/workflow/` remain the final authority when you need implementation-level confirmation.

Use Expert Manual and MMseqs2 Source Development Guide when you need strict reproducibility discipline or source-level debugging and extension guidance.

### Note

When narrative guidance and local command snapshots diverge, treat local snapshots and source code as canonical for exact behavior.

## 2. Performance Foundations: Algorithmic and Systems Acceleration in MMseqs2

MMseqs2 performance is a coupled system. Algorithmic pruning, kernel implementation, storage layout, indexing strategy, split policy, and execution topology all interact. This chapter unifies those layers so tuning decisions are made in the right order and interpreted with the correct assumptions.

### 2.1. Performance Model

MMseqs2 does not rely on one fast algorithm. It applies a cascade that removes expensive comparisons early, computes detailed alignment only for survivors, and keeps data movement predictable through explicit database contracts.

In practical terms, runtime and output behavior are jointly determined by three levers:

| Lever               | What It Changes                                      | Typical Mechanisms                                                      |
|---------------------|------------------------------------------------------|-------------------------------------------------------------------------|
| Candidate reduction | Number of sequence pairs that reach expensive stages | prefiltering, k-mer matching, diagonal filtering, clustering prefilters |
| Per-pair cost       | Compute cost of each surviving pair                  | ungapped rescoring, SIMD alignment modes, backtrace settings            |
| Data-movement cost  | Startup and merge overhead around kernels            | index reuse, preload mode, split policy, temporary I/O topology         |

The highest gains usually come from ordering these levers correctly: stabilize contracts and data movement first, then tune candidate volume, then tune expensive alignment settings.

## 2.2. Shared Comparison Backbone

Search and clustering workflows share a common execution shape. Workflows select paths, pre-filtering reduces candidate pairs, alignment or rescoring computes pair quality, and downstream modules convert those results into task artifacts.

For CPU search paths, the orchestration is explicit in `MMseqs2/data/workflow/blastp.sh`: prefilter then align, with optional iterative merge logic. Clustering orchestration (`MMseqs2/src/workflow/Cluster.cpp` and `MMseqs2/data/workflow/*.sh`) reuses the same primitives and then adds graph construction and cluster assignment.

Because this backbone is shared, upstream prefilter/index decisions usually have larger global impact than late-stage alignment-format tweaks.

## 2.3. Candidate Generation: K-mer Indexing and Expansion

The prefilter core (`MMseqs2/src/prefiltering/Prefiltering.cpp`, `QueryMatcher.cpp`, `IndexBuilder.cpp`) controls candidate growth through index-driven k-mer retrieval and thresholded similar-k-mer expansion.

For amino-acid searches, similar k-mers are generated from precomputed score matrices (`ExtendedSubstitutionMatrix`, `KmerGenerator`) rather than brute-force enumeration. In the main query loop (`QueryMatcher::match`), `MMseqs2` either looks up one exact k-mer index entry or expands into many score-qualified similar k-mers. This branch alone can change candidate fan-out by orders of magnitude and is often the largest runtime lever before alignment.

Composition-bias correction in `QueryMatcher::matchQuery` shifts effective k-mer thresholds per position before k-mer list generation. This means sensitivity and bias controls are coupled in practice: `-s`, `--k-score`, bias correction settings, and k-mer mode selections jointly determine lookup volume and therefore prefilter cost.

## 2.4. Reduced Alphabets and Information-Preserving Coupling

Reduced alphabets are a first-class acceleration path in `MMseqs2`. When amino-acid `--alphabet-size` is set below 21, prefiltering and linear-k-mer modules build a `ReducedMatrix` instead of the full substitution matrix (`Prefiltering::getSubstitutionMatrix`, `kmermatcher`, `kmerindexdb`, `kmersearch`, `alignbykmer`, `clusthash`).

The reduction is not a fixed hardcoded mapping. `ReducedMatrix` iteratively couples amino-acid states by maximizing retained mutual information in the substitution model, then rewrites residue-to-index mappings and the derived score matrix for the reduced state space. Operationally, this shrinks the k-mer combinatorial space ( $a^k$  term) used in index structures and similar-k-mer generation, which directly reduces memory and candidate-generation cost.

Workflow defaults use this intentionally. General search defaults keep full amino-acid alphabet size (`aa:21`), but linear clustering/search-oriented paths use smaller defaults (`aa:13` in `linclust`

family), and `clusthash` forces an even smaller amino-acid alphabet (`aa:3`) for fast redundancy pruning.

## 2.5. Exact, Similar, and Spaced K-mer Modes

MMseqs2 exposes both exact and non-exact k-mer matching. In `prefilter/search` CLI, `--exact-kmer-matching` forces exact lookup, while the default path expands similar k-mers under `--k-score` thresholding. In the current prefilter implementation, the same exact-only branch (`takeOnlyBestKmer`) is also activated for nucleotide-vs-nucleotide search, amino-acid query against profile target, and `--target-search-mode 1`.

Spaced k-mers are an additional speed/sensitivity control (`--spaced-kmer-mode`, `--spaced-kmer-pattern`). In linear modules, MMseqs2 also adapts k-mer behavior through automatic k-mer length/alphabet heuristics and nucleotide-specific adaptive k-mer length logic (`--adjust-kmer-len`) so the candidate generator stays tractable across identity regimes and sequence lengths.

When an index DB is reused, compatibility metadata checks (k-mer size, alphabet size, spaced-kmer mode/pattern, split assumptions) become part of performance correctness. A mismatch is not only a reproducibility risk; it also changes candidate-generation complexity and therefore runtime behavior.

## 2.6. Diagonal and Ungapped Filters

After raw seed hits are collected, MMseqs2 aggregates by target and diagonal and keeps strongest evidence per target (`QueryMatcher`). This compresses large hit lists into a tractable candidate set.

With diagonal scoring enabled, `UngappedAlignment` computes local ungapped scores on diagonals using SIMD paths (AVX2/SSE fallback in `MMseqs2/src/prefiltering/UngappedAlignment.cpp`). This stage is much cheaper than full Smith-Waterman and often removes weak candidates early enough to dominate total runtime savings.

### Performance

If runtime is high, reduce candidate volume and tune ungapped gates before tuning full alignment output richness.

## 2.7. SIMD Gapped Alignment and Output-Cost Economics

Full alignment (`MMseqs2/src/alignment/Alignment.cpp`) dispatches through `Matcher` to vectorized Striped Smith-Waterman (`StripedSmithWaterman.cpp`) for amino-acid/profile paths and banded nucleotide alignment for nucleotide paths.

The dynamic programming score is only part of cost. Coverage/identity calculations, alternative alignment handling, and backtrace generation can substantially increase per-pair cost and output size. Alignment mode is therefore both a semantic choice and a throughput choice.

Use richer alignment outputs only when downstream interpretation explicitly needs them.



## 2.8. Clustering-Specific Accelerators

Clustering workflows add graph-level acceleration on top of search-like filtering. Cascaded clustering intentionally starts with cheaper passes and increases sensitivity later (MMseqs2/src/workflow/Cluster.cpp). Early passes shrink graph density before expensive stages.

This staged design also changes k-mer representation on purpose. Cascaded clustering temporarily switches to linclust-oriented defaults (including reduced amino-acid alphabet) for early low-cost filtering, then restores broader settings for later steps. In single-step clustering, a dedicated clusthash redundancy pass uses an even smaller alphabet to remove near-duplicates cheaply before full prefilter/alignment work.

Single-step clustering scripts use redundancy shortcuts such as clusthash before full prefilter+alignment. Linclust (MMseqs2/src/workflow/Linclust.cpp) applies bounded-k-mer selection plus cheap filters (Hamming and ungapped rescoring) before full local alignment, which is why it scales near-linearly for very large sequence sets.

## 2.9. Masking and Composition-Bias Tradeoffs

Low-complexity masking and composition-bias correction are both speed and quality controls. Masking reduces repetitive false seed growth during indexing and matching. Composition-bias correction reduces spurious composition-driven matches in prefilter and alignment scoring.

Disabling these controls can improve throughput in some data regimes, but it can also destabilize precision and make run-to-run comparisons less interpretable if mode envelopes are not fixed.

## 2.10. GPU Comparison Backend

GPU acceleration is implemented in ungappedprefilter (MMseqs2/src/prefiltering/ungappedprefilter.cpp) with optional persistent server mode in gpuserver (MMseqs2/src/util/gpuserver.cpp).

In GPU mode, query profiles are prepared on CPU and scanned against GPU-resident target layouts. In server mode, target data remains resident and query exchange uses shared memory, reducing repeated startup overhead for low-latency repeated searches.

This backend targets search workflows; clustering workflows are not GPU-accelerated in the current architecture.

## 2.11. Internal Database and Storage Model

Most MMseqs2 modules exchange MMseqs2 DB artifacts rather than plain FASTA. Contiguous record files plus explicit index offsets reduce filesystem overhead and enable direct key-based access.

Core file roles are stable: <db> stores records, <db>.index maps IDs to offsets/lengths, and <db>.dbtype encodes contract type. Header and lookup sidecars (<db>\_h, <db>.lookup, taxonomy sidecars) separate sequence-heavy compute paths from annotation-heavy export paths.

This design is why MMseqs2 favors writing new DBs over in-place edits. Offsets are positional contracts; mutating payload layout in place would invalidate index semantics.

#### Note

Header and sequence separation is intentional. Compute-heavy kernels can avoid header I/O, while export and annotation stages opt in to sidecar reads.

### 2.12. Indexing and Data-Access Strategy

Precomputed indexes move cost from repeated startup into reusable artifacts. They pay off most when targets are stable across many runs. For short-lived targets, on-the-fly indexing can be cheaper than index management overhead.

Storage topology determines whether index reuse helps or hurts. On slower shared storage, startup can become I/O-bound even if compute kernels are fast. In those environments, local caching, preload controls, and resident-memory behavior may produce larger gains than threshold retuning.

`--db-load-mode` changes startup/throughput tradeoffs and should be selected as part of infrastructure design, not as an afterthought.

### 2.13. Memory Model and Split Tradeoffs

Prefilter structures are typically the largest memory consumers. Legacy MMseqs2 guidance models major prefilter terms as:

$$M = (7 * N * L + 8 * a^k) \text{ bytes}$$

with  $N$  sequence count,  $L$  mean length,  $a$  alphabet size, and  $k$  k-mer length. Operationally, memory grows with database scale plus alphabet/k-mer pointer terms.

When memory is constrained, MMseqs2 splits query or target space. Splitting lowers peak RAM but introduces extra merge and temporary I/O overhead. The penalty is often smaller at high sensitivity (compute-dominated runs) and larger at low sensitivity (I/O-dominated runs).

Set split policy as a resource-envelope decision first. Tune thresholds after split behavior is stable.

### 2.14. Parallel Execution Model

MMseqs2 parallelism is layered. Intra-node OpenMP paths accelerate kernels inside one process. Inter-node execution (MPI or scheduler partitioning) distributes chunks across machines.

Distributed throughput is constrained by I/O topology as much as CPU count. Target splitting may reduce per-node RAM but can increase merge overhead. Query splitting can simplify scheduling but may duplicate target-side work. Temporary directory placement can become the dominant bottleneck at high split counts.

#### Warning

Before attributing distributed slowdowns to compute, check shared-disk contention, split count, and merge pressure.

## 2.15. Practical Tuning Order

A robust tuning order for large runs is:

1. Validate DB contracts and sidecars.
2. Stabilize index and load strategy for your storage topology.
3. Set split-memory and temp-disk policy for resource limits.
4. Align parallel mode with hardware and scheduler realities.
5. Tune sensitivity and filtering thresholds last.

This sequence matches real production bottlenecks: contract and infrastructure instability usually dominates before score-threshold tuning becomes the true limiter.

## 2.16. Practical Playbooks

For repeated searches against a stable large target, prioritize index reuse and stable preload behavior before changing sensitivity. For memory-bound runs, tune split policy and merge overhead first. For near-identity mapping pipelines, favor ungapped/diagonal-centric paths and avoid expensive output modes that do not change decision quality.

For architecture and layer placement, use System Map; for task-oriented command selection, use Functional Modules Manual; for full topology evidence, use Dependency Map; and for advanced composition and reproducibility discipline, use Expert Manual.

# 3. System Map: Cascaded APIs and Module Connections

MMseqs2 commands are intentionally layered. Easy workflows and high-level entrypoints decide orchestration, while lower-level modules enforce data contracts and execute the expensive kernels. Understanding this layering is essential for both correctness and performance debugging: the command you invoke is often only the visible front of a deeper module chain.

The generated topology views, Command Reference Index and Dependency Map, provide the complete command-level map. This chapter explains how to interpret that map.

## 3.1. Layered API Model

At the top, workflow commands prioritize usability and sensible defaults. They hide many internal steps but still inherit all low-level behavior from their called modules. High-level API commands expose full DB-to-DB task execution and are typically the right interface for reproducible production pipelines.

Mid-level API commands are where core algorithmic work happens: prefiltering, alignment, clustering kernels, and related computational modules. Low-level API commands perform DB creation, conversion, indexing, filtering, and utility operations. These lower layers are frequently used in custom pipelines and debugging workflows where explicit control matters more than convenience.

### 3.2. Cascade Pattern in Practice

Most workflows follow a common shape: first reduce the candidate space, then score surviving pairs with progressively more expensive methods, then transform outputs into task-specific artifacts. Search and clustering differ in final objectives, but they share this backbone.

That shared backbone creates an important operational rule: changing a high-level command can silently switch lower-level modules and therefore change output semantics, runtime profile, or both. Dependency edges should therefore be read as behavior links, not merely as call graph arrows.

#### Note

Functional group and API layer are orthogonal. A command can belong to one functional group but live in a different API layer than neighboring commands in that group.

### 3.3. How Submodule Pages Encode This Map

Each entry in `submodules/*.md` now combines prose and compact metadata. The prose explains what a command does, how it participates in the cascade, and when to use it. The compact metadata table records API layer, category flags, and coupling counts so readers can quickly assess execution context.

To avoid duplication, full upstream/downstream edge lists and script evidence are centralized in Dependency Map.

### 3.4. Choosing Your Entry View

Start from Functional Modules Manual when your task intent is clear and you need command candidates quickly. Start from Dependency Map when behavior is surprising, performance is unstable, or you are composing a custom low-level pipeline and need exact topology evidence.

Use Performance Foundations with this chapter to connect architecture placement with storage, index, split, and parallel execution consequences.

## 4. Functional Modules Manual

This section is the task-oriented entry point to MMseqs2 modules. Each functional page groups commands by workflow intent, then links each command into API-layer and dependency context so you can trace from practical usage to implementation topology.

For production-scale work, read Performance Foundations before tuning command-level options. That chapter explains the algorithmic and infrastructure assumptions that dominate runtime and output behavior.

### 4.1. Functional Groups

Easy Workflows covers shortcut workflows over FASTA/FASTQ inputs for common tasks with minimal setup. Search Workflows covers homology search and mapping entrypoints. Clustering covers cascaded and linear clustering paths plus update operations.

Prefiltering and Alignment cover core compute kernels used inside search and clustering workflows. Profiles covers profile/MSA creation and conversion paths. Database Management covers creation, indexing, splitting, and storage operations required to keep DB contracts valid.

Result Handling, Sequence Manipulation, Taxonomy, Multi-hit, and Utilities cover downstream transforms, specialized analytics, and composition helpers for advanced pipelines. Functional pages answer what to run; generated maps answer how commands connect and where behavior is inherited from upstream orchestration. Use System Map for layered architecture framing, Command Reference Index for per-command CLI pages, and Dependency Map for complete upstream/downstream topology.

## 4.2. Easy Workflows

High-level shortcuts that operate directly on FASTA/FASTQ inputs and orchestrate MMseqs2 modules with practical defaults.

### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

### Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

### 4.2.1. easy-cluster

Slower, sensitive clustering.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

| Aspect                    | Value                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <pre>usage:          mmseqs          easy- cluster &lt;i:fastaFile1[.gz .bz2]&gt; ... &lt;i:fastaFileN[.gz .bz2]&gt; &lt;o:clusterPrefix&gt; &lt;tmpDir&gt; [options]</pre> |
| API layer                 | workflow                                                                                                                                                                    |
| Category flags            | COMMAND_EASY                                                                                                                                                                |
| Upstream command count    | 0                                                                                                                                                                           |
| Downstream command count  | 7                                                                                                                                                                           |
| Related functional groups | clustering, database, result_handling                                                                                                                                       |

| Aspect     | Value                       |
|------------|-----------------------------|
| References | Full CLI · Dependency entry |

#### 4.2.1.1. Key Options

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                          |
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                   | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically     |

#### 4.2.2. easy-linclud

Fast linear time cluster, less sensitive clustering.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

| Aspect                    | Value                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs easy-linclud <i:fastaFile1[.gz .bz2]> ... <i:fastaFileN[.gz .bz2]> <o:clusterPrefix> <tmpDir> [options] |
| API layer                 | workflow                                                                                                              |
| Category flags            | COMMAND_EASY                                                                                                          |
| Upstream command count    | 0                                                                                                                     |
| Downstream command count  | 7                                                                                                                     |
| Related functional groups | clustering, database, result_handling                                                                                 |
| References                | Full CLI · Dependency entry                                                                                           |

#### 4.2.2.1. Key Options

| Option                 | Purpose                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                                         |
| --alph-size            | Alphabet size (range 2-21)                                                                                   |
| --spaced-kmer-mode     | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern  | User-specified spaced k-mer pattern                                                                          |
| --mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob            | Mask sequences is probability is above threshold                                                             |

#### 4.2.3. easy-linsearch

Fast, less sensitive homology search.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

| Aspect                    | Value                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs easy-linsearch<br><i:queryFastaFile1[.gz .bz2]> ...<br><i:queryFastaFileN[.gz .bz2]><br><i:targetFastaFile[.gz .bz2]> <br><i:targetDB> <o:alignmentFile> <tmpDir><br>[options] |
| API layer                 | workflow                                                                                                                                                                                     |
| Category flags            | COMMAND_EASY   COMMAND_EXPERT                                                                                                                                                                |
| Upstream command count    | 0                                                                                                                                                                                            |
| Downstream command count  | 7                                                                                                                                                                                            |
| Related functional groups | database,<br>result_handling, search_workflows                                                                                                                                               |

| Aspect     | Value                       |
|------------|-----------------------------|
| References | Full CLI · Dependency entry |

#### 4.2.3.1. Key Options

| Option                 | Purpose                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                                         |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                                                |
| --mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob            | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case      | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat        | Repeat letters that occur > threshold in a row                                                               |

#### 4.2.4. easy-rbh

Find reciprocal best hit.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 4 downstream call(s).

| Aspect                   | Value                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Usage                    | usage: mmseqs easy-rbh <i:queryFastaFile1[.gz .bz2]> <i:targetFastaFile[.gz .bz2]>  <i:targetDB> <o:alignmentFile> <tmpDir> [options] |
| API layer                | workflow                                                                                                                              |
| Category flags           | COMMAND_EASY                                                                                                                          |
| Upstream command count   | 0                                                                                                                                     |
| Downstream command count | 4                                                                                                                                     |



| Aspect                    | Value                                          |
|---------------------------|------------------------------------------------|
| Related functional groups | database,<br>result_handling, search_workflows |
| References                | Full CLI · Dependency entry                    |

#### 4.2.4.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

#### 4.2.5. easy-search

Sensitive homology search.

Workflow-level entripoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

| Aspect         | Value                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage          | usage: mmseqs easy-search<br><i:queryFastaFile1[.gz .bz2]> ...<br><i:queryFastaFileN[.gz .bz2]> <br><i:stdin> <i:targetFastaFile[.gz]> <br><i:targetDB> <o:alignmentFile> <tmpDir><br>[options] |
| API layer      | workflow                                                                                                                                                                                        |
| Category flags | COMMAND_EASY                                                                                                                                                                                    |

| Aspect                    | Value                                          |
|---------------------------|------------------------------------------------|
| Upstream command count    | 0                                              |
| Downstream command count  | 7                                              |
| Related functional groups | database,<br>result_handling, search_workflows |
| References                | Full CLI · Dependency entry                    |

#### 4.2.5.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

#### 4.2.6. easy-taxonomy

Taxonomic classification.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 11 downstream call(s).

| Aspect    | Value                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| Usage     | usage: mmseqs easy-taxonomy <i:fastaFile1[.gz .bz2]> ...<br><i:fastaFileN[.gz .bz2]> <i:targetDB><br><o:taxReports> <tmpDir> [options] |
| API layer | workflow                                                                                                                               |

| Aspect                    | Value                                          |
|---------------------------|------------------------------------------------|
| Category flags            | COMMAND_EASY                                   |
| Upstream command count    | 0                                              |
| Downstream command count  | 11                                             |
| Related functional groups | database, result_handling, taxonomy, utilities |
| References                | Full CLI · Dependency entry                    |

#### 4.2.6.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

### 4.3. Search Workflows

Search and mapping workflows that compose prefiltering, alignment, and result conversion paths under different sensitivity and runtime envelopes.

#### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

#### 4.3.1. linsearch

Fast, less sensitive homology search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 2 upstream caller(s) and 9 downstream call(s).

| Aspect                    | Value                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs linsearch &lt;i:queryDB&gt; &lt;i:targetDB&gt; &lt;o:alignmentDB&gt; &lt;tmpDir&gt; [options]</code>                                          |
| API layer                 | <code>high_level_api</code>                                                                                                                                       |
| Category flags            | <code>COMMAND_MAIN COMMAND_EXPERT</code>                                                                                                                          |
| Upstream command count    | 2                                                                                                                                                                 |
| Downstream command count  | 9                                                                                                                                                                 |
| Related functional groups | <code>alignment,</code><br><code>database, easy_workflows, prefiltering,</code><br><code>result_handling, sequence_manipulation,</code><br><code>utilities</code> |
| References                | Full CLI · Dependency entry                                                                                                                                       |

##### 4.3.1.1. Key Options

| Option                              | Purpose                                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--comp-bias-corr</code>       | Correct for locally biased amino acid composition (range 0-1)                                                |
| <code>--comp-bias-corr-scale</code> | Correct for locally biased amino acid composition (range 0-1)                                                |
| <code>--add-self-matches</code>     | Artificially add entries of queries with themselves (for clustering)                                         |
| <code>--seed-sub-mat</code>         | Substitution matrix file for k-mer generation                                                                |
| <code>--mask</code>                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| <code>--mask-prob</code>            | Mask sequences is probability is above threshold                                                             |

| Option                         | Purpose                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------|
| <code>--mask-lower-case</code> | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region |
| <code>--mask-n-repeat</code>   | Repeat letters that occur > threshold in a row                                            |

#### 4.3.2. map

Map nearly identical sequences.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 0 upstream caller(s) and 1 downstream call(s).

| Aspect                    | Value                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs map &lt;i:queryDB&gt; &lt;i:targetDB&gt; &lt;o:alignmentDB&gt; &lt;tmpDir&gt; [options]</code> |
| API layer                 | <code>high_level_api</code>                                                                                        |
| Category flags            | <code>COMMAND_MAIN</code>                                                                                          |
| Upstream command count    | 0                                                                                                                  |
| Downstream command count  | 1                                                                                                                  |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                             |
| References                | Full CLI · Dependency entry                                                                                        |

##### 4.3.2.1. Key Options

| Option                            | Purpose                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------|
| <code>--seed-sub-mat</code>       | Substitution matrix file for k-mer generation                                          |
| <code>-s</code>                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| <code>-k</code>                   | k-mer length (0: automatically set to optimum)                                         |
| <code>--target-search-mode</code> | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| <code>--k-score</code>            | k-mer threshold for generating similar k-mer lists                                     |
| <code>--alph-size</code>          | Alphabet size (range 2-21)                                                             |
| <code>--max-seqs</code>           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |

| Option  | Purpose                                                                               |
|---------|---------------------------------------------------------------------------------------|
| --split | Split input into N equally distributed chunks.<br>0: set the best split automatically |

#### 4.3.3. rbh

Reciprocal best hit search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 1 upstream caller(s) and 6 downstream call(s).

| Aspect                    | Value                                                                               |
|---------------------------|-------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs rbh <i:queryDB><br><i:targetDB> <o:alignmentDB> <tmpDir><br>[options] |
| API layer                 | high_level_api                                                                      |
| Category flags            | COMMAND_MAIN                                                                        |
| Upstream command count    | 1                                                                                   |
| Downstream command count  | 6                                                                                   |
| Related functional groups | database, easy_workflows,<br>result_handling, utilities                             |
| References                | Full CLI · Dependency entry                                                         |

##### 4.3.3.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |

| Option                 | Purpose                                            |
|------------------------|----------------------------------------------------|
| <code>--k-score</code> | k-mer threshold for generating similar k-mer lists |

#### 4.3.4. search

Sensitive homology search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 8 upstream caller(s) and 22 downstream call(s).

| Aspect                    | Value                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs search &lt;i:queryDB&gt; &lt;i:targetDB&gt; &lt;o:alignmentDB&gt; &lt;tmpDir&gt; [options]</code>                             |
| API layer                 | <code>high_level_api</code>                                                                                                                       |
| Category flags            | <code>COMMAND_MAIN</code>                                                                                                                         |
| Upstream command count    | 8                                                                                                                                                 |
| Downstream command count  | 22                                                                                                                                                |
| Related functional groups | <code>alignment, clustering, database, easy_workflows, multi_hit, prefiltering, profiles, result_handling, sequence_manipulation, taxonomy</code> |
| References                | Full CLI · Dependency entry                                                                                                                       |

##### 4.3.4.1. Key Options

| Option                              | Purpose                                                              |
|-------------------------------------|----------------------------------------------------------------------|
| <code>--comp-bias-corr</code>       | Correct for locally biased amino acid composition (range 0-1)        |
| <code>--comp-bias-corr-scale</code> | Correct for locally biased amino acid composition (range 0-1)        |
| <code>--add-self-matches</code>     | Artificially add entries of queries with themselves (for clustering) |
| <code>--seed-sub-mat</code>         | Substitution matrix file for k-mer generation                        |
| <code>-s</code>                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| <code>-k</code>                     | k-mer length (0: automatically set to optimum)                       |

| Option                            | Purpose                                                 |
|-----------------------------------|---------------------------------------------------------|
| <code>--target-search-mode</code> | target search mode (0: regular k-mer, 1: similar k-mer) |
| <code>--k-score</code>            | k-mer threshold for generating similar k-mer lists      |

## 4.4. Clustering

Modules for cluster construction, incremental updates, and representative selection across cascaded and linear-time strategies.

### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

### Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

### 4.4.1. `clust`

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs clust &lt;i:sequenceDB&gt; &lt;i:resultDB&gt; &lt;o:clusterDB&gt; [options]</code> |
| API layer                 | <code>mid_level_api</code>                                                                             |
| Category flags            | <code>COMMAND_CLUSTER</code>                                                                           |
| Upstream command count    | 2                                                                                                      |
| Downstream command count  | 0                                                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                 |
| References                | Full CLI · Dependency entry                                                                            |



#### 4.4.1.1. Key Options

| Option                     | Purpose                                                                    |
|----------------------------|----------------------------------------------------------------------------|
| --cluster-mode             | 0: Set-Cover (greedy)                                                      |
| --max-iterations           | Maximum depth of breadth first search in connected component clustering    |
| --similarity-type          | Type of score used for clustering. 1: alignment score 2: sequence identity |
| --weights                  | Weights used for cluster prioritization                                    |
| --cluster-weight-threshold | Weight threshold used for cluster prioritization                           |
| --threads                  | Number of CPU-cores used (all by default)                                  |
| --compressed               | Write compressed output                                                    |
| -v                         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info              |

#### 4.4.2. cluster

Slower, sensitive clustering.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 18 downstream call(s).

| Aspect                    | Value                                                                               |
|---------------------------|-------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]               |
| API layer                 | high_level_api                                                                      |
| Category flags            | COMMAND_MAIN                                                                        |
| Upstream command count    | 2                                                                                   |
| Downstream command count  | 18                                                                                  |
| Related functional groups | alignment, database, easy_workflows, prefiltering, sequence_manipulation, utilities |
| References                | Full CLI · Dependency entry                                                         |

#### 4.4.2.1. Key Options

| Option         | Purpose                                       |
|----------------|-----------------------------------------------|
| --seed-sub-mat | Substitution matrix file for k-mer generation |

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                   | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically     |

#### 4.4.3. clusterupdate

Update previous clustering with new sequences.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 0 upstream caller(s) and 13 downstream call(s).

| Aspect                    | Value                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs clusterupdate <i:oldSequenceDB><br><i:newSequenceDB> <i:oldClustResultDB><br><o:newMappedSequenceDB><br><o:newClustResultDB> <tmpDir> [options] |
| API layer                 | high_level_api                                                                                                                                                   |
| Category flags            | COMMAND_MAIN                                                                                                                                                     |
| Upstream command count    | 0                                                                                                                                                                |
| Downstream command count  | 13                                                                                                                                                               |
| Related functional groups | database, result_handling,<br>search_workflows, utilities                                                                                                        |
| References                | Full CLI · Dependency entry                                                                                                                                      |

#### 4.4.3.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

#### 4.4.4. clusthash

Hash-based clustering of equal length sequences.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]       |
| API layer                 | mid_level_api                                                          |
| Category flags            | COMMAND_CLUSTER                                                        |
| Upstream command count    | 1                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.4.4.1. Key Options

| Option         | Purpose                                                                    |
|----------------|----------------------------------------------------------------------------|
| --alph-size    | Alphabet size (range 2-21)                                                 |
| --min-seq-id   | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| --sub-mat      | Substitution matrix file                                                   |
| --max-seq-len  | Maximum sequence length                                                    |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch            |
| --threads      | Number of CPU-cores used (all by default)                                  |
| --compressed   | Write compressed output                                                    |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info              |

#### 4.4.5. linclust

Fast, less sensitive clustering.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 8 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options] |
| API layer                 | high_level_api                                                         |
| Category flags            | COMMAND_MAIN                                                           |
| Upstream command count    | 2                                                                      |
| Downstream command count  | 8                                                                      |
| Related functional groups | alignment, database, easy_workflows, prefiltering, utilities           |
| References                | Full CLI · Dependency entry                                            |

#### 4.4.5.1. Key Options

| Option           | Purpose                                                       |
|------------------|---------------------------------------------------------------|
| --comp-bias-corr | Correct for locally biased amino acid composition (range 0-1) |

| Option                 | Purpose                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                                         |
| --alph-size            | Alphabet size (range 2-21)                                                                                   |
| --spaced-kmer-mode     | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern  | User-specified spaced k-mer pattern                                                                          |
| --mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob            | Mask sequences is probability is above threshold                                                             |

#### 4.4.6. mergeclusters

Merge multiple cascaded clustering steps.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs mergeclusters <i:sequenceDB><br><o:clusterDB> <i:clusterDB1> ...<br><i:clusterDBn> [options] |
| API layer                 | mid_level_api                                                                                                 |
| Category flags            | COMMAND_CLUSTER                                                                                               |
| Upstream command count    | 2                                                                                                             |
| Downstream command count  | 0                                                                                                             |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                        |
| References                | Full CLI · Dependency entry                                                                                   |

#### 4.4.6.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.4.7. pickconsensusrep

Select new representatives for each cluster based on consensus.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

| Aspect                    | Value                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs pickconsensusrep<br><inputDB(s)> <outputDB> [options]<br>(source-derived synopsis; run mmseqs<br>pickconsensusrep for exact syntax) |
| API layer                 | mid_level_api                                                                                                                                     |
| Category flags            | COMMAND_CLUSTER                                                                                                                                   |
| Upstream command count    | 0                                                                                                                                                 |
| Downstream command count  | 7                                                                                                                                                 |
| Related functional groups | alignment, database, profiles,<br>result_handling, utilities                                                                                      |
| References                | Full CLI · Dependency entry                                                                                                                       |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

### 4.5. Prefiltering

Candidate-generation modules that prune comparison space before expensive alignment kernels.

#### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### 4.5.1. countkmer

Count k-mers.

Low-level DB or utility command used for composition and contract enforcement. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs countkmer <DB> [args] [options] (source-derived synopsis; run mmseqs countkmer for exact syntax) |
| API layer                 | low_level_api                                                                                                  |
| Category flags            | COMMAND_SPECIAL                                                                                                |
| Upstream command count    | 0                                                                                                              |
| Downstream command count  | 0                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                         |
| References                | Full CLI · Dependency entry                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.5.2. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs gappedprefilter <i:queryDB><br><i:targetDB> <o:prefilterDB> [options] |
| API layer                 | mid_level_api                                                                          |
| Category flags            | COMMAND_PREFILTER                                                                      |
| Upstream command count    | 0                                                                                      |
| Downstream command count  | 0                                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                 |
| References                | Full CLI · Dependency entry                                                            |

#### 4.5.2.1. Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --min-ungapped-score   | Accept only matches with ungapped alignment score above threshold                      |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --gap-open             | Gap open cost                                                                          |
| --gap-extend           | Gap extension cost                                                                     |
| -e                     | List matches below this E-value (range 0.0-inf)                                        |
| -c                     | List matches above this fraction of aligned (covered) residues (see --cov-mode)        |

#### 4.5.3. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                    |
|---------------------------|--------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs kmermatcher <i:sequenceDB><br><o:prefilterDB> [options] |
| API layer                 | mid_level_api                                                            |
| Category flags            | COMMAND_PREFILTER                                                        |
| Upstream command count    | 1                                                                        |
| Downstream command count  | 0                                                                        |
| Related functional groups | clustering                                                               |
| References                | Full CLI · Dependency entry                                              |

#### 4.5.3.1. Key Options

| Option      | Purpose                    |
|-------------|----------------------------|
| --alph-size | Alphabet size (range 2-21) |



| Option                | Purpose                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                    |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                                                             |
| --mask                | Mask sequences in prefilter stage with tantan:<br>0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob           | Mask sequences is probablity is above threshold                                                                 |
| --mask-lower-case     | Lowercase letters will be excluded from k-mer search<br>0: include region, 1: exclude region                    |
| --mask-n-repeat       | Repeat letters that occure > threshold in a rwo                                                                 |
| -k                    | k-mer length (0: automatically set to optimum)                                                                  |

#### 4.5.4. kmersearch

Find bottom-m-hashed k-mer matches between target and query DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                |
|---------------------------|--------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs kmersearch <i:queryDB><br><i:kmerIndexDB> <o:prefilterDB><br>[options] |
| API layer                 | mid_level_api                                                                        |
| Category flags            | COMMAND_PREFILTER                                                                    |
| Upstream command count    | 1                                                                                    |
| Downstream command count  | 0                                                                                    |
| Related functional groups | search_workflows                                                                     |
| References                | Full CLI · Dependency entry                                                          |

##### 4.5.4.1. Key Options

| Option         | Purpose                                       |
|----------------|-----------------------------------------------|
| --seed-sub-mat | Substitution matrix file for k-mer generation |

| Option               | Purpose                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| --mask               | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob          | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case    | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat      | Repeat letters that occur > threshold in a row                                                               |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| --cov-mode           | 0: coverage of query and target                                                                              |
| -c                   | List matches above this fraction of aligned (covered) residues (see --cov-mode)                              |

#### 4.5.5. prefilter

Double consecutive diagonal k-mer search.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                      |
|---------------------------|----------------------------------------------------------------------------|
| Usage                     | usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options] |
| API layer                 | mid_level_api                                                              |
| Category flags            | COMMAND_PREFILTER                                                          |
| Upstream command count    | 3                                                                          |
| Downstream command count  | 0                                                                          |
| Related functional groups | clustering, search_workflows, taxonomy                                     |
| References                | Full CLI · Dependency entry                                                |

##### 4.5.5.1. Key Options

| Option         | Purpose                                          |
|----------------|--------------------------------------------------|
| --seed-sub-mat | Substitution matrix file for k-mer generation    |
| -s             | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive |

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| -k                   | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically     |

#### 4.5.6. ungappedprefilter

Optimal diagonal score search.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs ungappedprefilter <i:queryDB><br><i:targetDB> <o:prefilterDB> [options] |
| API layer                 | mid_level_api                                                                            |
| Category flags            | COMMAND_PREFILTER                                                                        |
| Upstream command count    | 1                                                                                        |
| Downstream command count  | 0                                                                                        |
| Related functional groups | search_workflows                                                                         |
| References                | Full CLI · Dependency entry                                                              |

##### 4.5.6.1. Key Options

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

| Option                            | Purpose                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| <code>--min-ungapped-score</code> | Accept only matches with ungapped alignment score above threshold                            |
| <code>--max-seqs</code>           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)       |
| <code>-c</code>                   | List matches above this fraction of aligned (covered) residues (see <code>-cov-mode</code> ) |
| <code>-e</code>                   | List matches below this E-value (range 0.0-inf)                                              |
| <code>--cov-mode</code>           | 0: coverage of query and target                                                              |
| <code>--taxon-list</code>         | Taxonomy ID, possibly multiple values separated by ‘;                                        |

## 4.6. Alignment

Alignment and rescoring modules used after prefiltering to compute pair quality and coordinates.

### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

### 4.6.1. align

Optimal gapped local alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs align <i:queryDB> <i:targetDB><br><i:resultDB> <o:alignmentDB> [options] |
| API layer                 | mid_level_api                                                                             |
| Category flags            | COMMAND_ALIGNMENT                                                                         |
| Upstream command count    | 5                                                                                         |
| Downstream command count  | 0                                                                                         |
| Related functional groups | clustering, search_workflows                                                              |
| References                | Full CLI · Dependency entry                                                               |

#### 4.6.1.1. Key Options

| Option                  | Purpose                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr        | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --comp-bias-corr-scale  | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --add-self-matches      | Artificially add entries of queries with themselves (for clustering)                                                     |
| -a                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |
| --alignment-mode        | How to compute the alignment:                                                                                            |
| --alignment-output-mode | How to compute the alignment:                                                                                            |
| --wrapped-scoring       | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| -e                      | List matches below this E-value (range 0.0-inf)                                                                          |

#### 4.6.2. alignall

Within-result all-vs-all gapped local alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                        |
|---------------------------|------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options] |
| API layer                 | mid_level_api                                                                |
| Category flags            | COMMAND_ALIGNMENT                                                            |
| Upstream command count    | 0                                                                            |
| Downstream command count  | 0                                                                            |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.       |
| References                | Full CLI · Dependency entry                                                  |

#### 4.6.2.1. Key Options

| Option                              | Purpose                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------|
| <code>--comp-bias-corr</code>       | Correct for locally biased amino acid composition (range 0-1)               |
| <code>--comp-bias-corr-scale</code> | Correct for locally biased amino acid composition (range 0-1)               |
| <code>--add-self-matches</code>     | Artificially add entries of queries with themselves (for clustering)        |
| <code>-a</code>                     | Add backtrace string (convert to alignments with mmseqs convertalis module) |
| <code>--alignment-mode</code>       | How to compute the alignment:                                               |
| <code>-e</code>                     | List matches below this E-value (range 0.0-inf)                             |
| <code>--min-seq-id</code>           | List matches above this sequence identity (for clustering) (range 0.0-1.0)  |
| <code>--min-aln-len</code>          | Minimum alignment length (range 0-INT_MAX)                                  |

#### 4.6.3. alignbykmer

Heuristic gapped local k-mer based alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options] |
| API layer                 | mid_level_api                                                                          |
| Category flags            | COMMAND_ALIGNMENT                                                                      |
| Upstream command count    | 0                                                                                      |
| Downstream command count  | 0                                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                 |
| References                | Full CLI · Dependency entry                                                            |

#### 4.6.3.1. Key Options

| Option                | Purpose                                                                         |
|-----------------------|---------------------------------------------------------------------------------|
| -k                    | k-mer length (0: automatically set to optimum)                                  |
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                    |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                             |
| --alph-size           | Alphabet size (range 2-21)                                                      |
| --add-self-matches    | Artificially add entries of queries with themselves (for clustering)            |
| -c                    | List matches above this fraction of aligned (covered) residues (see --cov-mode) |
| -e                    | List matches below this E-value (range 0.0-inf)                                 |
| --cov-mode            | 0: coverage of query and target                                                 |

#### 4.6.4. expandaln

Expand an alignment result based on another.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB ca3mDB> <o:alignmentDB> [options] |
| API layer                 | mid_level_api                                                                                               |
| Category flags            | COMMAND_PROFILE_PROFILE                                                                                     |
| Upstream command count    | 1                                                                                                           |
| Downstream command count  | 0                                                                                                           |
| Related functional groups | search_workflows                                                                                            |
| References                | Full CLI · Dependency entry                                                                                 |

##### 4.6.4.1. Key Options

| Option           | Purpose                                                       |
|------------------|---------------------------------------------------------------|
| --comp-bias-corr | Correct for locally biased amino acid composition (range 0-1) |

| Option                 | Purpose                                                                        |
|------------------------|--------------------------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                  |
| --gap-open             | Gap open cost                                                                  |
| --gap-extend           | Gap extension cost                                                             |
| --score-bias           | Score bias when computing SW alignment (in bits)                               |
| -e                     | List matches below this E-value (range 0.0-inf)                                |
| --min-seq-id           | List matches above this sequence identity (for clustering) (range 0.0-1.0)     |
| -c                     | List matches above this fraction of aligned (covered) residues (see -cov-mode) |

#### 4.6.5. fwbw

Forward Backward Alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs fwbw <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs fwbw for exact syntax) |
| API layer                 | mid_level_api                                                                                                    |
| Category flags            | COMMAND_ALIGNMENT                                                                                                |
| Upstream command count    | 0                                                                                                                |
| Downstream command count  | 0                                                                                                                |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                           |
| References                | Full CLI · Dependency entry                                                                                      |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.6.6. offsetalignment

Offset alignment by ORF start position.



Low-level DB or utility command used for composition and contract enforcement. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs offsetalignment<br><i:queryDB> <i:queryOrfDB> <i:targetDB><br><i:targetOrfDB> <i:alnDB> <o:alnDB><br>[options] |
| API layer                 | low_level_api                                                                                                                |
| Category flags            | COMMAND_RESULT                                                                                                               |
| Upstream command count    | 3                                                                                                                            |
| Downstream command count  | 0                                                                                                                            |
| Related functional groups | clustering, search_workflows                                                                                                 |
| References                | Full CLI · Dependency entry                                                                                                  |

#### 4.6.6.1. Key Options

| Option             | Purpose                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------|
| --search-type      | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment |
| --threads          | Number of CPU-cores used (all by default)                                                           |
| --compressed       | Write compressed output                                                                             |
| --db-load-mode     | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                                     |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                       |
| --chain-alignments | Chain overlapping alignments                                                                        |
| --merge-query      | Combine ORFs/split sequences to a single entry                                                      |

#### 4.6.7. proteinaln2nucl

Transform protein alignments to nucleotide alignments.

Low-level DB or utility command used for composition and contract enforcement. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs proteinaln2nucl<br><i:nuclQueryDB> <i:nuclTargetDB><br><i:aaQueryDB> <i:aaTargetDB> <i:alnDB><br><o:alnDB> [options] |
| API layer                 | low_level_api                                                                                                                      |
| Category flags            | COMMAND_RESULT                                                                                                                     |
| Upstream command count    | 0                                                                                                                                  |
| Downstream command count  | 0                                                                                                                                  |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                             |
| References                | Full CLI · Dependency entry                                                                                                        |

#### 4.6.7.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --gap-open   | Gap open cost                                                 |
| --gap-extend | Gap extension cost                                            |
| --sub-mat    | Substitution matrix file                                      |
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.6.8. rescorediagonal

Compute sequence identity for diagonal.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

| Aspect                   | Value                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Usage                    | usage: mmseqs rescorediagonal<br><i:queryDB> <i:targetDB><br><i:prefilterDB> <o:resultDB> [options] |
| API layer                | mid_level_api                                                                                       |
| Category flags           | COMMAND_ALIGNMENT                                                                                   |
| Upstream command count   | 5                                                                                                   |
| Downstream command count | 0                                                                                                   |

| Aspect                    | Value                                  |
|---------------------------|----------------------------------------|
| Related functional groups | clustering, search_workflows, taxonomy |
| References                | Full CLI · Dependency entry            |

#### 4.6.8.1. Key Options

| Option                          | Purpose                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>--add-self-matches</code> | Artificially add entries of queries with themselves (for clustering)                                                     |
| <code>--wrapped-scoring</code>  | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| <code>-e</code>                 | List matches below this E-value (range 0.0-inf)                                                                          |
| <code>-c</code>                 | List matches above this fraction of aligned (covered) residues (see <code>--cov-mode</code> )                            |
| <code>-a</code>                 | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |
| <code>--cov-mode</code>         | 0: coverage of query and target                                                                                          |
| <code>--min-seq-id</code>       | List matches above this sequence identity (for clustering) (range 0.0-1.0)                                               |
| <code>--min-aln-len</code>      | Minimum alignment length (range 0-INT_MAX)                                                                               |

#### 4.6.9. transitivealign

Transfer alignments via transitivity.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                 | Value                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                  | usage: mmseqs transitivealign<br><inputDB(s)> <outputDB> [options]<br>(source-derived synopsis; run mmseqs<br>transitivealign for exact syntax) |
| API layer              | mid_level_api                                                                                                                                   |
| Category flags         | COMMAND_ALIGNMENT                                                                                                                               |
| Upstream command count | 0                                                                                                                                               |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

## 4.7. Profiles

Profile and MSA modules for profile construction, conversion, and profile-driven search workflows.

### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

### 4.7.1. convertca3m

Convert a cA3M DB to a result DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs convertca3m <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs convertca3m for exact syntax) |
| API layer                 | mid_level_api                                                                                                                  |
| Category flags            | COMMAND_PROFILE_PROFILE                                                                                                        |
| Upstream command count    | 0                                                                                                                              |
| Downstream command count  | 0                                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                         |
| References                | Full CLI · Dependency entry                                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

### 4.7.2. convertmsa

Convert Stockholm/PFAM MSA file to a MSA DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                   |
|---------------------------|-------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs convertmsa <i:msaFile.sto[.gz]><br><o:msaDB> [options] |
| API layer                 | low_level_api                                                           |
| Category flags            | COMMAND_DATABASE_CREATION                                               |
| Upstream command count    | 1                                                                       |
| Downstream command count  | 0                                                                       |
| Related functional groups | database                                                                |
| References                | Full CLI · Dependency entry                                             |

#### 4.7.2.1. Key Options

| Option             | Purpose                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --identifier-field | Field from STOCKHOLM comments for choosing the MSA identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of the first sequence will become the identifier |
| --compressed       | Write compressed output                                                                                                                                                          |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                                                                    |

#### 4.7.3. convertprofiledb

Convert a HH-suite HHM DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect    | Value                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Usage     | usage: mmseqs convertprofiledb <DB><br>[args] [options] (source-derived synopsis;<br>run mmseqs convertprofiledb for exact syn-<br>tax) |
| API layer | low_level_api                                                                                                                           |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Category flags            | COMMAND_PROFILE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.4. **expand2profile**

Expand an alignment result based on another and create a profile.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs expand2profile<br><inputDB(s)> <outputDB> [options]<br>(source-derived synopsis; run mmseqs<br>expand2profile for exact syntax) |
| API layer                 | mid_level_api                                                                                                                                 |
| Category flags            | COMMAND_PROFILE_PROFILE                                                                                                                       |
| Upstream command count    | 1                                                                                                                                             |
| Downstream command count  | 0                                                                                                                                             |
| Related functional groups | search_workflows                                                                                                                              |
| References                | Full CLI · Dependency entry                                                                                                                   |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.5. **msa2profile**

Convert a MSA DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                          |
|---------------------------|----------------------------------------------------------------|
| Usage                     | usage: mmseqs msa2profile <i:msaDB><br><o:profileDB> [options] |
| API layer                 | low_level_api                                                  |
| Category flags            | COMMAND_PROFILE<br>  COMMAND_DATABASE_CREATION                 |
| Upstream command count    | 2                                                              |
| Downstream command count  | 0                                                              |
| Related functional groups | clustering, database                                           |
| References                | Full CLI · Dependency entry                                    |

#### 4.7.5.1. Key Options

| Option                 | Purpose                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --gap-open             | Gap open cost                                                                                                                            |
| --gap-extend           | Gap extension cost                                                                                                                       |
| --match-mode           | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept |
| --match-ratio          | Columns that have a residue in this ratio of all sequences are kept                                                                      |
| --pseudo-cnt-mode      | use 0: substitution-matrix or 1: context-specific pseudocounts                                                                           |
| --pca                  | Pseudo count admixture strength                                                                                                          |

#### 4.7.6. msa2result

Convert a MSA DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                   |
|---------------------------|-------------------------------------------------------------------------|
| Usage                     | usage: mmseqs msa2result <i:msaDB><br><o:seqDB> <o:profileDB> [options] |
| API layer                 | low_level_api                                                           |
| Category flags            | COMMAND_PROFILE   COMMAND_EXPERT                                        |
| Upstream command count    | 0                                                                       |
| Downstream command count  | 0                                                                       |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.  |
| References                | Full CLI · Dependency entry                                             |

#### 4.7.6.1. Key Options

| Option                 | Purpose                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --gap-open             | Gap open cost                                                                                                                            |
| --gap-extend           | Gap extension cost                                                                                                                       |
| --match-mode           | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept |
| --match-ratio          | Columns that have a residue in this ratio of all sequences are kept                                                                      |
| --pseudo-cnt-mode      | use 0: substitution-matrix or 1: context-specific pseudocounts                                                                           |
| --pca                  | Pseudo count admixture strength                                                                                                          |

#### 4.7.7. pairaln

Pair sequences to match best protein A and B from a species.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).



| Aspect                    | Value                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs pairaln <DB> [args] [options] (source-derived synopsis; run mmseqs pairaln for exact syntax) |
| API layer                 | low_level_api                                                                                              |
| Category flags            | COMMAND_EXPERT                                                                                             |
| Upstream command count    | 0                                                                                                          |
| Downstream command count  | 0                                                                                                          |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                     |
| References                | Full CLI · Dependency entry                                                                                |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.8. profile2consensus

Extract consensus sequence DB from a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs profile2consensus <DB> [args] [options] (source-derived synopsis; run mmseqs profile2consensus for exact syntax) |
| API layer                 | low_level_api                                                                                                                  |
| Category flags            | COMMAND_PROFILE                                                                                                                |
| Upstream command count    | 1                                                                                                                              |
| Downstream command count  | 0                                                                                                                              |
| Related functional groups | search_workflows                                                                                                               |
| References                | Full CLI · Dependency entry                                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.9. profile2neff

Convert a profile DB to a tab-separated list of Neff scores.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs profile2neff <DB> [args] [options] (source-derived synopsis; run mmseqs profile2neff for exact syntax) |
| API layer                 | low_level_api                                                                                                        |
| Category flags            | COMMAND_PROFILE                                                                                                      |
| Upstream command count    | 0                                                                                                                    |
| Downstream command count  | 0                                                                                                                    |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                               |
| References                | Full CLI · Dependency entry                                                                                          |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.10. profile2pssm

Convert a profile DB to a tab-separated PSSM file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs profile2pssm <DB> [args] [options] (source-derived synopsis; run mmseqs profile2pssm for exact syntax) |
| API layer                 | low_level_api                                                                                                        |
| Category flags            | COMMAND_PROFILE                                                                                                      |
| Upstream command count    | 0                                                                                                                    |
| Downstream command count  | 0                                                                                                                    |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                               |
| References                | Full CLI · Dependency entry                                                                                          |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.11. profile2repseq

Extract representative sequence DB from a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs profile2repseq <DB><br>[args] [options] (source-derived synopsis;<br>run mmseqs profile2repseq for exact syntax) |
| API layer                 | low_level_api                                                                                                                  |
| Category flags            | COMMAND_PROFILE                                                                                                                |
| Upstream command count    | 0                                                                                                                              |
| Downstream command count  | 0                                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the<br>current dependency map.                                                      |
| References                | Full CLI · Dependency entry                                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.12. result2profile

Compute profile DB from a result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs result2profile<br><i:queryDB> <i:targetDB> <i:resultDB><br><o:profileDB> [options] |
| API layer                 | low_level_api                                                                                    |
| Category flags            | COMMAND_PROFILE                                                                                  |
| Upstream command count    | 1                                                                                                |
| Downstream command count  | 0                                                                                                |
| Related functional groups | search_workflows                                                                                 |
| References                | Full CLI · Dependency entry                                                                      |

#### 4.7.12.1. Key Options

| Option                 | Purpose                                                                |
|------------------------|------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)          |
| -e                     | List matches below this E-value (range 0.0-inf)                        |
| --gap-open             | Gap open cost                                                          |
| --gap-extend           | Gap extension cost                                                     |
| --mask-profile         | Mask query sequence of profile using tantan [0,1]                      |
| --e-profile            | Include sequences matches with < E-value thr. into the profile (>=0.0) |
| --wg                   | Use global sequence weighting for profile calculation                  |

#### 4.7.13. sequence2profile

Turn sequence into profile by adding context specific pseudo counts.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs sequence2profile <DB> [args] [options] (source-derived synopsis; run mmseqs sequence2profile for exact syntax) |
| API layer                 | low_level_api                                                                                                                |
| Category flags            | COMMAND_PROFILE                                                                                                              |
| Upstream command count    | 0                                                                                                                            |
| Downstream command count  | 0                                                                                                                            |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                       |
| References                | Full CLI · Dependency entry                                                                                                  |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.7.14. tsv2exprofiledb

Create an expandable profile db from TSV files.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 5 downstream call(s).

| Aspect                    | Value                                                                          |
|---------------------------|--------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs tsv2exprofiledb<br><i:tsvFilesBase> <o:exprofileDB><br>[options] |
| API layer                 | mid_level_api                                                                  |
| Category flags            | COMMAND_PROFILE_PROFILE                                                        |
| Upstream command count    | 0                                                                              |
| Downstream command count  | 5                                                                              |
| Related functional groups | database, utilities                                                            |
| References                | Full CLI · Dependency entry                                                    |

##### 4.7.14.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --gpu        | Use GPU (CUDA) if possible                                    |
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## 4.8. Database Management

Database lifecycle modules for creation, indexing, splitting, merging, and contract-preserving transforms.

### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### 4.8.1. aliasdb

Create relative symlink of DB to another name in the same folder.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                  |
|---------------------------|--------------------------------------------------------|
| Usage                     | usage: mmseqs aliasdb <i:srcDB><br><o:dstDB> [options] |
| API layer                 | low_level_api                                          |
| Category flags            | COMMAND_STORAGE                                        |
| Upstream command count    | 1                                                      |
| Downstream command count  | 0                                                      |
| Related functional groups | profiles                                               |
| References                | Full CLI · Dependency entry                            |

#### 4.8.1.1. Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.2. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                     |
|---------------------------|-----------------------------------------------------------|
| Usage                     | usage: mmseqs concatdbs <i:DB> <i:DB><br><o:DB> [options] |
| API layer                 | low_level_api                                             |
| Category flags            | COMMAND_SET                                               |
| Upstream command count    | 3                                                         |
| Downstream command count  | 0                                                         |
| Related functional groups | clustering, search_workflows                              |
| References                | Full CLI · Dependency entry                               |

#### 4.8.2.1. Key Options

| Option           | Purpose                                                                                    |
|------------------|--------------------------------------------------------------------------------------------|
| - -preserve-keys | The keys of the two DB should be distinct, and they will be preserved in the concatenation |

| Option              | Purpose                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| --take-larger-entry | Only keep the larger entry (dataSize >) in the concatenation, both databases need the same keys in the index |
| --compressed        | Write compressed output                                                                                      |
| --threads           | Number of CPU-cores used (all by default)                                                                    |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                |

#### 4.8.3. cpdb

Copy a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]                       |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_STORAGE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

##### 4.8.3.1. Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.4. createdb

Convert FASTA/Q file(s) to a sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 8 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs<br>createdb <i:fastaFile1[.gz .bz2]> ...<br><i:fastaFileN[.gz .bz2]> <i:stdin><br><o:sequenceDB> [options] |
| API layer                 | low_level_api                                                                                                            |
| Category flags            | COMMAND_DATABASE_CREATION                                                                                                |
| Upstream command count    | 8                                                                                                                        |
| Downstream command count  | 0                                                                                                                        |
| Related functional groups | easy_workflows, multi_hit                                                                                                |
| References                | Full CLI · Dependency entry                                                                                              |

#### 4.8.4.1. Key Options

| Option          | Purpose                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------|
| --dbtype        | Database type 0: auto, 1: amino acid 2: nucleotides                                                     |
| --shuffle       | Shuffle input database                                                                                  |
| --createdb-mode | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) |
| --id-offset     | Numeric ids in index file are offset by this value                                                      |
| --compressed    | Write compressed output                                                                                 |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                           |
| --write-lookup  | write .lookup file containing mapping from internal id, fasta id and file number                        |

#### 4.8.5. createindex

Store precomputed index on disk to reduce search overhead.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 4 downstream call(s).



| Aspect                    | Value                                                          |
|---------------------------|----------------------------------------------------------------|
| Usage                     | usage: mmseqs createindex<br><i:sequenceDB> <tmpDir> [options] |
| API layer                 | low_level_api                                                  |
| Category flags            | COMMAND_DATABASE_CREATION                                      |
| Upstream command count    | 0                                                              |
| Downstream command count  | 4                                                              |
| Related functional groups | sequence_manipulation                                          |
| References                | Full CLI · Dependency entry                                    |

#### 4.8.5.1. Key Options

| Option                  | Purpose                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| - -seed-sub-mat         | Substitution matrix file for k-mer generation                                                                |
| -k                      | k-mer length (0: automatically set to optimum)                                                               |
| - -alph-size            | Alphabet size (range 2-21)                                                                                   |
| - -comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                |
| - -comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| - -max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)                       |
| - -mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| - -mask-prob            | Mask sequences is probability is above threshold                                                             |

#### 4.8.6. createlinindex

Create linsearch index.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 4 downstream call(s).

| Aspect                    | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| Usage                     | usage: mmseqs createlinindex<br><i:sequenceDB> <tmpDir> [options] |
| API layer                 | low_level_api                                                     |
| Category flags            | COMMAND_DATABASE_CREATION<br>  COMMAND_EXPERT                     |
| Upstream command count    | 2                                                                 |
| Downstream command count  | 4                                                                 |
| Related functional groups | easy_workflows, sequence_manipulation                             |
| References                | Full CLI · Dependency entry                                       |

#### 4.8.6.1. Key Options

| Option               | Purpose                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                                                |
| -k                   | k-mer length (0: automatically set to optimum)                                                               |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| --alph-size          | Alphabet size (range 2-21)                                                                                   |
| --mask               | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob          | Mask sequences is probablity is above threshold                                                              |
| --mask-lower-case    | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat      | Repeat letters that occure > threshold in a two                                                              |

#### 4.8.7. createsubdb

Create a subset of a DB from list of DB keys.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                     |
|---------------------------|---------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createsubdb<br><i:subsetFile DB> <i:DB> <o:DB><br>[options] |
| API layer                 | low_level_api                                                             |
| Category flags            | COMMAND_SET                                                               |
| Upstream command count    | 5                                                                         |
| Downstream command count  | 0                                                                         |
| Related functional groups | clustering, search_workflows, taxonomy                                    |
| References                | Full CLI · Dependency entry                                               |

#### 4.8.7.1. Key Options

| Option       | Purpose                                                                     |
|--------------|-----------------------------------------------------------------------------|
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index                   |
| --id-mode    | Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info               |

#### 4.8.8. databases

List and download databases.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 8 downstream call(s).

| Aspect                    | Value                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs databases <DB> [args]<br>[options] (source-derived synopsis; run<br>mmseqs databases for exact syntax) |
| API layer                 | low_level_api                                                                                                        |
| Category flags            | COMMAND_DATABASE_CREATION                                                                                            |
| Upstream command count    | 0                                                                                                                    |
| Downstream command count  | 8                                                                                                                    |
| Related functional groups | profiles, taxonomy, utilities                                                                                        |

| Aspect     | Value                       |
|------------|-----------------------------|
| References | Full CLI · Dependency entry |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.8.9. db2tar

Archive contents of a DB to a tar archive.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs db2tar <i:DB><br><o:tar[.gz]> [options]                  |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_DATABASE_CREATION<br>  COMMAND_EXPERT                          |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

##### 4.8.9.1. Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.10. lndb

Symlink a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect         | Value                                               |
|----------------|-----------------------------------------------------|
| Usage          | usage: mmseqs lndb <i:srcDB> <o:dstDB><br>[options] |
| API layer      | low_level_api                                       |
| Category flags | COMMAND_STORAGE                                     |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.8.10.1. Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.11. mergedbs

Merge entries from multiple DBs.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                              |
|---------------------------|--------------------------------------------------------------------|
| Usage                     | usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options] |
| API layer                 | low_level_api                                                      |
| Category flags            | COMMAND_SET                                                        |
| Upstream command count    | 4                                                                  |
| Downstream command count  | 0                                                                  |
| Related functional groups | clustering, search_workflows                                       |
| References                | Full CLI · Dependency entry                                        |

#### 4.8.11.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --prefixes   | Comma separated list of prefixes for each entry               |

| Option                          | Purpose                                             |
|---------------------------------|-----------------------------------------------------|
| <code>--merge-stop-empty</code> | Don't continue merging entries after an empty entry |

#### 4.8.12. mvdb

Move a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                            |
|---------------------------|--------------------------------------------------|
| Usage                     | usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options] |
| API layer                 | low_level_api                                    |
| Category flags            | COMMAND_STORAGE                                  |
| Upstream command count    | 5                                                |
| Downstream command count  | 0                                                |
| Related functional groups | clustering, profiles, search_workflows, taxonomy |
| References                | Full CLI · Dependency entry                      |

##### 4.8.12.1. Key Options

| Option          | Purpose                                                       |
|-----------------|---------------------------------------------------------------|
| <code>-v</code> | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.13. renamedbkeys

Create a new DB with original keys renamed.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect         | Value                                                                  |
|----------------|------------------------------------------------------------------------|
| Usage          | usage: mmseqs renamedbkeys <i:idMapFile stdin> <i:DB> <o:DB> [options] |
| API layer      | low_level_api                                                          |
| Category flags | COMMAND_DB                                                             |

| Aspect                    | Value                       |
|---------------------------|-----------------------------|
| Upstream command count    | 2                           |
| Downstream command count  | 0                           |
| Related functional groups | clustering                  |
| References                | Full CLI · Dependency entry |

#### 4.8.13.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index     |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.14. rmdb

Remove a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 19 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs rmdb <DB> [args] [options]<br>(source-derived synopsis; run mmseqs rmdb for exact syntax) |
| API layer                 | low_level_api                                                                                           |
| Category flags            | COMMAND_STORAGE                                                                                         |
| Upstream command count    | 19                                                                                                      |
| Downstream command count  | 0                                                                                                       |
| Related functional groups | clustering, easy_workflows, multi_hit, profiles, search_workflows, taxonomy                             |
| References                | Full CLI · Dependency entry                                                                             |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.8.15. splitdb

Split DB into subsets.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs splitdb <i:DB> <o:DB> [options]                          |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_SET                                                            |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.8.15.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --split      | Split input into N equally distributed chunks                 |
| --split-aa   | Try to find the best split boundaries by entry lengths        |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.8.16. splitsequence

Split sequences by length.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                 | Value                                                               |
|------------------------|---------------------------------------------------------------------|
| Usage                  | usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options] |
| API layer              | low_level_api                                                       |
| Category flags         | COMMAND_SEQUENCE                                                    |
| Upstream command count | 3                                                                   |



| Aspect                    | Value                       |
|---------------------------|-----------------------------|
| Downstream command count  | 0                           |
| Related functional groups | search_workflows            |
| References                | Full CLI · Dependency entry |

#### 4.8.16.1. Key Options

| Option                | Purpose                                                                  |
|-----------------------|--------------------------------------------------------------------------|
| --sequence-overlap    | Overlap between sequences                                                |
| --sequence-split-mode | Sequence split mode 0: copy data, 1: soft link data and write new index, |
| --headers-split-mode  | Header split mode: 0: split position, 1: original header                 |
| --max-seq-len         | Maximum sequence length                                                  |
| --threads             | Number of CPU-cores used (all by default)                                |
| --compressed          | Write compressed output                                                  |
| -v                    | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info            |
| --create-lookup       | Create database lookup file (can be very large)                          |

#### 4.8.17. subtractdbs

Remove all entries from first DB occurring in second DB by key.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs subtractdbs<br><i:resultDBLeft> <i:resultDBRight><br><o:resultDB> [options] |
| API layer                 | low_level_api                                                                             |
| Category flags            | COMMAND_SET                                                                               |
| Upstream command count    | 2                                                                                         |
| Downstream command count  | 0                                                                                         |
| Related functional groups | clustering, search_workflows                                                              |
| References                | Full CLI · Dependency entry                                                               |

#### 4.8.17.1. Key Options

| Option       | Purpose                                                                |
|--------------|------------------------------------------------------------------------|
| -e           | List matches below this E-value (range 0.0-inf)                        |
| --e-profile  | Include sequences matches with < E-value thr. into the profile (>=0.0) |
| --threads    | Number of CPU-cores used (all by default)                              |
| --compressed | Write compressed output                                                |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info          |

#### 4.8.18. swapdb

Transpose DB with integer values in first column.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                    |
|---------------------------|----------------------------------------------------------|
| Usage                     | usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options] |
| API layer                 | low_level_api                                            |
| Category flags            | COMMAND_DB                                               |
| Upstream command count    | 4                                                        |
| Downstream command count  | 0                                                        |
| Related functional groups | clustering, multi_hit, taxonomy                          |
| References                | Full CLI · Dependency entry                              |

#### 4.8.18.1. Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --threads            | Number of CPU-cores used (all by default)                                                    |
| --compressed         | Write compressed output                                                                      |
| -v                   | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                |

#### 4.8.19. tar2db

Convert content of tar archives to any DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                        |
|---------------------------|------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs tar2db <i:tar[.gz]> ...<br><i:tar[.gz]> <o:resultDB> [options] |
| API layer                 | low_level_api                                                                |
| Category flags            | COMMAND_DATABASE_CREATION<br>  COMMAND_EXPERT                                |
| Upstream command count    | 1                                                                            |
| Downstream command count  | 0                                                                            |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.       |
| References                | Full CLI · Dependency entry                                                  |

##### 4.8.19.1. Key Options

| Option          | Purpose                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --output-dbtype | Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Pre-filtering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| --tar-include   | Include file names based on this regex                                                                                                                                                                                                                                                                                                        |
| --tar-exclude   | Exclude file names based on this regex                                                                                                                                                                                                                                                                                                        |
| --compressed    | Write compressed output                                                                                                                                                                                                                                                                                                                       |
| --threads       | Number of CPU-cores used (all by default)                                                                                                                                                                                                                                                                                                     |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                                                                                                                                                                                                                                 |

#### 4.8.20. tsv2db

Convert a TSV file to any DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                            |
|---------------------------|----------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs tsv2db &lt;i:tsvFile&gt; &lt;o:resultDB&gt; [options]</code> |
| API layer                 | <code>low_level_api</code>                                                       |
| Category flags            | <code>COMMAND_DATABASE_CREATION</code><br><code>  COMMAND_EXPERT</code>          |
| Upstream command count    | 4                                                                                |
| Downstream command count  | 0                                                                                |
| Related functional groups | <code>clustering, multi_hit, profiles</code>                                     |
| References                | Full CLI · Dependency entry                                                      |

#### 4.8.20.1. Key Options

| Option                          | Purpose                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--add-self-matches</code> | Artificially add entries of queries with themselves (for clustering)                                                                                                                                                                                                                                                                          |
| <code>--output-dbtype</code>    | Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Pre-filtering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| <code>--compressed</code>       | Write compressed output                                                                                                                                                                                                                                                                                                                       |
| <code>-v</code>                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                                                                                                                                                                                                                                 |

### 4.9. Sequence Manipulation

Sequence-level transform modules for ORFs, frames, masking, coordinate transforms, and related preprocessing.

**Note**

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

### Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

#### 4.9.1. extractalignedregion

Extract aligned sequence region from query.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs extractalignedregion<br><i:queryDB> <i:targetDB> <i:resultDB><br><o:sequenceDB> [options] |
| API layer                 | low_level_api                                                                                           |
| Category flags            | COMMAND_SEQUENCE                                                                                        |
| Upstream command count    | 0                                                                                                       |
| Downstream command count  | 0                                                                                                       |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                  |
| References                | Full CLI · Dependency entry                                                                             |

##### 4.9.1.1. Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --extract-mode | Extract from 1: Query, 2: Target                                |
| --compressed   | Write compressed output                                         |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads      | Number of CPU-cores used (all by default)                       |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### 4.9.2. extractframes

Extract frames from a nucleotide sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs extractframes<br><i:sequenceDB> <o:sequenceDB> [options] |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_SEQUENCE                                                       |
| Upstream command count    | 4                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | clustering, database, search_workflows                                 |
| References                | Full CLI · Dependency entry                                            |

##### 4.9.2.1. Key Options

| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --forward-frames    | Comma-separated list of frames on the forward strand to be extracted                                                                |
| --reverse-frames    | Comma-separated list of frames on the reverse strand to be extracted                                                                |
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --translate         | Translate ORF to amino acid                                                                                                         |
| --threads           | Number of CPU-cores used (all by default)                                                                                           |
| --compressed        | Write compressed output                                                                                                             |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                       |
| --create-lookup     | Create database lookup file (can be very large)                                                                                     |

#### 4.9.3. extractorfs

Six-frame extraction of open reading frames.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 6 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs extractorfs &lt;i:sequenceDB&gt; &lt;o:sequenceDB&gt; [options]</code> |
| API layer                 | <code>low_level_api</code>                                                                 |
| Category flags            | <code>COMMAND_SEQUENCE</code>                                                              |
| Upstream command count    | 6                                                                                          |
| Downstream command count  | 0                                                                                          |
| Related functional groups | database, multi_hit, search_workflows, taxonomy                                            |
| References                | Full CLI · Dependency entry                                                                |

#### 4.9.3.1. Key Options

| Option                           | Purpose                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>--min-length</code>        | Minimum codon number in open reading frames                                                                                     |
| <code>--max-length</code>        | Maximum codon number in open reading frames                                                                                     |
| <code>--max-gaps</code>          | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected                                 |
| <code>--contig-start-mode</code> | Contig start can be 0: incomplete, 1: complete, 2: both                                                                         |
| <code>--contig-end-mode</code>   | Contig end can be 0: incomplete, 1: complete, 2: both                                                                           |
| <code>--orf-start-mode</code>    | Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle) |
| <code>--forward-frames</code>    | Comma-separated list of frames on the forward strand to be extracted                                                            |
| <code>--reverse-frames</code>    | Comma-separated list of frames on the reverse strand to be extracted                                                            |

#### 4.9.4. masksequence

Soft mask sequence DB using tantan.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]     |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_SEQUENCE                                                       |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

##### 4.9.4.1. Key Options

| Option            | Purpose                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| --mask            | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob       | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat   | Repeat letters that occur > threshold in a row                                                               |
| --threads         | Number of CPU-cores used (all by default)                                                                    |
| --compressed      | Write compressed output                                                                                      |
| -v                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                |

#### 4.9.5. orftocontig

Write ORF locations in alignment format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).



| Aspect                    | Value                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs<br>orftocontig <i:contigsSequenceDB><br><i:extractedOrfsHeadersDB><br><o:orfsAlignedToContigDB> [options] |
| API layer                 | low_level_api                                                                                                           |
| Category flags            | COMMAND_SEQUENCE                                                                                                        |
| Upstream command count    | 1                                                                                                                       |
| Downstream command count  | 0                                                                                                                       |
| Related functional groups | multi_hit                                                                                                               |
| References                | Full CLI · Dependency entry                                                                                             |

#### 4.9.5.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.9.6. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                              |
|---------------------------|------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs recoverlongestorf<br><i:orfDB> <i:resultDB> <o:tsvFile><br>[options] |
| API layer                 | low_level_api                                                                      |
| Category flags            | COMMAND_EXPERT                                                                     |
| Upstream command count    | 1                                                                                  |
| Downstream command count  | 0                                                                                  |
| Related functional groups | taxonomy                                                                           |
| References                | Full CLI · Dependency entry                                                        |

#### 4.9.6.1. Key Options

| Option     | Purpose                                                       |
|------------|---------------------------------------------------------------|
| - -threads | Number of CPU-cores used (all by default)                     |
| -v         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.9.7. reverseseq

Reverse (without complement) sequences.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                     |
|---------------------------|---------------------------------------------------------------------------|
| Usage                     | usage: mmseqs reverseseq <i:sequenceDB><br><o:revSequenceDB> [options]    |
| API layer                 | low_level_api                                                             |
| Category flags            | COMMAND_SEQUENCE                                                          |
| Upstream command count    | 0                                                                         |
| Downstream command count  | 0                                                                         |
| Related functional groups | No direct cross-group coupling detected in the<br>current dependency map. |
| References                | Full CLI · Dependency entry                                               |

#### 4.9.7.1. Key Options

| Option        | Purpose                                                       |
|---------------|---------------------------------------------------------------|
| - -threads    | Number of CPU-cores used (all by default)                     |
| - -compressed | Write compressed output                                       |
| -v            | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.9.8. translateaa

Translate proteins to lexicographically lowest codons.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs translateaa<br><i:sequenceDB> <o:sequenceDB> [options]   |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_SEQUENCE                                                       |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.9.8.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.9.9. translatenucs

Translate nucleotides to proteins.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs translatenucs<br><i:sequenceDB> <o:sequenceDB> [options] |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_SEQUENCE                                                       |
| Upstream command count    | 1                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | multi_hit                                                              |
| References                | Full CLI · Dependency entry                                            |

#### 4.9.9.1. Key Options

| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --add-orf-stop      | Add stop codon '*' at complete start and end                                                                                        |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                       |
| --compressed        | Write compressed output                                                                                                             |
| --threads           | Number of CPU-cores used (all by default)                                                                                           |

### 4.10. Result Handling

Modules that filter, summarize, reshape, and export result databases for downstream analysis.

#### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

#### 4.10.1. convert2fasta

Convert sequence DB to FASTA format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect         | Value                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Usage          | usage: mmseqs convert2fasta <DB> [args]<br>[options] (source-derived synopsis; run<br>mmseqs convert2fasta for exact syntax) |
| API layer      | low_level_api                                                                                                                |
| Category flags | COMMAND_FORMAT_CONVERSION                                                                                                    |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.10.2. convertalis

Convert alignment DB to BLAST-tab, SAM or custom format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs convertalis <i:queryDb><br><i:targetDb> <i:alignmentDB><br><o:alignmentFile> [options] |
| API layer                 | low_level_api                                                                                        |
| Category flags            | COMMAND_FORMAT_CONVERSION                                                                            |
| Upstream command count    | 4                                                                                                    |
| Downstream command count  | 0                                                                                                    |
| Related functional groups | easy_workflows                                                                                       |
| References                | Full CLI · Dependency entry                                                                          |

##### 4.10.2.1. Key Options

| Option          | Purpose                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| --gap-open      | Gap open cost                                                                                                                      |
| --gap-extend    | Gap extension cost                                                                                                                 |
| --format-mode   | Output format:                                                                                                                     |
| --format-output | Choose comma separated list of<br>output columns from: query,tar-<br>get,evalue,gapopen,pident,fident,nident,qs-<br>tart,qend,qlen |

| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --search-type       | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment                                 |
| --sub-mat           | Substitution matrix file                                                                                                            |
| --db-load-mode      | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                                                                     |

#### 4.10.3. createseqfiledb

Create a DB of unaligned FASTA entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createseqfiledb<br><i:sequenceDB> <i:resultDB> <o:fastaDB><br>[options] |
| API layer                 | low_level_api                                                                         |
| Category flags            | COMMAND_FORMAT_CONVERSION<br>  COMMAND_EXPERT                                         |
| Upstream command count    | 2                                                                                     |
| Downstream command count  | 0                                                                                     |
| Related functional groups | easy_workflows                                                                        |
| References                | Full CLI · Dependency entry                                                           |

##### 4.10.3.1. Key Options

| Option          | Purpose                                           |
|-----------------|---------------------------------------------------|
| --min-sequences | Minimum number of sequences a cluster may contain |
| --max-sequences | Maximum number of sequences a cluster may contain |

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --hh-format    | Format entries to use with hhsuite (for single-ton clusters)    |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads      | Number of CPU-cores used (all by default)                       |
| --compressed   | Write compressed output                                         |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### 4.10.4. createtsv

Convert result DB to tab-separated flat file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile> [options] |
| API layer                 | low_level_api                                                                         |
| Category flags            | COMMAND_FORMAT_CONVERSION                                                             |
| Upstream command count    | 3                                                                                     |
| Downstream command count  | 0                                                                                     |
| Related functional groups | easy_workflows                                                                        |
| References                | Full CLI · Dependency entry                                                           |

##### 4.10.4.1. Key Options

| Option              | Purpose                                                                    |
|---------------------|----------------------------------------------------------------------------|
| --first-seq-as-repr | Use the first sequence of the clustering result as representative sequence |
| --target-column     | Select a target column (default 1), 0 if no target id exists               |
| --full-header       | Replace DB ID by its corresponding Full Header                             |

| Option                     | Purpose                                                       |
|----------------------------|---------------------------------------------------------------|
| <code>--idx-seq-src</code> | 0: auto, 1: split/translated sequences, 2: input sequences    |
| <code>--threads</code>     | Number of CPU-cores used (all by default)                     |
| <code>--compressed</code>  | Write compressed output                                       |
| <code>-v</code>            | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| <code>--db-output</code>   | Return a result DB instead of a text file                     |

#### 4.10.5. **extractdomains**

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs extractdomains &lt;DB&gt; [args] [options] (source-derived synopsis; run mmseqs extractdomains for exact syntax)</code> |
| API layer                 | <code>low_level_api</code>                                                                                                                  |
| Category flags            | <code>COMMAND_SPECIAL</code>                                                                                                                |
| Upstream command count    | 0                                                                                                                                           |
| Downstream command count  | 0                                                                                                                                           |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                                      |
| References                | Full CLI · Dependency entry                                                                                                                 |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.10.6. **filterresult**

Pairwise alignment result filter.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).



| Aspect                    | Value                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs filterresult <i:queryDB><br><i:targetDB> <i:resultDB> <o:resultDB><br>[options] |
| API layer                 | low_level_api                                                                                 |
| Category flags            | COMMAND_RESULT                                                                                |
| Upstream command count    | 1                                                                                             |
| Downstream command count  | 0                                                                                             |
| Related functional groups | search_workflows                                                                              |
| References                | Full CLI · Dependency entry                                                                   |

#### 4.10.6.1. Key Options

| Option                 | Purpose                                                                         |
|------------------------|---------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                   |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                   |
| --add-self-matches     | keep the query (representative) sequence                                        |
| --gap-open             | Gap open cost                                                                   |
| --gap-extend           | Gap extension cost                                                              |
| --filter-min-enable    | Only filter MSAs with more than N sequences, 0 always filters                   |
| --max-seq-id           | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] |
| --qid                  | Reduce diversity of output MSAs using min.seq. identity with query sequences    |

#### 4.10.7. result2dnamsa

Compute MSA DB with out insertions in the query for DNA sequences.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs result2dnamsa <i:queryDB><br><i:targetDB> <i:resultDB> <o:msaDB><br>[options] |
| API layer                 | low_level_api                                                                               |
| Category flags            | COMMAND_RESULT                                                                              |
| Upstream command count    | 0                                                                                           |
| Downstream command count  | 0                                                                                           |
| Related functional groups | No direct cross-group coupling detected in the<br>current dependency map.                   |
| References                | Full CLI · Dependency entry                                                                 |

#### 4.10.7.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --skip-query | Skip the query sequence                                       |

#### 4.10.8. result2flat

Create flat file by adding FASTA headers to DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs result2flat <i:queryDB><br><i:targetDB> <i:resultDB> <o:fastaDB><br>[options] |
| API layer                 | low_level_api                                                                               |
| Category flags            | COMMAND_FORMAT_CONVERSION<br>  COMMAND_EXPERT                                               |
| Upstream command count    | 2                                                                                           |
| Downstream command count  | 0                                                                                           |
| Related functional groups | easy_workflows                                                                              |
| References                | Full CLI · Dependency entry                                                                 |

#### 4.10.8.1. Key Options

| Option                          | Purpose                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>--use-fasta-header</code> | Use the id parsed from the fasta header as the index key instead of using incrementing numeric identifiers |
| <code>-v</code>                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                              |

#### 4.10.9. **result2msa**

Compute MSA DB from a result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs result2msa &lt;i:queryDB&gt; &lt;i:targetDB&gt; &lt;i:resultDB&gt; &lt;o:msaDB&gt; [options]</code> |
| API layer                 | <code>low_level_api</code>                                                                                              |
| Category flags            | <code>COMMAND_RESULT</code>                                                                                             |
| Upstream command count    | 1                                                                                                                       |
| Downstream command count  | 0                                                                                                                       |
| Related functional groups | <code>clustering</code>                                                                                                 |
| References                | Full CLI · Dependency entry                                                                                             |

#### 4.10.9.1. Key Options

| Option                              | Purpose                                                       |
|-------------------------------------|---------------------------------------------------------------|
| <code>--comp-bias-corr</code>       | Correct for locally biased amino acid composition (range 0-1) |
| <code>--comp-bias-corr-scale</code> | Correct for locally biased amino acid composition (range 0-1) |
| <code>--gap-open</code>             | Gap open cost                                                 |
| <code>--gap-extend</code>           | Gap extension cost                                            |
| <code>--filter-msa</code>           | Filter msa: 0: do not filter, 1: filter                       |
| <code>--filter-min-enable</code>    | Only filter MSAs with more than N sequences, 0 always filters |

| Option                    | Purpose                                                                         |
|---------------------------|---------------------------------------------------------------------------------|
| <code>--max-seq-id</code> | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] |
| <code>--qid</code>        | Reduce diversity of output MSAs using min.seq. identity with query sequences    |

#### 4.10.10. **result2rbh**

Filter a merged result DB to retain only reciprocal best hits.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                        |
|---------------------------|--------------------------------------------------------------|
| Usage                     | usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options] |
| API layer                 | low_level_api                                                |
| Category flags            | COMMAND_RESULT                                               |
| Upstream command count    | 1                                                            |
| Downstream command count  | 0                                                            |
| Related functional groups | search_workflows                                             |
| References                | Full CLI · Dependency entry                                  |

##### 4.10.10.1. Key Options

| Option                    | Purpose                                                       |
|---------------------------|---------------------------------------------------------------|
| <code>--threads</code>    | Number of CPU-cores used (all by default)                     |
| <code>--compressed</code> | Write compressed output                                       |
| <code>-v</code>           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.10.11. **result2repseq**

Get representative sequences from result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs result2repseq <i:sequenceDB><br><i:resultDB> <o:sequenceDb> [options] |
| API layer                 | low_level_api                                                                          |
| Category flags            | COMMAND_RESULT                                                                         |
| Upstream command count    | 3                                                                                      |
| Downstream command count  | 0                                                                                      |
| Related functional groups | clustering, easy_workflows                                                             |
| References                | Full CLI · Dependency entry                                                            |

#### 4.10.11.1. Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --compressed   | Write compressed output                                         |
| --threads      | Number of CPU-cores used (all by default)                       |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### 4.10.12. result2stats

Compute statistics for each entry in a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs result2stats <i:queryDB><br><i:targetDB> <i:resultDB> <o:statsDB><br>[options] |
| API layer                 | low_level_api                                                                                |
| Category flags            | COMMAND_RESULT                                                                               |
| Upstream command count    | 2                                                                                            |
| Downstream command count  | 0                                                                                            |
| Related functional groups | multi_hit, search_workflows                                                                  |

| Aspect     | Value                       |
|------------|-----------------------------|
| References | Full CLI · Dependency entry |

#### 4.10.12.1. Key Options

| Option       | Purpose                                                                  |
|--------------|--------------------------------------------------------------------------|
| --stat       | One of: linecount, mean, min, max, doolittle, charges, seqlen, firstline |
| --tsv        | Return output in TSV format                                              |
| --compressed | Write compressed output                                                  |
| --threads    | Number of CPU-cores used (all by default)                                |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info            |

#### 4.10.13. sortresult

Sort a result DB in the same order as the prefilter or align module.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]          |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_RESULT                                                         |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.10.13.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.10.14. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                        |
|---------------------------|------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs summarizealis<br><i:alignmentDB> <o:summerizedDB><br>[options] |
| API layer                 | low_level_api                                                                |
| Category flags            | COMMAND_RESULT                                                               |
| Upstream command count    | 1                                                                            |
| Downstream command count  | 0                                                                            |
| Related functional groups | easy_workflows                                                               |
| References                | Full CLI · Dependency entry                                                  |

##### 4.10.14.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.10.15. summarizeheaders

Summarize FASTA headers of result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                 | Value                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Usage                  | usage: mmseqs summarizeheaders <DB><br>[args] [options] (source-derived synopsis;<br>run mmseqs summarizeheaders for exact syn-<br>tax) |
| API layer              | low_level_api                                                                                                                           |
| Category flags         | COMMAND_SPECIAL                                                                                                                         |
| Upstream command count | 0                                                                                                                                       |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.10.16. summarizeresult

Extract annotations from alignment DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                         |
|---------------------------|-------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs summarizeresult<br><i:alignmentDB> <o:alignmentDB><br>[options] |
| API layer                 | low_level_api                                                                 |
| Category flags            | COMMAND_RESULT                                                                |
| Upstream command count    | 2                                                                             |
| Downstream command count  | 0                                                                             |
| Related functional groups | easy_workflows                                                                |
| References                | Full CLI · Dependency entry                                                   |

##### 4.10.16.1. Key Options

| Option       | Purpose                                                                        |
|--------------|--------------------------------------------------------------------------------|
| -a           | Add backtrace string (convert to alignments with mmseqs convertalis module)    |
| -c           | List matches above this fraction of aligned (covered) residues (see -cov-mode) |
| --overlap    | Maximum overlap of covered regions                                             |
| --threads    | Number of CPU-cores used (all by default)                                      |
| --compressed | Write compressed output                                                        |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                  |



#### 4.10.17. swapresults

Transpose prefilter/alignment DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options] |
| API layer                 | low_level_api                                                                          |
| Category flags            | COMMAND_RESULT                                                                         |
| Upstream command count    | 4                                                                                      |
| Downstream command count  | 0                                                                                      |
| Related functional groups | easy_workflows, search_workflows                                                       |
| References                | Full CLI · Dependency entry                                                            |

##### 4.10.17.1. Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| -e                   | List matches below this E-value (range 0.0-inf)                                              |
| --gap-open           | Gap open cost                                                                                |
| --gap-extend         | Gap extension cost                                                                           |
| --sub-mat            | Substitution matrix file                                                                     |
| --threads            | Number of CPU-cores used (all by default)                                                    |
| --compressed         | Write compressed output                                                                      |
| --db-load-mode       | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                              |

#### 4.11. Multi-hit

Set-based modules that aggregate sequence-level evidence into per-set statistics and outputs.

##### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### 4.11.1. **besthitperset**

For each set of sequences compute the best element and update p-value.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| Usage                     | <code>usage:<br/>mmseqs besthitperset &lt;i:targetSetDB&gt;<br/>&lt;i:resultDB&gt; &lt;o:resultDB&gt; [options]</code> |
| API layer                 | <code>high_level_api</code>                                                                                            |
| Category flags            | <code>COMMAND_MULTIHIT</code>                                                                                          |
| Upstream command count    | 1                                                                                                                      |
| Downstream command count  | 0                                                                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                 |
| References                | Full CLI · Dependency entry                                                                                            |

##### 4.11.1.1. Key Options

| Option                         | Purpose                                                                  |
|--------------------------------|--------------------------------------------------------------------------|
| <code>--simple-best-hit</code> | Update the p-value by a single best hit, or by best and second best hits |
| <code>--threads</code>         | Number of CPU-cores used (all by default)                                |
| <code>--compressed</code>      | Write compressed output                                                  |
| <code>-v</code>                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info            |

#### 4.11.2. **combinepvalperset**

For each set compute the combined p-value.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect | Value                                                                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage  | <code>usage:<br/>mmseqs combinepvalperset &lt;i:querySetDB&gt;<br/>&lt;i:targetSetDB&gt; &lt;i:resultDB&gt; &lt;o:pvalDB&gt;<br/>[options]</code> |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| API layer                 | high_level_api                                                         |
| Category flags            | COMMAND_MULTIHIT                                                       |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.11.2.1. Key Options

| Option             | Purpose                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| --alpha            | Set alpha for combining p-values during aggregation                                                                    |
| --aggregation-mode | Combined P-values computed from 0: multi-hit, 1: minimum of all P-values, 2: product-of-P-values, 3: truncated product |
| --threads          | Number of CPU-cores used (all by default)                                                                              |
| --compressed       | Write compressed output                                                                                                |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                          |

#### 4.11.3. mergeresultsbyset

Merge results from multiple ORFs back to their respective contig.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| Usage                     | usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options] |
| API layer                 | high_level_api                                                    |
| Category flags            | COMMAND_MULTIHIT                                                  |
| Upstream command count    | 2                                                                 |
| Downstream command count  | 0                                                                 |
| Related functional groups | taxonomy                                                          |

| Aspect     | Value                       |
|------------|-----------------------------|
| References | Full CLI · Dependency entry |

#### 4.11.3.1. Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads      | Number of CPU-cores used (all by default)                       |
| --compressed   | Write compressed output                                         |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### 4.11.4. multihitdb

Create sequence DB for multi hit searches.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 8 downstream call(s).

| Aspect                    | Value                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs multihitdb <i:fastaFile1[.gz bz2]> ... <i:fastaFileN[.gz bz2]><br><o:setDB> <tmpDir> [options] |
| API layer                 | high_level_api                                                                                                  |
| Category flags            | COMMAND_MULTIHIT                                                                                                |
| Upstream command count    | 0                                                                                                               |
| Downstream command count  | 8                                                                                                               |
| Related functional groups | database, result_handling,<br>sequence_manipulation, utilities                                                  |
| References                | Full CLI · Dependency entry                                                                                     |

#### 4.11.4.1. Key Options

| Option    | Purpose                                             |
|-----------|-----------------------------------------------------|
| --dbtype  | Database type 0: auto, 1: amino acid 2: nucleotides |
| --shuffle | Shuffle input database                              |

| Option              | Purpose                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------|
| --createdb-mode     | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) |
| --id-offset         | Numeric ids in index file are offset by this value                                                      |
| --min-length        | Minimum codon number in open reading frames                                                             |
| --max-length        | Maximum codon number in open reading frames                                                             |
| --max-gaps          | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected         |
| --contig-start-mode | Contig start can be 0: incomplete, 1: complete, 2: both                                                 |

#### 4.11.5. multihitsearch

Search with a grouped set of sequences against another grouped set.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 4 downstream call(s).

| Aspect                    | Value                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs multihitsearch <i:querySetDB><br><i:targetSetDB> <o:resultDB> <tmpDir><br>[options] |
| API layer                 | high_level_api                                                                                       |
| Category flags            | COMMAND_MULTIHIT                                                                                     |
| Upstream command count    | 0                                                                                                    |
| Downstream command count  | 4                                                                                                    |
| Related functional groups | database, search_workflows                                                                           |
| References                | Full CLI · Dependency entry                                                                          |

#### 4.11.5.1. Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

#### 4.12. Taxonomy

Modules for taxonomy database preparation, assignment, filtering, and report generation.

##### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

##### Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

##### 4.12.1. addtaxonomy

Add taxonomic labels to result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs addtaxonomy <DB> [args]<br>[options] (source-derived synopsis; run<br>mmseqs addtaxonomy for exact syntax) |
| API layer                 | low_level_api                                                                                                            |
| Category flags            | COMMAND_TAXONOMY   COMMAND_EXPERT                                                                                        |
| Upstream command count    | 1                                                                                                                        |
| Downstream command count  | 0                                                                                                                        |
| Related functional groups | easy_workflows                                                                                                           |
| References                | Full CLI · Dependency entry                                                                                              |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.2. **aggregatetax**

Aggregate multiple taxon labels to a single label.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs aggregatetax <DB> [args]<br>[options] (source-derived synopsis; run<br>mmseqs aggregatetax for exact syntax) |
| API layer                 | low_level_api                                                                                                              |
| Category flags            | COMMAND_TAXONOMY                                                                                                           |
| Upstream command count    | 1                                                                                                                          |
| Downstream command count  | 0                                                                                                                          |
| Related functional groups | No direct cross-group coupling detected in the<br>current dependency map.                                                  |
| References                | Full CLI · Dependency entry                                                                                                |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.3. **aggregatetaxweights**

Aggregate multiple taxon labels to a single label.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs aggregatetaxweights <DB> [args] [options] (source-derived synopsis; run mmseqs aggregatetaxweights for exact syntax) |
| API layer                 | low_level_api                                                                                                                      |
| Category flags            | COMMAND_TAXONOMY                                                                                                                   |
| Upstream command count    | 1                                                                                                                                  |
| Downstream command count  | 0                                                                                                                                  |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                             |
| References                | Full CLI · Dependency entry                                                                                                        |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.4. createbintaxmapping

Create binary taxonomy mapping from tabular taxonomy mapping.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createbintaxmapping <DB> [args] [options] (source-derived synopsis; run mmseqs createbintaxmapping for exact syntax) |
| API layer                 | low_level_api                                                                                                                      |
| Category flags            | COMMAND_TAXONOMY   COMMAND_EXPERT                                                                                                  |
| Upstream command count    | 0                                                                                                                                  |
| Downstream command count  | 0                                                                                                                                  |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                             |
| References                | Full CLI · Dependency entry                                                                                                        |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.5. createbintaxonomy

Create binary taxonomy from NCBI input.



Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createbintaxonomy <DB> [args] [options] (source-derived synopsis; run mmseqs createbintaxonomy for exact syntax) |
| API layer                 | low_level_api                                                                                                                  |
| Category flags            | COMMAND_TAXONOMY   COMMAND_EXPERT                                                                                              |
| Upstream command count    | 1                                                                                                                              |
| Downstream command count  | 0                                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                         |
| References                | Full CLI · Dependency entry                                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.6. createdmptaxonomy

Create dmp files from binary taxonomy.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs createdmptaxonomy <DB> [args] [options] (source-derived synopsis; run mmseqs createdmptaxonomy for exact syntax) |
| API layer                 | low_level_api                                                                                                                  |
| Category flags            | COMMAND_TAXONOMY   COMMAND_EXPERT                                                                                              |
| Upstream command count    | 0                                                                                                                              |
| Downstream command count  | 0                                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                         |
| References                | Full CLI · Dependency entry                                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.7. createtaxdb

Add taxonomic labels to sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 1 downstream call(s).

| Aspect                    | Value                                                          |
|---------------------------|----------------------------------------------------------------|
| Usage                     | usage: mmseqs createtaxdb<br><i:sequenceDB> <tmpDir> [options] |
| API layer                 | low_level_api                                                  |
| Category flags            | COMMAND_TAXONOMY                                               |
| Upstream command count    | 1                                                              |
| Downstream command count  | 1                                                              |
| Related functional groups | database                                                       |
| References                | Full CLI · Dependency entry                                    |

##### 4.12.7.1. Key Options

| Option             | Purpose                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| --ncbi-tax-dump    | tax dump directory. The tax dump can be downloaded here “ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz” |
| --tax-mapping-file | File to map sequence identifier to taxonomical identifier                                                        |
| --tax-mapping-mode | Map taxonomy based on sequence database<br>0: .lookup file 1: .source file                                       |
| --tax-db-mode      | Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin)                  |
| --threads          | Number of CPU-cores used (all by default)                                                                        |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                    |

#### 4.12.8. filtertaxdb

Filter taxonomy result database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]   |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_TAXONOMY                                                       |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.12.8.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --taxon-list | Taxonomy ID, possibly multiple values separated by ‘,’        |
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.12.9. filtertaxseqdb

Filter taxonomy sequence database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                 | Value                                                            |
|------------------------|------------------------------------------------------------------|
| Usage                  | usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options] |
| API layer              | low_level_api                                                    |
| Category flags         | COMMAND_TAXONOMY                                                 |
| Upstream command count | 0                                                                |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.12.9.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --taxon-list | Taxonomy ID, possibly multiple values separated by ‘,’        |
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index     |
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.12.10. lca

Compute the lowest common ancestor.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                            |
|---------------------------|------------------------------------------------------------------|
| Usage                     | usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options] |
| API layer                 | low_level_api                                                    |
| Category flags            | COMMAND_TAXONOMY                                                 |
| Upstream command count    | 2                                                                |
| Downstream command count  | 0                                                                |
| Related functional groups | easy_workflows                                                   |
| References                | Full CLI · Dependency entry                                      |

#### 4.12.10.1. Key Options

| Option        | Purpose                                                            |
|---------------|--------------------------------------------------------------------|
| --lca-ranks   | Add column with specified ranks (',' separated)                    |
| --blacklist   | Comma separated list of ignored taxa in LCA computation            |
| --tax-lineage | 0: don't show, 1: add all lineage names, 2: add all lineage taxids |
| --compressed  | Write compressed output                                            |
| --threads     | Number of CPU-cores used (all by default)                          |
| -v            | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info      |

#### 4.12.11. lcaalign

Efficient gapped alignment for lca computation.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs lcaalign<br><i:queryDB> <i:targetDB> <i:resultDB><br><o:alignmentDB> [options] |
| API layer                 | low_level_api                                                                                |
| Category flags            | COMMAND_TAXONOMY                                                                             |
| Upstream command count    | 1                                                                                            |
| Downstream command count  | 0                                                                                            |
| Related functional groups | search_workflows                                                                             |
| References                | Full CLI · Dependency entry                                                                  |

#### 4.12.11.1. Key Options

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

| Option                               | Purpose                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>--add-self-matches</code>      | Artificially add entries of queries with themselves (for clustering)                                                     |
| <code>-a</code>                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |
| <code>--alignment-mode</code>        | How to compute the alignment:                                                                                            |
| <code>--alignment-output-mode</code> | How to compute the alignment:                                                                                            |
| <code>--wrapped-scoring</code>       | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| <code>-e</code>                      | List matches below this E-value (range 0.0-inf)                                                                          |

#### 4.12.12. **majoritylca**

Compute the lowest common ancestor using majority voting.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs majoritylca <DB> [args] [options] (source-derived synopsis; run mmseqs majoritylca for exact syntax) |
| API layer                 | low_level_api                                                                                                      |
| Category flags            | COMMAND_TAXONOMY   COMMAND_EXPERT                                                                                  |
| Upstream command count    | 0                                                                                                                  |
| Downstream command count  | 0                                                                                                                  |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                             |
| References                | Full CLI · Dependency entry                                                                                        |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.13. **nrtotaxmapping**

Create taxonomy mapping for NR database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs nrtotaxmapping <DB> [args] [options] (source-derived synopsis; run mmseqs nrtotaxmapping for exact syntax) |
| API layer                 | low_level_api                                                                                                            |
| Category flags            | COMMAND_SPECIAL                                                                                                          |
| Upstream command count    | 1                                                                                                                        |
| Downstream command count  | 0                                                                                                                        |
| Related functional groups | database                                                                                                                 |
| References                | Full CLI · Dependency entry                                                                                              |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.12.14. taxonomy

Taxonomic classification.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 2 upstream caller(s) and 15 downstream call(s).

| Aspect                    | Value                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]                                    |
| API layer                 | high_level_api                                                                                                   |
| Category flags            | COMMAND_MAIN                                                                                                     |
| Upstream command count    | 2                                                                                                                |
| Downstream command count  | 15                                                                                                               |
| Related functional groups | alignment, database, easy_workflows, multi_hit, prefiltering, search_workflows, sequence_manipulation, utilities |
| References                | Full CLI · Dependency entry                                                                                      |

##### 4.12.14.1. Key Options

| Option           | Purpose                                                       |
|------------------|---------------------------------------------------------------|
| --comp-bias-corr | Correct for locally biased amino acid composition (range 0-1) |

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |

#### 4.12.15. taxonomyreport

Create a taxonomy report in Kraken or Krona format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage:<br>mmseqs taxonomyreport <i:seqTaxDB><br><i:taxResultDB/resultDB/sequenceDB><br><o:taxonomyReport> [options] |
| API layer                 | low_level_api                                                                                                       |
| Category flags            | COMMAND_TAXONOMY<br>  COMMAND_FORMAT_CONVERSION                                                                     |
| Upstream command count    | 1                                                                                                                   |
| Downstream command count  | 0                                                                                                                   |
| Related functional groups | easy_workflows                                                                                                      |
| References                | Full CLI · Dependency entry                                                                                         |

##### 4.12.15.1. Key Options

| Option        | Purpose                                 |
|---------------|-----------------------------------------|
| --report-mode | Taxonomy report mode 0: Kraken 1: Krona |



| Option    | Purpose                                                       |
|-----------|---------------------------------------------------------------|
| --threads | Number of CPU-cores used (all by default)                     |
| -v        | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### 4.13. Utilities

General-purpose helpers and special-purpose composition commands used in advanced pipelines.

#### Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

#### 4.13.1. apply

Execute given program on each DB entry.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                     |
|---------------------------|---------------------------------------------------------------------------|
| Usage                     | usage: mmseqs apply <i:DB> <o:DB> --<br>program [args...] [options]       |
| API layer                 | low_level_api                                                             |
| Category flags            | COMMAND_DB                                                                |
| Upstream command count    | 0                                                                         |
| Downstream command count  | 0                                                                         |
| Related functional groups | No direct cross-group coupling detected in the<br>current dependency map. |
| References                | Full CLI · Dependency entry                                               |

##### 4.13.1.1. Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.13.2. compress

Compress DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                          |
|---------------------------|------------------------------------------------|
| Usage                     | usage: mmseqs compress <i:DB> <o:DB> [options] |
| API layer                 | low_level_api                                  |
| Category flags            | COMMAND_STORAGE                                |
| Upstream command count    | 1                                              |
| Downstream command count  | 0                                              |
| Related functional groups | profiles                                       |
| References                | Full CLI · Dependency entry                    |

#### 4.13.2.1. Key Options

| Option     | Purpose                                                       |
|------------|---------------------------------------------------------------|
| - -threads | Number of CPU-cores used (all by default)                     |
| -v         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.13.3. convertkb

Convert UniProtKB data to a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs convertkb <DB> [args] [options] (source-derived synopsis; run mmseqs convertkb for exact syntax) |
| API layer                 | low_level_api                                                                                                  |
| Category flags            | COMMAND_SPECIAL                                                                                                |
| Upstream command count    | 0                                                                                                              |
| Downstream command count  | 0                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                         |
| References                | Full CLI · Dependency entry                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.13.4. decompress

Decompress DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs decompress <i:DB> <o:DB> [options]                       |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_STORAGE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

##### 4.13.4.1. Key Options

| Option     | Purpose                                                       |
|------------|---------------------------------------------------------------|
| - -threads | Number of CPU-cores used (all by default)                     |
| -v         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.13.5. diffseqdbs

Compute diff of two sequence DBs.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

| Aspect                 | Value                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------|
| Usage                  | usage: mmseqs diffseqdbs <DB> [args] [options] (source-derived synopsis; run mmseqs diffseqdbs for exact syntax) |
| API layer              | low_level_api                                                                                                    |
| Category flags         | COMMAND_SPECIAL                                                                                                  |
| Upstream command count | 1                                                                                                                |

| Aspect                    | Value                       |
|---------------------------|-----------------------------|
| Downstream command count  | 0                           |
| Related functional groups | clustering                  |
| References                | Full CLI · Dependency entry |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.13.6. filterdb

DB filtering by given conditions.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 8 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                |
|---------------------------|----------------------------------------------------------------------|
| Usage                     | usage: mmseqs filterdb <i:resultDB><br><o:resultDB> [options]        |
| API layer                 | low_level_api                                                        |
| Category flags            | COMMAND_DB                                                           |
| Upstream command count    | 8                                                                    |
| Downstream command count  | 0                                                                    |
| Related functional groups | clustering, easy_workflows, multi_hit,<br>search_workflows, taxonomy |
| References                | Full CLI · Dependency entry                                          |

##### 4.13.6.1. Key Options

| Option              | Purpose                                                                   |
|---------------------|---------------------------------------------------------------------------|
| --add-self-matches  | Artificially add entries of queries with themselves (for clustering)      |
| --filter-expression | Specify a mathematical expression to filter lines                         |
| --filter-column     | column                                                                    |
| --column-to-take    | column to take in join mode. If -1, the whole line is taken               |
| --filter-regex      | Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9]) |

| Option                         | Purpose                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--positive-filter</code> | Used in conjunction with <code>--filter-file</code> . If true, out = in filter ; if false, out = in - filter |
| <code>--filter-file</code>     | Specify a file that contains the filtering elements                                                          |
| <code>--beats-first</code>     | Filter by comparing each entry to the first entry                                                            |

#### 4.13.7. gff2db

Extract regions from a sequence database based on a GFF3 file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs gff2db <DB> [args] [options] (source-derived synopsis; run mmseqs gff2db for exact syntax) |
| API layer                 | low_level_api                                                                                            |
| Category flags            | COMMAND_SPECIAL                                                                                          |
| Upstream command count    | 0                                                                                                        |
| Downstream command count  | 0                                                                                                        |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                   |
| References                | Full CLI · Dependency entry                                                                              |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.13.8. gpuserver

Start a GPU server.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect    | Value                                    |
|-----------|------------------------------------------|
| Usage     | usage: mmseqs gpuserver <i:DB> [options] |
| API layer | low_level_api                            |

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Category flags            | COMMAND_STORAGE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.13.8.1. Key Options

| Option                        | Purpose                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------|
| <code>--max-seqs</code>       | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| <code>--prefilter-mode</code> | prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped          |
| <code>--gpu</code>            | Use GPU (CUDA) if possible                                                             |
| <code>--db-load-mode</code>   | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                        |

#### 4.13.9. maskbygff

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs maskbygff <DB> [args] [options] (source-derived synopsis; run mmseqs maskbygff for exact syntax) |
| API layer                 | low_level_api                                                                                                  |
| Category flags            | COMMAND_SPECIAL                                                                                                |
| Upstream command count    | 0                                                                                                              |
| Downstream command count  | 0                                                                                                              |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                         |
| References                | Full CLI · Dependency entry                                                                                    |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.13.10. **prefixid**

For each entry in a DB prepend the entry key to the entry itself.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                          |
|---------------------------|------------------------------------------------|
| Usage                     | usage: mmseqs prefixid <i:DB> <o:DB> [options] |
| API layer                 | low_level_api                                  |
| Category flags            | COMMAND_DB                                     |
| Upstream command count    | 3                                              |
| Downstream command count  | 0                                              |
| Related functional groups | clustering, database                           |
| References                | Full CLI · Dependency entry                    |

##### 4.13.10.1. Key Options

| Option          | Purpose                                                                 |
|-----------------|-------------------------------------------------------------------------|
| - -prefix       | Use this prefix for all entries                                         |
| - -mapping-file | Specify a file that translates the keys of a DB to new keys, TSV format |
| - -tsv          | Return output in TSV format                                             |
| - -threads      | Number of CPU-cores used (all by default)                               |
| - -compressed   | Write compressed output                                                 |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

#### 4.13.11. **setextendeddbtype**

Write an extended DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs setextendeddbtype <i:DB> [options]                       |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_DB                                                             |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.13.11.1. Key Options

| Option            | Purpose                                                                 |
|-------------------|-------------------------------------------------------------------------|
| --extended-dbtype | Set extended dbtype 1: compressed, 2: need src, 4: context pseudoe cnts |
| -v                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

#### 4.13.12. suffixid

For each entry in a DB append the entry key to the entry itself.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]             |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_DB                                                             |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |



#### 4.13.12.1. Key Options

| Option         | Purpose                                                                 |
|----------------|-------------------------------------------------------------------------|
| --prefix       | Use this prefix for all entries                                         |
| --mapping-file | Specify a file that translates the keys of a DB to new keys, TSV format |
| --tsv          | Return output in TSV format                                             |
| --threads      | Number of CPU-cores used (all by default)                               |
| --compressed   | Write compressed output                                                 |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

#### 4.13.13. summarizetabs

Extract annotations from HHblits BLAST-tab-formatted results.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| Usage                     | usage: mmseqs summarizetabs <DB> [args] [options] (source-derived synopsis; run mmseqs summarizetabs for exact syntax) |
| API layer                 | low_level_api                                                                                                          |
| Category flags            | COMMAND_SPECIAL                                                                                                        |
| Upstream command count    | 0                                                                                                                      |
| Downstream command count  | 0                                                                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map.                                                 |
| References                | Full CLI · Dependency entry                                                                                            |

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

#### 4.13.14. touchdb

Preload DB into memory (page cache).

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs touchdb <i:DB> [options]                                 |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_STORAGE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.13.14.1. Key Options

| Option     | Purpose                                                       |
|------------|---------------------------------------------------------------|
| - -threads | Number of CPU-cores used (all by default)                     |
| -v         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### 4.13.15. unpackdb

Unpack a DB into separate files.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | usage: mmseqs unpackdb <i:DB> <o:outDir> [options]                     |
| API layer                 | low_level_api                                                          |
| Category flags            | COMMAND_STORAGE                                                        |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.13.15.1. Key Options

| Option                          | Purpose                                                                        |
|---------------------------------|--------------------------------------------------------------------------------|
| <code>--unpack-name-mode</code> | Name unpacked files by 0: DB key, 1: accession (through <code>.lookup</code> ) |
| <code>--unpack-suffix</code>    | File suffix for unpacked files.                                                |
| <code>--threads</code>          | Number of CPU-cores used (all by default)                                      |
| <code>-v</code>                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                  |

#### 4.13.16. view

Print DB entries given in `--id-list` to stdout.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

| Aspect                    | Value                                                                  |
|---------------------------|------------------------------------------------------------------------|
| Usage                     | <code>usage: mmseqs view &lt;i:DB&gt; [options]</code>                 |
| API layer                 | <code>low_level_api</code>                                             |
| Category flags            | <code>COMMAND_DB</code>                                                |
| Upstream command count    | 0                                                                      |
| Downstream command count  | 0                                                                      |
| Related functional groups | No direct cross-group coupling detected in the current dependency map. |
| References                | Full CLI · Dependency entry                                            |

#### 4.13.16.1. Key Options

| Option                        | Purpose                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------|
| <code>--id-list</code>        | Entries to be printed separated by ‘;’                                                     |
| <code>--id-mode</code>        | Select DB entries based on 0: database keys, 1: FASTA identifiers ( <code>.lookup</code> ) |
| <code>--idx-entry-type</code> | 0: sequence, 1: src sequence, 2: header, 3: src header                                     |
| <code>-v</code>               | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                              |

## 5. Expert Manual

This chapter covers advanced MMseqs2 operation for users who build custom pipelines, run large production workloads, or debug behavior across layered workflows. It assumes familiarity with System Map and Performance Foundations.

### 5.1. Expert Operating Discipline

At expert scale, the common failure mode is not command syntax. It is uncontrolled assumptions. Pipelines drift when DB contracts change silently, when modes are mixed across experiments, or when topology changes are not tracked.

A disciplined workflow starts by fixing contracts and mode envelopes before threshold tuning. DB type compatibility, sidecar completeness, and mode stability should be treated as required preconditions for meaningful performance or quality comparison.

### 5.2. Contract Enforcement in Custom Pipelines

Every module boundary should be treated as an explicit contract surface. Structural contracts define whether downstream modules can parse and trust the input DB type and index structures. Metadata contracts define whether required sidecars (headers, lookup maps, taxonomy files, mapping tables) are present. Semantic contracts define whether output fields still mean the same thing after mode and filter choices.

A frequent anti-pattern is adjusting scoring thresholds while one of these contracts is already broken. That often yields plausible-looking output with unstable interpretation.

#### Warning

When a downstream result looks wrong, verify structural, metadata, and semantic contracts first. Threshold tuning should be the last step.

### 5.3. Reproducibility Controls

Reproducibility requires pinning an execution envelope, not only command names. You should lock alignment and rescoring modes, filter gates, profile-construction controls, taxonomy extraction/aggregation policy, and runtime envelope settings such as split mode, load mode, and parallel topology.

In practice, most hard-to-explain result deltas come from one unpinned control in that envelope. Record these controls with every benchmark or production run so regression triage is possible.

### 5.4. Performance Triage for Advanced Workloads

When large jobs become slow or unstable, triage in infrastructure-first order: startup/index behavior, split and merge overhead, distributed I/O topology, then mode/filter output inflation. This ordering matches how MMseqs2 workloads typically fail at scale.

#### Performance

In production, index/load/split decisions usually produce larger runtime swings than late-stage threshold changes.

## 5.5. MMseqs2 Source Development Guide

This section is for MMseqs2 source development. The authoritative behavior definition lives in MMseqs2/ source files; documentation markdown is a derived layer for users.

### 5.5.1. Source-First Trace Strategy

When you change a command, start from command registration in MMseqs2/src/MMseqsBase.cpp and confirm declarations in MMseqs2/src/CommandDeclarations.h. This tells you visibility, category flags, and baseline intent.

When you change workflow behavior, inspect MMseqs2/src/workflow/\*.cpp and the corresponding script glue in MMseqs2/data/workflow/\*.sh. Most user-visible behavior differences come from this orchestration boundary rather than isolated kernel changes.

When you change algorithmic behavior, move to the corresponding kernel directories: prefiltering, alignment, clustering, linclust, taxonomy, and multihit. Then verify where those kernels are invoked from workflow entrypoints so you can reason about downstream impact.

### 5.5.2. Source Tree Responsibilities

MMseqs2/src/workflow/ is the orchestration layer for end-to-end workflows. MMseqs2/src/prefiltering/, MMseqs2/src/alignment/, MMseqs2/src/clustering/, and MMseqs2/src/linclust/ host core compute modules. MMseqs2/src/taxonomy/ and MMseqs2/src/multihit/ cover specialized assignment and set-aggregation logic.

MMseqs2/src/util/ and MMseqs2/src/commons/ provide DB transforms, exports, shared parameter logic, and low-level I/O abstractions. If behavior shifts unexpectedly across many commands, inspect commons first; it is often the shared root cause.

### 5.5.3. Practical Debug Workflow

Use a repeatable debug loop:

1. Locate command registration and category metadata.
2. Trace workflow composition and parameter propagation.
3. Inspect the target kernel implementation.
4. Validate DB type and sidecar boundary assumptions.
5. Re-check output semantics in result/export transforms.

Representative commands for this loop:

```
• rg -n '<command>' MMseqs2/src/MMseqsBase.cpp
• rg -n 'createParameterString\('par\.' MMseqs2/src/workflow/*.cpp
• rg -n '<symbol>' MMseqs2/src/
  {prefiltering,alignment,clustering,linclust,taxonomy,multihit}/*.cpp
```

- `rg -n 'DBReader|DBWriter|DBTYPE' MMseqs2/src/{workflow,util,commons}/*.{h,cpp}`
- `rg -n 'MMSEQS|RUNNER|\$\{.*_PAR\}' MMseqs2/data/workflow/*.sh`

After source validation, regenerate documentation artifacts so command pages and module pages remain synchronized with implementation.

Use Performance Foundations for storage/index/split/parallel mechanics, System Map and Dependency Map for architecture and topology debugging, Functional Modules Manual for task-first command selection, and this section (MMseqs2 Source Development Guide) for source-first tracing paths.

## 6. MMseqs2 Command Reference Index

This index is generated from command metadata, dependency topology, and local CLI help snapshots where available. Use it to move quickly from command name to functional placement and detailed page content.

| Metric                                      | Value |
|---------------------------------------------|-------|
| Total visible commands                      | 128   |
| Commands with help snapshots                | 97    |
| Commands with source-derived fallback pages | 31    |

### Note

Some commands currently use source-derived fallback text because local help snapshots were not present in `mmseqs_help_output`.

| Command                          | Snapshot status         |
|----------------------------------|-------------------------|
| <code>addtaxonomy</code>         | source-derived fallback |
| <code>aggregatetax</code>        | source-derived fallback |
| <code>aggregatetaxweights</code> | source-derived fallback |
| <code>convert2fasta</code>       | source-derived fallback |
| <code>convertca3m</code>         | source-derived fallback |
| <code>convertkb</code>           | source-derived fallback |
| <code>convertprofiledb</code>    | source-derived fallback |
| <code>countkmer</code>           | source-derived fallback |
| <code>createbintaxmapping</code> | source-derived fallback |
| <code>createbintaxonomy</code>   | source-derived fallback |
| <code>createdmptaxonomy</code>   | source-derived fallback |

| Command           | Snapshot status         |
|-------------------|-------------------------|
| databases         | source-derived fallback |
| diffseqdbs        | source-derived fallback |
| expand2profile    | source-derived fallback |
| extractdomains    | source-derived fallback |
| fwbw              | source-derived fallback |
| gff2db            | source-derived fallback |
| majoritylca       | source-derived fallback |
| maskbygff         | source-derived fallback |
| nrtotaxmapping    | source-derived fallback |
| pairaln           | source-derived fallback |
| pickconsensusrep  | source-derived fallback |
| profile2consensus | source-derived fallback |
| profile2neff      | source-derived fallback |
| profile2pssm      | source-derived fallback |
| profile2repseq    | source-derived fallback |
| rmdb              | source-derived fallback |
| sequence2profile  | source-derived fallback |
| summarizeheaders  | source-derived fallback |
| summarizetabs     | source-derived fallback |
| transitivealign   | source-derived fallback |

Primary topology view: Dependency map.

## 6.1. Command Group Map

| Group            | Command count | Commands                                                                         |
|------------------|---------------|----------------------------------------------------------------------------------|
| easy_workflows   | 6             | easy-cluster, easy-linclud, easy-linsearch, easy-rbh, easy-search, easy-taxonomy |
| search_workflows | 4             | linsearch, map, rbh, search                                                      |
| clustering       | 7             | clust, cluster, clusterupdate, clusthash,                                        |

| Group           | Command count | Commands                                                                                                                                                                                                                                       |
|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefiltering    | 6             | linclust, mergeclusters,<br>pickconsensusrep<br><br>countkmer,<br>gappedprefilter,<br>kmermatcher, kmersearch,<br>prefilter,<br>ungappedprefilter                                                                                              |
| alignment       | 9             | align, alignall,<br>alignbykmer, expandaln,<br>fwbw, offsetalignment,<br>proteinaln2nucl,<br>rescorediagonal,<br>transitivealign                                                                                                               |
| profiles        | 14            | convertca3m, convertmsa,<br>convertprofiledb,<br>expand2profile,<br>msa2profile, msa2result,<br>pairaln,<br>profile2consensus,<br>profile2neff,<br>profile2pssm,<br>profile2repseq,<br>result2profile,<br>sequence2profile,<br>tsv2exprofiledb |
| database        | 20            | aliasdb, concatdbs, cpdb,<br>createdb, createindex,<br>createlinindex,<br>createsubdb, databases,<br>db2tar, lndb, mergedbs,<br>mvdb, renamedbkeys, rmdb,<br>splitdb, splitsequence,<br>subtractdbs, swapdb,<br>tar2db, tsv2db                 |
| result_handling | 17            | convert2fasta,<br>convertalis,<br>createseqfiledb,<br>createtsv, extractdomains,<br>filterresult,                                                                                                                                              |



| Group                 | Command count | Commands                                                                                                                                                                                                                                             |
|-----------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |               | result2dnamsa,<br>result2flat, result2msa,<br>result2rbh, result2repseq,<br>result2stats,<br>sortresult, summarizealis,<br>summarizeheaders,<br>summarizeresult,<br>swapresults                                                                      |
| sequence_manipulation | 9             | extractalignedregion,<br>extractframes,<br>extractorfs, masksequence,<br>orftocontig,<br>recoverlongestorf,<br>reverseseq, translateaa,<br>translatenucs                                                                                             |
| taxonomy              | 15            | addtaxonomy, aggregatetax,<br>aggregatetaxweights,<br>createbintaxmapping,<br>createbintaxonomy,<br>createdmptaxonomy,<br>createtaxdb, filtertaxdb,<br>filtertaxseqdb, lca,<br>lcaalign, majoritylca,<br>nrtotaxmapping, taxonomy,<br>taxonomyreport |
| multi_hit             | 5             | besthitperset,<br>combinepvalperset,<br>mergeresultsbyset,<br>multihitdb, multihitsearch                                                                                                                                                             |
| utilities             | 16            | apply, compress, convertkb,<br>decompress, diffseqdb,<br>filterdb,<br>gff2db, gpuserver,<br>maskbygff, prefixid,<br>setextendeddbtype,<br>suffixid, summarizetabs,<br>touchdb, unpackdb, view                                                        |

## 6.2. Command Pages

Full command pages follow below. Each page keeps local usage/options snapshots plus dependency links.

### 6.2.1. **addtaxonomy**

Add taxonomic labels to result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### Usage

usage: mmseqs addtaxonomy <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs addtaxonomy locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

##### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.2. **aggregatetax**

Aggregate multiple taxon labels to a single label.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs aggregatetax <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs aggregatetax` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.3. `aggregatetaxweights`

Aggregate multiple taxon labels to a single label.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs aggregatetaxweights <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs aggregatetaxweights` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.4. **aliasdb**

Create relative symlink of DB to another name in the same folder.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs aliasdb <i:srcDB> <o:dstDB> [options]

#### Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### Full CLI Help Snapshot

usage: mmseqs aliasdb <i:srcDB> <o:dstDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.5. **alignall**

Within-result all-vs-all gapped local alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

## Usage

usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]

## Key Options

| Option                 | Purpose                                                                         |
|------------------------|---------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                   |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                   |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)            |
| -a                     | Add backtrace string (convert to alignments with mmseqs convertalis module)     |
| --alignment-mode       | How to compute the alignment:                                                   |
| -e                     | List matches below this E-value (range 0.0-inf)                                 |
| --min-seq-id           | List matches above this sequence identity (for clustering) (range 0.0-1.0)      |
| --min-aln-len          | Minimum alignment length (range 0-INT_MAX)                                      |
| --seq-id-mode          | 0: alignment length 1: shorter, 2: longer sequence                              |
| -c                     | List matches above this fraction of aligned (covered) residues (see --cov-mode) |
| --cov-mode             | 0: coverage of query and target                                                 |
| --score-bias           | Score bias when computing SW alignment (in bits)                                |

## Full CLI Help Snapshot

usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition (range 0-1) [1]

--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition (range 0-1) [1.000]

```

--add-self-matches BOOL      Artificially add entries of queries with themselves
(for clustering) [0]
align:
-a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
--alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
-e DOUBLE                     List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT            List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT             Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT             0: alignment length 1: shorter, 2: longer sequence
[0]
-c FLOAT                      List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
--cov-mode INT                0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the other
seq. length [0]
--score-bias FLOAT            Score bias when computing SW alignment (in bits)
[0.000]
--gap-open TWIN               Gap open cost [aa:11,nucl:5]
--gap-extend TWIN             Gap extension cost [aa:1,nucl:2]
--zdrop INT                   Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
common:
--sub-mat TWIN                 Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT              Maximum sequence length [65535]
--db-load-mode INT             Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT               Write compressed output [0]
-v INT                         Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.6. alignbykmer

Heuristic gapped local k-mer based alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

#### Usage

```
usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
```

#### Key Options

| Option                | Purpose                                                                        |
|-----------------------|--------------------------------------------------------------------------------|
| -k                    | k-mer length (0: automatically set to optimum)                                 |
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                   |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                            |
| --alph-size           | Alphabet size (range 2-21)                                                     |
| --add-self-matches    | Artificially add entries of queries with themselves (for clustering)           |
| -c                    | List matches above this fraction of aligned (covered) residues (see -cov-mode) |
| -e                    | List matches below this E-value (range 0.0-inf)                                |
| --cov-mode            | 0: coverage of query and target                                                |
| --min-seq-id          | List matches above this sequence identity (for clustering) (range 0.0-1.0)     |
| --min-aln-len         | Minimum alignment length (range 0-INT_MAX)                                     |

| Option       | Purpose            |
|--------------|--------------------|
| --gap-open   | Gap open cost      |
| --gap-extend | Gap extension cost |

## Full CLI Help Snapshot

Rescore diagonals.

usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>  
[options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

-k INT k-mer length (0: automatically set to optimum) [0]

--spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use

spaced k-mers [1]

--spaced-kmer-pattern STR User-specified spaced k-mer pattern []

--alph-size TWIN Alphabet size (range 2-21) [aa:21,nucl:5]

--add-self-matches BOOL Artificially add entries of queries with themselves

(for clustering) [0]

align:

-c FLOAT List matches above this fraction of aligned  
(covered) residues (see --cov-mode) [0.000]

-e DOUBLE List matches below this E-value (range 0.0-inf)  
[1.000E-03]

--cov-mode INT 0: coverage of query and target  
1: coverage of target  
2: coverage of query  
3: target seq. length has to be at least x% of query

length

4: query seq. length has to be at least x% of target

length

5: short seq. needs to be at least x% of the other

seq. length [0]

--min-seq-id FLOAT List matches above this sequence identity (for  
clustering) (range 0.0-1.0) [0.000]

--min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

common:

--sub-mat TWIN Substitution matrix file  
[aa:blosum62.out,nucl:nucleotide.out]

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings,

3: +info [3]

expert:

--filter-hits BOOL Filter hits by seq.id. and coverage [0]

references:



- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.7. align

Optimal gapped local alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

### Usage

usage: mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]

### Key Options

| Option                  | Purpose                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr        | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --comp-bias-corr-scale  | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --add-self-matches      | Artificially add entries of queries with themselves (for clustering)                                                     |
| -a                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |
| --alignment-mode        | How to compute the alignment:                                                                                            |
| --alignment-output-mode | How to compute the alignment:                                                                                            |
| --wrapped-scoring       | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| -e                      | List matches below this E-value (range 0.0-inf)                                                                          |
| --min-seq-id            | List matches above this sequence identity (for clustering) (range 0.0-1.0)                                               |
| --min-aln-len           | Minimum alignment length (range 0-INT_MAX)                                                                               |

| Option        | Purpose                                            |
|---------------|----------------------------------------------------|
| --seq-id-mode | 0: alignment length 1: shorter, 2: longer sequence |
| --alt-ali     | Show up to this many alternative alignments        |

### Full CLI Help Snapshot

```
usage: mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Maria Hauser
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
  --alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
  --wrapped-scoring BOOL       Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --min-seq-id FLOAT           List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT            0: alignment length 1: shorter, 2: longer sequence
[0]
  --alt-ali INT                Show up to this many alternative alignments [0]
  -c FLOAT                     List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
  --cov-mode INT               0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
```

```

query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the other
seq. length [0]
--max-rejected INT              Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
--max-accept INT               Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT             Score bias when computing SW alignment (in bits)
[0.000]
--realign BOOL                 Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT     Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT         Maximum number of results to return in realignment
[2147483647]
--corr-score-weight FLOAT      Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN                Gap open cost [aa:11,nucl:5]
--gap-extend TWIN              Gap extension cost [aa:1,nucl:2]
--zdrop INT                    Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
common:
--sub-mat TWIN                 Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT              Maximum sequence length [65535]
--db-load-mode INT             Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT               Write compressed output [0]
-v INT                         Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

### 6.2.8. apply

Execute given program on each DB entry.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

```
usage: mmseqs apply <i:DB> <o:DB> -- program [args...] [options]
```

## Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## Full CLI Help Snapshot

```
usage: mmseqs apply <i:DB> <o:DB> -- program [args...] [options]
  By Milot Mirdita <milot@mirdita.de>
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

examples:
# Gather all sequences from a cluster DB
mmseqs createseqfiledb sequenceDB clusterDB unalignedDB --min-sequences 2
# Build MSAs with Clustal-Omega
mmseqs apply unalignedDB msadb -- clustalo -i - -o stdout --threads=1

# Count lines in each DB entry inefficiently (result2stats is way faster)
mmseqs apply DB wcDB -- awk '{ counter++; } END { print counter; }'
```

## references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.9. besthitperset

For each set of sequences compute the best element and update p-value.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi\_hit.

## Usage

```
usage: mmseqs besthitperset <i:targetSetDB> <i:resultDB> <o:resultDB> [options]
```

## Key Options

| Option            | Purpose                                                                  |
|-------------------|--------------------------------------------------------------------------|
| --simple-best-hit | Update the p-value by a single best hit, or by best and second best hits |
| --threads         | Number of CPU-cores used (all by default)                                |
| --compressed      | Write compressed output                                                  |
| -v                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info            |

## Full CLI Help Snapshot

```
usage: mmseqs besthitperset <i:targetSetDB> <i:resultDB> <o:resultDB> [options]
```

By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>

options: misc:

```
--simple-best-hit BOOL Update the p-value by a single best hit, or by best and second best hits [1]
```

common:

```
--threads INT          Number of CPU-cores used (all by default) [10]
```

```
--compressed INT       Write compressed output [0]
```

```
-v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

references:

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
```

### 6.2.10. cluster

Slower, sensitive clustering.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2

upstream caller(s) and 18 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

## Usage

usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]

## Key Options

| Option                 | Purpose                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------|
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                                |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                             |
| -k                     | k-mer length (0: automatically set to optimum)                                               |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)                                      |
| --k-score              | k-mer threshold for generating similar k-mer lists                                           |
| --alph-size            | Alphabet size (range 2-21)                                                                   |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)       |
| --split                | Split input into N equally distributed chunks. 0: set the best split automatically           |
| --split-mode           | 0: split target db; 1: split query db; 2: auto, depending on main memory                     |
| --split-memory-limit   | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                |

## Full CLI Help Snapshot

usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr> & Lars von den Driesch

options: prefilter:

--seed-sub-mat TWIN Substitution matrix file for k-mer generation

```

[aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT          Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
-k INT            k-mer length (0: automatically set to optimum)
[0]
--target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
--k-score TWIN              k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT              Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [20]
--split INT                 Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT            0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--comp-bias-corr INT        Correct for locally biased amino acid
composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT Correct for locally biased amino acid
composition (range 0-1) [1.000]
--diag-score BOOL          Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT   Extract only exact k-mers for matching (range
0-1) [0]
--mask INT                 Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT          Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT      Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT        Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT    Accept only matches with ungapped alignment
score above threshold [15]
--add-self-matches BOOL    Artificially add entries of queries with
themselves (for clustering) [0]
--spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR   User-specified spaced k-mer pattern []
--local-tmp STR             Path where some of the temporary files will be
created []
align:
-c FLOAT                  List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
--cov-mode INT             0: coverage of query and target
1: coverage of target
2: coverage of query

```

|                                                                                |                                                 |
|--------------------------------------------------------------------------------|-------------------------------------------------|
| query length                                                                   | 3: target seq. length has to be at least x% of  |
| target length                                                                  | 4: query seq. length has to be at least x% of   |
| other seq. length [0]                                                          | 5: short seq. needs to be at least x% of the    |
| -a BOOL                                                                        | Add backtrace string (convert to alignments     |
| with mmseqs convertalis module) [0]                                            |                                                 |
| --alignment-mode INT                                                           | How to compute the alignment:                   |
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment [3]                  |
| --alignment-output-mode INT                                                    | How to compute the alignment:                   |
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment                      |
|                                                                                | 5: score only (output) cluster format [0]       |
| --wrapped-scoring BOOL                                                         | Double the (nucleotide) query sequence during   |
| the scoring process to allow wrapped diagonal scoring around end and start [0] |                                                 |
| -e DOUBLE                                                                      | List matches below this E-value (range 0.0-inf) |
| [1.000E-03]                                                                    |                                                 |
| --min-seq-id FLOAT                                                             | List matches above this sequence identity (for  |
| clustering) (range 0.0-1.0) [0.000]                                            |                                                 |
| --min-aln-len INT                                                              | Minimum alignment length (range 0-INT_MAX) [0]  |
| --seq-id-mode INT                                                              | 0: alignment length 1: shorter, 2: longer       |
| sequence [0]                                                                   |                                                 |
| --alt-ali INT                                                                  | Show up to this many alternative alignments [0] |
| --max-rejected INT                                                             | Maximum rejected alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --max-accept INT                                                               | Maximum accepted alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --score-bias FLOAT                                                             | Score bias when computing SW alignment (in      |
| bits) [0.000]                                                                  |                                                 |
| --realign BOOL                                                                 | Compute more conservative, shorter alignments   |
| (scores and E-values not changed) [0]                                          |                                                 |
| --realign-score-bias FLOAT                                                     | Additional bias when computing realignment      |
| [-0.200]                                                                       |                                                 |
| --realign-max-seqs INT                                                         | Maximum number of results to return in          |
| realignment [2147483647]                                                       |                                                 |
| --corr-score-weight FLOAT                                                      | Weight of backtrace correlation score that is   |
| added to the alignment score [0.000]                                           |                                                 |
| --gap-open TWIN                                                                | Gap open cost [aa:11,nucl:5]                    |
| --gap-extend TWIN                                                              | Gap extension cost [aa:1,nucl:2]                |
| --zdrop INT                                                                    | Maximal allowed difference between score values |
| before alignment is truncated (nucleotide alignment only) [40]                 |                                                 |



```

clust:
  --cluster-mode INT          0: Set-Cover (greedy)
                              1: Connected component (BLASTclust)
                              2,3: Greedy clustering by sequence length

(CDHIT) [0]
  --max-iterations INT       Maximum depth of breadth first search in
connected component clustering [1000]
  --similarity-type INT      Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
  --single-step-clustering BOOL Switch from cascaded to simple clustering
workflow [0]
  --cluster-steps INT        Cascaded clustering steps from 1 to -s [3]
  --cluster-reassign BOOL    Cascaded clustering can cluster sequence that
do not fulfill the clustering criteria.
                              Cluster reassignment corrects these errors [0]

kmermatcher:
  --weights STR              Weights used for cluster prioritization []
  --cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
  --kmer-per-seq INT         k-mers per sequence [21]
  --kmer-per-seq-scale TWIN   Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
  --adjust-kmer-len BOOL     Adjust k-mer length based on specificity (only
for nucleotides) [0]
  --hash-shift INT           Shift k-mer hash initialization [67]
  --include-only-extendable BOOL Include only extendable [0]
  --ignore-multi-kmer BOOL   Skip k-mers occurring multiple times (>=2) [0]

profile:
  --pca                      Pseudo count admixture strength []
  --pcb                      Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []

misc:
  --taxon-list STR           Taxonomy ID, possibly multiple values separated
by ',' []
  --rescore-mode INT         Rescore diagonals with:
                              0: Hamming distance
                              1: local alignment (score only)
                              2: local alignment
                              3: global alignment
                              4: longest alignment fulfilling window quality

criterion [0]
common:
  --sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT          Maximum sequence length [65535]
  --db-load-mode INT         Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT           Write compressed output [0]

```

```

-v INT                                Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--remove-tmp-files BOOL                Delete temporary files [0]
--force-reuse BOOL                     Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--mpi-runner STR                       Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL                     Filter hits by seq.id. and coverage [0]
--sort-results INT                     Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]

```

examples:

```

# Cascaded clustering of FASTA file
mmseqs cluster sequenceDB clusterDB tmp

```

```

#                               --cov-mode
# Sequence                      0    1    2
# Q: MAVGTACRPA  60%  IGN  60%
# T: -AVGTAC---  60% 100%  IGN
# Cutoff -c 0.7   -    +    -
#          -c 0.6   +    +    +

```

```

# Cascaded clustering with reassignment
# - Corrects criteria-violations of cascaded merging
# - Produces more clusters and is a bit slower
mmseqs cluster sequenceDB clusterDB tmp --cluster-reassign

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep clustering and searching of large protein sequence sets. *Bioinformatics*, 32(9), 1323-1330 (2016)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

### 6.2.11. clusterupdate

Update previous clustering with new sequences.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 0 upstream caller(s) and 13 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

## Usage

```
usage:      mmseqs      clusterupdate      <i:oldSequenceDB>      <i:newSequenceDB>
<i:oldClustResultDB>      <o:newMappedSequenceDB>      <o:newClustResultDB>      <tmpDir>
[options]
```

## Key Options

| Option                 | Purpose                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                         |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                                |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                             |
| -k                     | k-mer length (0: automatically set to optimum)                                               |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)                                      |
| --k-score              | k-mer threshold for generating similar k-mer lists                                           |
| --alph-size            | Alphabet size (range 2-21)                                                                   |
| --split                | Split input into N equally distributed chunks. 0: set the best split automatically           |
| --split-mode           | 0: split target db; 1: split query db; 2: auto, depending on main memory                     |
| --split-memory-limit   | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |

## Full CLI Help Snapshot

```
usage: mmseqs clusterupdate <i:oldSequenceDB> <i:newSequenceDB>
<i:oldClustResultDB> <o:newMappedSequenceDB> <o:newClustResultDB> <tmpDir>
[options]
  By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
```

```

--comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
--add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
--seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
-k INT                       k-mer length (0: automatically set to optimum)
[0]
--target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
--k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
--split INT                  Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL           Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT    Extract only exact k-mers for matching (range
0-1) [0]
--mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT         Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT     Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT       0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR    User-specified spaced k-mer pattern []
--local-tmp STR              Path where some of the temporary files will be
created []
--disk-space-limit BYTE     Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                      Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT         How to compute the alignment:

```

|                                                                                |                                                 |
|--------------------------------------------------------------------------------|-------------------------------------------------|
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment [3]                  |
| --alignment-output-mode INT                                                    | How to compute the alignment:                   |
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment                      |
|                                                                                | 5: score only (output) cluster format [0]       |
| --wrapped-scoring BOOL                                                         | Double the (nucleotide) query sequence during   |
| the scoring process to allow wrapped diagonal scoring around end and start [0] |                                                 |
| -e DOUBLE                                                                      | List matches below this E-value (range 0.0-inf) |
| [1.000E-03]                                                                    |                                                 |
| --min-seq-id FLOAT                                                             | List matches above this sequence identity (for  |
| clustering) (range 0.0-1.0) [0.000]                                            |                                                 |
| --min-aln-len INT                                                              | Minimum alignment length (range 0-INT_MAX) [0]  |
| --seq-id-mode INT                                                              | 0: alignment length 1: shorter, 2: longer       |
| sequence [0]                                                                   |                                                 |
| --alt-ali INT                                                                  | Show up to this many alternative alignments [0] |
| -c FLOAT                                                                       | List matches above this fraction of aligned     |
| (covered) residues (see --cov-mode) [0.000]                                    |                                                 |
| --cov-mode INT                                                                 | 0: coverage of query and target                 |
|                                                                                | 1: coverage of target                           |
|                                                                                | 2: coverage of query                            |
|                                                                                | 3: target seq. length has to be at least x% of  |
| query length                                                                   |                                                 |
|                                                                                | 4: query seq. length has to be at least x% of   |
| target length                                                                  |                                                 |
|                                                                                | 5: short seq. needs to be at least x% of the    |
| other seq. length [0]                                                          |                                                 |
| --max-rejected INT                                                             | Maximum rejected alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --max-accept INT                                                               | Maximum accepted alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --score-bias FLOAT                                                             | Score bias when computing SW alignment (in      |
| bits) [0.000]                                                                  |                                                 |
| --realign BOOL                                                                 | Compute more conservative, shorter alignments   |
| (scores and E-values not changed) [0]                                          |                                                 |
| --realign-score-bias FLOAT                                                     | Additional bias when computing realignment      |
| [-0.200]                                                                       |                                                 |
| --realign-max-seqs INT                                                         | Maximum number of results to return in          |
| realignment [2147483647]                                                       |                                                 |
| --corr-score-weight FLOAT                                                      | Weight of backtrace correlation score that is   |
| added to the alignment score [0.000]                                           |                                                 |
| --gap-open TWIN                                                                | Gap open cost [aa:11,nuc:5]                     |
| --gap-extend TWIN                                                              | Gap extension cost [aa:1,nuc:2]                 |

```

--zdrop INT                                Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
--exhaustive-search-filter INT              Filter result during search: 0: do not filter,
1: filter [0]
clust:
--cluster-mode INT                          0: Set-Cover (greedy)
                                              1: Connected component (BLASTclust)
                                              2,3: Greedy clustering by sequence length

(CDHIT) [0]
--max-iterations INT                       Maximum depth of breadth first search in
connected component clustering [1000]
--similarity-type INT                       Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
--single-step-clustering BOOL              Switch from cascaded to simple clustering
workflow [0]
--cluster-steps INT                         Cascaded clustering steps from 1 to -s [3]
--cluster-reassign BOOL                    Cascaded clustering can cluster sequence that
do not fulfill the clustering criteria.
Cluster reassignment corrects these errors [0]
kmermatcher:
--weights STR                              Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT            Weight threshold used for cluster prioritization
[0.900]
--kmer-per-seq INT                         k-mers per sequence [21]
--kmer-per-seq-scale TWIN                   Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL                     Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT                           Shift k-mer hash initialization [67]
--include-only-extendable BOOL              Include only extendable [0]
--ignore-multi-kmer BOOL                   Skip k-mers occurring multiple times (>=2) [0]
profile:
--pca                                       Pseudo count admixture strength []
--pcb                                       Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT                         Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE                         Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg BOOL                                  Use global sequence weighting for profile
calculation [0]
--filter-msa INT                           Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT                     Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT                         Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                                  Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple

```

thresholds:  
 E.g.: 0.15,0.30,0.50 to defines filter buckets  
 of ]0.15-0.30] and ]0.30-0.50] [0.0]

--qsc FLOAT                    Reduce diversity of output MSAs using min.  
 score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT                    Filter output MSAs using min. fraction of query  
 residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT                    Filter MSAs by selecting most diverse set of  
 sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT           use 0: substitution-matrix or 1: context-  
 specific pseudocounts [0]

--profile-output-mode INT       Profile output mode: 0: binary log-odds 1:  
 human-readable frequencies [0]

--num-iterations INT           Number of iterative profile search iterations  
 [1]

--exhaustive-search BOOL       For bigger profile DB, run iteratively the  
 search by greedily swapping the search results [0]

--lca-search BOOL              Efficient search for LCA candidates [0]

misc:

--taxon-list STR               Taxonomy ID, possibly multiple values separated  
 by ',' []

--prefilter-mode INT           prefilter mode: 0: kmer/ungapped 1: ungapped,  
 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT            Rescore diagonals with:  
                               0: Hamming distance  
                               1: local alignment (score only)  
                               2: local alignment  
                               3: global alignment  
                               4: longest alignment fulfilling window quality

criterion [0]

--allow-deletion BOOL           Allow deletions in a MSA [0]

--min-length INT               Minimum codon number in open reading frames  
 [30]

--max-length INT               Maximum codon number in open reading frames  
 [32734]

--max-gaps INT                Maximum number of codons with gaps or unknown  
 residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT        Contig start can be 0: incomplete, 1: complete,  
 2: both [2]

--contig-end-mode INT         Contig end can be 0: incomplete, 1: complete,  
 2: both [2]

--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1:  
 from any to stop, 2: from last encountered start to stop (no start in the middle)  
 [1]

--forward-frames STR           Comma-separated list of frames on the forward  
 strand to be extracted [1,2,3]

--reverse-frames STR           Comma-separated list of frames on the reverse  
 strand to be extracted [1,2,3]

--translation-table INT        1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3)

YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE  
 9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11)  
 PROKARYOTE, 12) ALT\_YEAST, 13) ASCIDIAN\_MITOCHONDRIAL, 14)  
 ALT\_FLATWORM\_MITOCHONDRIAL  
 15) BLEPHARISMA, 16)  
 CHLOROPHYCEAN\_MITOCHONDRIAL, 21) TREMATODE\_MITOCHONDRIAL, 22)  
 SCENEDESMUS\_MITOCHONDRIAL  
 23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24)  
 PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)  
 KARYORELICT, 28) CONDYLOSTOMA  
 29) MESODINIUM, 30) PERTRICH, 31)  
 BLASTOCRITHIDIA [1]  
 --translate INT Translate ORF to amino acid [0]  
 --use-all-table-starts BOOL Use all alternatives for a start codon in the  
 genetic table, if false - only ATG (AUG) [0]  
 --id-offset INT Numeric ids in index file are offset by this  
 value [0]  
 --sequence-overlap INT Overlap between sequences [0]  
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link  
 data and write new index, [1]  
 --headers-split-mode INT Header split mode: 0: split position, 1:  
 original header [0]  
 --search-type INT Search type 0: auto 1: amino acid, 2:  
 translated, 3: nucleotide, 4: translated nucleotide alignment [0]  
 --start-sens FLOAT Start sensitivity [4.000]  
 --sens-steps INT Number of search steps performed from --start-  
 sens to -s [1]  
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs,  
 1: full reading frames [0]  
 --use-seq-id BOOL Sequence ID (Uniprot, GenBank, ...) is used for  
 identifying matches between the old and the new DB [0]  
 --recover-deleted BOOL Find and recover deleted sequences during  
 updating of clustering [0]  
 common:  
 --sub-mat TWIN Substitution matrix file  
 [aa:blosum62.out,nucl:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:  
 mmap, 3: mmap+touch [0]  
 --threads INT Number of CPU-cores used (all by default) [10]  
 --compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:  
 +warnings, 3: +info [3]  
 --mpi-runner STR Use MPI on compute cluster with this MPI  
 command (e.g. "mpirun -np 42") []  
 --force-reuse BOOL Reuse tmp file in tmp/latest folder ignoring  
 parameters and version changes [0]  
 --remove-tmp-files BOOL Delete temporary files [0]  
 expert:



```

--filter-hits BOOL           Filter hits by seq.id. and coverage [0]
--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
                             (Alignment) or seq.id. (Hamming) [0]
--create-lookup INT         Create database lookup file (can be very large)
                             [0]
--chain-alignments INT       Chain overlapping alignments [0]
--merge-query INT           Combine ORFs/split sequences to a single entry
                             [1]
--strand INT                Strand selection only works for DNA/DNA search
                             0: reverse, 1: forward, 2: both [1]

```

examples:

```

# Update clustering workflow
# Perform initial clustering of 5000 sequences
mmseqs createdb <(head -n 10000 examples/DB.fasta) sequenceDB
mmseqs cluster sequenceDB clusterDB tmp

# Use-case 1: Update by only adding sequences
mmseqs createdb examples/QUERY.fasta addedSequenceDB
mmseqs concatdbs sequenceDB addedSequenceDB allSequenceDB
mmseqs concatdbs sequenceDB_h addedSequenceDB_h allSequenceDB_h
mmseqs clusterupdate sequenceDB allSequenceDB clusterDB newSequenceDB
newClusterDB tmp

# Use-case 2: Update clustering with deletions)
# Create a FASTA file missing 500 of the original sequences and 2500 new ones
mmseqs createdb <(tail -n +1001 examples/DB.fasta | head -n 15000)
updateSequenceDB
mmseqs clusterupdate sequenceDB updateSequenceDB clusterDB newSequenceDB
newClusterDB tmp

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep clustering and searching of large protein sequence sets. *Bioinformatics*, 32(9), 1323-1330 (2016)

### 6.2.12. clusthash

Hash-based clustering of equal length sequences.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

## Usage

usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]

## Key Options

| Option         | Purpose                                                                    |
|----------------|----------------------------------------------------------------------------|
| --alph-size    | Alphabet size (range 2-21)                                                 |
| --min-seq-id   | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| --sub-mat      | Substitution matrix file                                                   |
| --max-seq-len  | Maximum sequence length                                                    |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch            |
| --threads      | Number of CPU-cores used (all by default)                                  |
| --compressed   | Write compressed output                                                    |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info              |

## Full CLI Help Snapshot

usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--alph-size TWIN Alphabet size (range 2-21) [aa:3,nucl:5]

align:

--min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.990]

common:

--sub-mat TWIN Substitution matrix file [aa:blosum62.out,nucl:nucleotide.out]

--max-seq-len INT Maximum sequence length [65535]

--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching

for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.13. clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

#### Usage

usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]

#### Key Options

| Option                     | Purpose                                                                    |
|----------------------------|----------------------------------------------------------------------------|
| --cluster-mode             | 0: Set-Cover (greedy)                                                      |
| --max-iterations           | Maximum depth of breadth first search in connected component clustering    |
| --similarity-type          | Type of score used for clustering. 1: alignment score 2: sequence identity |
| --weights                  | Weights used for cluster prioritization                                    |
| --cluster-weight-threshold | Weight threshold used for cluster prioritization                           |
| --threads                  | Number of CPU-cores used (all by default)                                  |
| --compressed               | Write compressed output                                                    |
| -v                         | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info              |

#### Full CLI Help Snapshot

usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr> & Lars von den Driesch & Maria Hauser

options: clust:

--cluster-mode INT            0: Set-Cover (greedy)  
                                 1: Connected component (BLASTclust)  
                                 2,3: Greedy clustering by sequence length

```
(CDHIT) [0]
--max-iterations INT           Maximum depth of breadth first search in
connected component clustering [1000]
--similarity-type INT          Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
kmermatcher:
--weights STR                  Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
common:
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT              Write compressed output [0]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep clustering and searching of large protein sequence sets. *Bioinformatics*, 32(9), 1323-1330 (2016)

#### 6.2.14. combinepvalperset

For each set compute the combined p-value.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi\_hit.

#### Usage

```
usage: mmseqs combinepvalperset <i:querySetDB> <i:targetSetDB> <i:resultDB>
<o:pvalDB> [options]
```

#### Key Options

| Option  | Purpose                                             |
|---------|-----------------------------------------------------|
| --alpha | Set alpha for combining p-values during aggregation |

| Option             | Purpose                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| --aggregation-mode | Combined P-values computed from 0: multi-hit, 1: minimum of all P-values, 2: product-of-P-values, 3: truncated product |
| --threads          | Number of CPU-cores used (all by default)                                                                              |
| --compressed       | Write compressed output                                                                                                |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                          |

### Full CLI Help Snapshot

```
usage: mmseqs combinepvalperset <i:querySetDB> <i:targetSetDB> <i:resultDB>
<o:pvalDB> [options]
  By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>
options: misc:
  --alpha FLOAT          Set alpha for combining p-values during aggregation
  [1.000]
  --aggregation-mode INT  Combined P-values computed from 0: multi-hit, 1:
  minimum of all P-values, 2: product-of-P-values, 3: truncated product [0]
common:
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
  +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
```

#### 6.2.15. compress

Compress DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

#### Usage

```
usage: mmseqs compress <i:DB> <o:DB> [options]
```

## Key Options

| Option    | Purpose                                                       |
|-----------|---------------------------------------------------------------|
| --threads | Number of CPU-cores used (all by default)                     |
| -v        | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## Full CLI Help Snapshot

```
usage: mmseqs compress <i:DB> <o:DB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: common:
  --threads INT    Number of CPU-cores used (all by default) [10]
  -v INT           Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.16. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

```
usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]
```

## Key Options

| Option              | Purpose                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| --preserve-keys     | The keys of the two DB should be distinct, and they will be preserved in the concatenation                   |
| --take-larger-entry | Only keep the larger entry (dataSize >) in the concatenation, both databases need the same keys in the index |

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]
  By Clovis Galiez, Eli Levy Karin & Martin Steinegger
  (martin.steinegger@snu.ac.kr)
options: misc:
  --preserve-keys BOOL      The keys of the two DB should be distinct, and they
will be preserved in the concatenation [0]
  --take-larger-entry BOOL  Only keep the larger entry (dataSize >) in the
concatenation, both databases need the same keys in the index [0]
common:
  --compressed INT          Write compressed output [0]
  --threads INT             Number of CPU-cores used (all by default) [1]
  -v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
```

#### examples:

```
# Download two sequences databases and concat them
mmseqs databases PDB pdbDB tmp
mmseqs UniProtKB/Swiss-Prot swissprotDB tmp
# Works only single threaded since seq. and header DB need the same ordering
mmseqs concatdbs pdbDB swissprotDB pdbAndSwissprotDB --threads 1
mmseqs concatdbs pdbDB_h swissprotDB_h pdbAndSwissprotDB_h --threads 1
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.17. convert2fasta

Convert sequence DB to FASTA format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

### Usage

```
usage: mmseqs convert2fasta <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run mmseqs convert2fasta locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.18. convertalis

Convert alignment DB to BLAST-tab, SAM or custom format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

### Usage

```
usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB>
<o:alignmentFile> [options]
```

### Key Options

| Option           | Purpose                                                        |
|------------------|----------------------------------------------------------------|
| - -gap-open      | Gap open cost                                                  |
| - -gap-extend    | Gap extension cost                                             |
| - -format-mode   | Output format:                                                 |
| - -format-output | Choose comma separated list of output columns from: query,tar- |



| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                     | get,evaluate,gapopen,pident,fident,nident,qstart,qend,qlen                                                                          |
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --search-type       | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment                                 |
| --sub-mat           | Substitution matrix file                                                                                                            |
| --db-load-mode      | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                                                                     |
| --threads           | Number of CPU-cores used (all by default)                                                                                           |
| --compressed        | Write compressed output                                                                                                             |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                       |
| --db-output         | Return a result DB instead of a text file                                                                                           |

### Full CLI Help Snapshot

usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB>

<o:alignmentFile> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: align:

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

misc:

--format-mode INT Output format:  
0: BLAST-TAB  
1: SAM  
2: BLAST-TAB + query/db length  
3: Pretty HTML  
4: BLAST-TAB + column headers  
BLAST-TAB (0) and BLAST-TAB + column headers (4)

support custom output formats (--format-output) [0]

--format-output STR Choose comma separated list of output columns from:

query,target,evaluate,gapopen,pident,fident,nident,qstart,qend,qlen

tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c

qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos  
[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evaluate,bits]

```

--translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
                           9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
                           15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL, 21)
TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
                           23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
                           29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA [1]
--search-type INT          Search type 0: auto 1: amino acid, 2: translated, 3:
nucleotide, 4: translated nucleotide alignment [0]
common:
--sub-mat TWIN              Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT          Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
--threads INT               Number of CPU-cores used (all by default) [10]
--compressed INT            Write compressed output [0]
-v INT                      Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
--db-output B00L            Return a result DB instead of a text file [0]

examples:
# Create output in BLAST M8 format (12 columns):
# (1,2) identifiers for query and target sequences/profiles,
# (3) sequence identity, (4) alignment length, (5) number of mismatches,
# (6) number of gap openings, (7-8, 9-10) alignment start and end-position in
query and in target,
# (11) E-value, and (12) bit score
mmseqs convertalis queryDB targetDB result.m8

# Create a TSV containing pairwise alignments
mmseqs convertalis queryDB targetDB result.tsv --format-output
query,target,qaln,taln

# Annotate a alignment result with taxonomy information from targetDB
mmseqs convertalis queryDB targetDB result.tsv --format-output
query,target,taxid,taxname,taxlineage

Create SAM output
mmseqs convertalis queryDB targetDB result.sam --format-mode 1

# Create a TSV containing which query file a result comes from
mmseqs createdb euk_queries.fasta bac_queries.fasta queryDB
mmseqs convertalis queryDB targetDB result.tsv --format-output qset,query,target

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

#### 6.2.19. **convertca3m**

Convert a cA3M DB to a result DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

```
usage: mmseqs convertca3m <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs convertca3m` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

##### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.20. **convertkb**

Convert UniProtKB data to a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

### Usage

```
usage: mmseqs convertkb <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs convertkb` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.21. convertmsa

Convert Stockholm/PFAM MSA file to a MSA DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

### Usage

```
usage: mmseqs convertmsa <i:msaFile.sto[.gz]> <o:msaDB> [options]
```

### Key Options

| Option                          | Purpose                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--identifier-field</code> | Field from STOCKHOLM comments for choosing the MSA identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of the first sequence will become the identifier |
| <code>--compressed</code>       | Write compressed output                                                                                                                                                          |

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs convertmsa <i:msaFile.sto[.gz]> <o:msaDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: common:
  --identifier-field INT    Field from STOCKHOLM comments for choosing the MSA
                           identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of
                           the first sequence will become the identifier [1]
  --compressed INT         Write compressed output [0]
  -v INT                    Verbose level: 0: quiet, 1: +errors, 2: +warnings, 3:
                           +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
    for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
    (2017)
  - Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app
    for fast, interactive sequence searches. Bioinformatics, 35(16), 2856-2858 (2019)
```

#### 6.2.22. convertprofiledb

Convert a HH-suite HHM DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

```
usage: mmseqs convertprofiledb <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs convertprofiledb` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.23. countkmer

Count k-mers.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

#### Usage

```
usage: mmseqs countkmer <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs countkmer` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.24. cpdb

Copy a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

### Usage

```
usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]
```

### Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: common:
```

```
-v INT    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
```

### 6.2.25. createbintaxmapping

Create binary taxonomy mapping from tabular taxonomy mapping.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs createbintaxmapping <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs createbintaxmapping` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

## Full CLI Help Snapshot

### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.26. **createbintaxonomy**

Create binary taxonomy from NCBI input.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs createbintaxonomy <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs createbintaxonomy` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

## Full CLI Help Snapshot

### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.27. **createdb**

Convert FASTA/Q file(s) to a sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 8 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.



Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

```
usage: mmseqs createdb <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>|
<i:stdin> <o:sequenceDB> [options]
```

## Key Options

| Option          | Purpose                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------|
| --dbtype        | Database type 0: auto, 1: amino acid 2: nucleotides                                                     |
| --shuffle       | Shuffle input database                                                                                  |
| --createdb-mode | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) |
| --id-offset     | Numeric ids in index file are offset by this value                                                      |
| --compressed    | Write compressed output                                                                                 |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                           |
| --write-lookup  | write .lookup file containing mapping from internal id, fasta id and file number                        |

## Full CLI Help Snapshot

```
usage: mmseqs createdb <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>|
<i:stdin> <o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --dbtype INT          Database type 0: auto, 1: amino acid 2: nucleotides [0]
  --shuffle BOOL        Shuffle input database [1]
  --createdb-mode INT   Createdb mode 0: copy data, 1: soft link data and write
new index (works only with single line fasta/q) [0]
  --id-offset INT       Numeric ids in index file are offset by this value [0]
common:
  --compressed INT      Write compressed output [0]
  -v INT                Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
expert:
  --write-lookup INT     write .lookup file containing mapping from internal id,
fasta id and file number [1]
```

examples:

```
# Create a sequence database from multiple FASTA files
mmseqs createdb file1.fa file2.fa.gz file3.fa sequenceDB

# Create a seqDB from stdin
cat seq.fasta | mmseqs createdb stdin sequenceDB

# Create a seqDB by indexing existing FASTA/Q (for single line fasta entries
only)
mmseqs createdb seq.fasta sequenceDB --createdb-mode 1
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.28. createdmptaxonomy

Create dmp files from binary taxonomy.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### Usage

usage: mmseqs createdmptaxonomy <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs createdmptaxonomy locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

##### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.29. createindex

Store precomputed index on disk to reduce search overhead.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

```
usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]
```

#### Key Options

| Option                 | Purpose                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                                                |
| -k                     | k-mer length (0: automatically set to optimum)                                                               |
| --alph-size            | Alphabet size (range 2-21)                                                                                   |
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)                       |
| --mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob            | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case      | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat        | Repeat letters that occur > threshold in a row                                                               |
| --spaced-kmer-mode     | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |

| Option                | Purpose                             |
|-----------------------|-------------------------------------|
| --spaced-kmer-pattern | User-specified spaced k-mer pattern |

## Full CLI Help Snapshot

usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

```
--seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
-k INT                        k-mer length (0: automatically set to optimum) [0]
--alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
--comp-bias-corr INT         Correct for locally biased amino acid composition
(range 0-1) [1]
--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
--max-seqs INT               Maximum results per query sequence allowed to pass
the prefilter (affects sensitivity) [300]
--mask INT                   Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT         Repeat letters that occure > threshold in a rwo [0]
--spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR    User-specified spaced k-mer pattern []
-s FLOAT                    Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
[7.500]
--k-score TWIN              k-mer threshold for generating similar k-mer lists
[seq:0,prof:0]
--split INT                 Split input into N equally distributed chunks. 0:
set the best split automatically [0]
--split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
misc:
--check-compatible INT      0: Always recreate index, 1: Check if recreating
index is needed, 2: Fail if index is incompatible [0]
--search-type INT          Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
--min-length INT           Minimum codon number in open reading frames [30]
--max-length INT           Maximum codon number in open reading frames [32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT     Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT      Contig end can be 0: incomplete, 1: complete, 2:
```

```

both [2]
  --orf-start-mode INT          Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
  --forward-frames STR          Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
  --reverse-frames STR          Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
  --translation-table INT        1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA

[1]
  --translate INT                Translate ORF to amino acid [0]
  --use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
  --id-offset INT                Numeric ids in index file are offset by this value
[0]
  --sequence-overlap INT          Overlap between sequences [0]
  --sequence-split-mode INT        Sequence split mode 0: copy data, 1: soft link data
and write new index, [1]
  --headers-split-mode INT        Header split mode: 0: split position, 1: original
header [0]
  --translation-mode INT          Translation AA seq from nucleotide by 0: ORFs, 1:
full reading frames [0]
common:
  --max-seq-len INT              Maximum sequence length [65535]
  -v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
  --threads INT                  Number of CPU-cores used (all by default) [10]
  --compressed INT                Write compressed output [0]
  --remove-tmp-files BOOL        Delete temporary files [0]
expert:
  --index-subset INT              Create specialized index with subset of entries
0: normal index
1: index without headers
2: index without prefiltering data
4: index without aln (for cluster db)
Flags can be combined bit wise [0]
  --create-lookup INT            Create database lookup file (can be very large) [0]
  --strand INT                    Strand selection only works for DNA/DNA search 0:
reverse, 1: forward, 2: both [1]

```

examples:

```
# Create protein sequence index
mmseqs createindex sequenceDB tmp

# Create TBLASTX/N index from nucleotide sequences
mmseqs createindex sequenceDB tmp --search-type 2

# Create BLASTN index from nucleotide sequences
mmseqs createindex sequenceDB tmp --search-type 3
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

### 6.2.30. createlinindex

Create linsearch index.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs createlinindex <i:sequenceDB> <tmpDir> [options]

#### Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                                |
| -k                   | k-mer length (0: automatically set to optimum)                                               |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --alph-size          | Alphabet size (range 2-21)                                                                   |

| Option                | Purpose                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| --mask                | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob           | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case     | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat       | Repeat letters that occur > threshold in a row                                                               |
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                                                          |
| --min-seq-id          | List matches above this sequence identity (for clustering) (range 0.0-1.0)                                   |
| --hash-shift          | Shift k-mer hash initialization                                                                              |

### Full CLI Help Snapshot

```
usage: mmseqs createlindex <i:sequenceDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
                               [aa:blosum62.out,nucl:nucleotide.out]
  -k INT                      k-mer length (0: automatically set to optimum) [0]
  --split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M, 1G.
                               Default (0) to all available system memory [0]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --mask INT                  Mask sequences in prefilter stage with tantan: 0:
                               w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT           Mask sequences is probability is above threshold
                               [0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
                               search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT         Repeat letters that occur > threshold in a row [0]
  --spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
                               spaced k-mers [0]
  --spaced-kmer-pattern STR    User-specified spaced k-mer pattern []
align:
  --min-seq-id FLOAT          List matches above this sequence identity (for
                               clustering) (range 0.0-1.0) [0.000]
kmermatcher:
  --hash-shift INT            Shift k-mer hash initialization [67]
  --kmer-per-seq INT          k-mers per sequence [21]
  --kmer-per-seq-scale TWIN    Scale k-mer per sequence based on sequence length
```

```

as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL      Adjust k-mer length based on specificity (only for
nucleotides) [0]
--ignore-multi-kmer BOOL    Skip k-mers occurring multiple times (>=2) [0]
misc:
--check-compatible INT      0: Always recreate index, 1: Check if recreating
index is needed, 2: Fail if index is incompatible [0]
--search-type INT          Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
--min-length INT           Minimum codon number in open reading frames [30]
--max-length INT           Maximum codon number in open reading frames [32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT     Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT       Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT        Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR        Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR        Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT     1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT            Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT            Numeric ids in index file are offset by this value
[0]
common:
--max-seq-len INT          Maximum sequence length [65535]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--create-lookup INT        Create database lookup file (can be very large) [0]

```



references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

### 6.2.31. createseqfiledb

Create a DB of unaligned FASTA entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]

#### Key Options

| Option          | Purpose                                                         |
|-----------------|-----------------------------------------------------------------|
| --min-sequences | Minimum number of sequences a cluster may contain               |
| --max-sequences | Maximum number of sequences a cluster may contain               |
| --hh-format     | Format entries to use with hhsuite (for singleton clusters)     |
| --db-load-mode  | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads       | Number of CPU-cores used (all by default)                       |
| --compressed    | Write compressed output                                         |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### Full CLI Help Snapshot

usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]  
By Milot Mirdita <milot@mirdita.de>

```

options: misc:
  --min-sequences INT    Minimum number of sequences a cluster may contain [1]
  --max-sequences INT    Maximum number of sequences a cluster may contain
[2147483647]
  --hh-format BOOL      Format entries to use with hhsuite (for singleton
clusters) [0]
common:
  --db-load-mode INT     Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

```

#### examples:

```

# Gather all sequences from a cluster DB
mmseqs createseqfiledb sequenceDB clusterDB unalignedDB --min-sequences 2
# Build MSAs with Clustal-Omega
mmseqs apply unalignedDB msaDB -- clustalo -i - -o stdout --threads=1

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.32. createsubdb

Create a subset of a DB from list of DB keys.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs createsubdb <i:subsetFile|DB> <i:DB> <o:DB> [options]

#### Key Options

| Option       | Purpose                                                   |
|--------------|-----------------------------------------------------------|
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index |

| Option                 | Purpose                                                                     |
|------------------------|-----------------------------------------------------------------------------|
| <code>--id-mode</code> | Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) |
| <code>-v</code>        | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info               |

### Full CLI Help Snapshot

```
usage: mmseqs createsubdb <i:subsetFile|DB> <i:DB> <o:DB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --subdb-mode INT    Subdb mode 0: copy data 1: soft link data and write index [0]
  --id-mode INT       Select DB entries based on 0: database keys, 1: FASTA
identifiers (.lookup) [0]
common:
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

#### examples:

```
# Create a new sequenceDB from sequenceDB entries with keys 1, 2 and 3
mmseqs createsubdb <(printf '1
2
3
') sequenceDB oneTwoThreeDB
```

```
# Create a new sequence database with representatives of clusterDB
mmseqs cluster sequenceDB clusterDB tmp
mmseqs createsubdb clusterDB sequenceDB representativesDB
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.33. createtaxdb

Add taxonomic labels to sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 1 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

## Usage

usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]

## Key Options

| Option             | Purpose                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| --ncbi-tax-dump    | tax dump directory. The tax dump can be downloaded here "ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz" |
| --tax-mapping-file | File to map sequence identifier to taxonomical identifier                                                        |
| --tax-mapping-mode | Map taxonomy based on sequence database<br>0: .lookup file 1: .source file                                       |
| --tax-db-mode      | Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin)                  |
| --threads          | Number of CPU-cores used (all by default)                                                                        |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                    |

## Full CLI Help Snapshot

usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--ncbi-tax-dump STR NCBI tax dump directory. The tax dump can be downloaded here "ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz" []

--tax-mapping-file STR File to map sequence identifier to taxonomical identifier []

--tax-mapping-mode INT Map taxonomy based on sequence database 0: .lookup file 1: .source file [0]

--tax-db-mode INT Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin) [1]

common:

--threads INT Number of CPU-cores used (all by default) [10]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btbl184 (2021)

### 6.2.34. createtsv

Convert result DB to tab-separated flat file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

### Usage

```
usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile>
[options]
```

### Key Options

| Option                           | Purpose                                                                    |
|----------------------------------|----------------------------------------------------------------------------|
| <code>--first-seq-as-repr</code> | Use the first sequence of the clustering result as representative sequence |
| <code>--target-column</code>     | Select a target column (default 1), 0 if no target id exists               |
| <code>--full-header</code>       | Replace DB ID by its corresponding Full Header                             |
| <code>--idx-seq-src</code>       | 0: auto, 1: split/translated sequences, 2: input sequences                 |
| <code>--threads</code>           | Number of CPU-cores used (all by default)                                  |
| <code>--compressed</code>        | Write compressed output                                                    |
| <code>-v</code>                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info              |
| <code>--db-output</code>         | Return a result DB instead of a text file                                  |

### Full CLI Help Snapshot

```
usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --first-seq-as-repr BOOL  Use the first sequence of the clustering result as
                             representative sequence [0]
  --target-column INT       Select a target column (default 1), 0 if no target id
                             exists [1]
  --full-header BOOL        Replace DB ID by its corresponding Full Header [0]
  --idx-seq-src INT         0: auto, 1: split/translated sequences, 2: input
```

```

sequences [0]
common:
  --threads INT           Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
  --db-output BOOL       Return a result DB instead of a text file [0]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.35. databases

List and download databases.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 8 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

```
usage: mmseqs databases <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs databases` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

##### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.36. db2tar

Archive contents of a DB to a tar archive.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

### Usage

```
usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]
```

### Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: common:
```

```
-v INT  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

```
examples:
```

```
# Create a tar from a MSA DB
```

```
mmseqs db2tar msaDB archive.tar.gz
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
```

### 6.2.37. decompress

Decompress DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

usage: mmseqs decompress <i:DB> <o:DB> [options]

## Key Options

| Option    | Purpose                                                       |
|-----------|---------------------------------------------------------------|
| --threads | Number of CPU-cores used (all by default)                     |
| -v        | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## Full CLI Help Snapshot

usage: mmseqs decompress <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--threads INT Number of CPU-cores used (all by default) [10]  
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info  
[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.38. diffseqdbs

Compute diff of two sequence DBs.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

usage: mmseqs diffseqdbs <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs diffseqdbs locally for exact positional arguments in your build.

## Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

## Full CLI Help Snapshot



### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.39. easy-cluster

Slower, sensitive clustering.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy\_workflows.

### Usage

```
usage: mmseqs easy-cluster <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]
```

### Key Options

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                          |
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                   | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically     |
| --split-mode         | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

| Option                 | Purpose                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------|
| --split-memory-limit   | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                |

### Full CLI Help Snapshot

```
usage: mmseqs easy-cluster <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN              k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT              Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [20]
  --split INT                 Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT            0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --comp-bias-corr INT        Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --diag-score BOOL           Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT   Extract only exact k-mers for matching (range
0-1) [0]
  --mask INT                  Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
```

|                                                                                |                                                 |
|--------------------------------------------------------------------------------|-------------------------------------------------|
| <code>--mask-n-repeat INT</code>                                               | Repeat letters that occur > threshold in a row  |
| <code>[0]</code>                                                               |                                                 |
| <code>--min-ungapped-score INT</code>                                          | Accept only matches with ungapped alignment     |
| score above threshold [15]                                                     |                                                 |
| <code>--add-self-matches BOOL</code>                                           | Artificially add entries of queries with        |
| themselves (for clustering) [0]                                                |                                                 |
| <code>--spaced-kmer-mode INT</code>                                            | 0: use consecutive positions in k-mers; 1: use  |
| spaced k-mers [1]                                                              |                                                 |
| <code>--spaced-kmer-pattern STR</code>                                         | User-specified spaced k-mer pattern []          |
| <code>--local-tmp STR</code>                                                   | Path where some of the temporary files will be  |
| created []                                                                     |                                                 |
| <code>align:</code>                                                            |                                                 |
| <code>-c FLOAT</code>                                                          | List matches above this fraction of aligned     |
| (covered) residues (see <code>--cov-mode</code> ) [0.800]                      |                                                 |
| <code>--cov-mode INT</code>                                                    | 0: coverage of query and target                 |
|                                                                                | 1: coverage of target                           |
|                                                                                | 2: coverage of query                            |
|                                                                                | 3: target seq. length has to be at least x% of  |
| query length                                                                   |                                                 |
|                                                                                | 4: query seq. length has to be at least x% of   |
| target length                                                                  |                                                 |
|                                                                                | 5: short seq. needs to be at least x% of the    |
| other seq. length [0]                                                          |                                                 |
| <code>-a BOOL</code>                                                           | Add backtrace string (convert to alignments     |
| with <code>mmseqs convertalis</code> module) [0]                               |                                                 |
| <code>--alignment-mode INT</code>                                              | How to compute the alignment:                   |
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment [3]                  |
| <code>--alignment-output-mode INT</code>                                       | How to compute the alignment:                   |
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment                      |
|                                                                                | 5: score only (output) cluster format [0]       |
| <code>--wrapped-scoring BOOL</code>                                            | Double the (nucleotide) query sequence during   |
| the scoring process to allow wrapped diagonal scoring around end and start [0] |                                                 |
| <code>-e DOUBLE</code>                                                         | List matches below this E-value (range 0.0-inf) |
| [1.000E-03]                                                                    |                                                 |
| <code>--min-seq-id FLOAT</code>                                                | List matches above this sequence identity (for  |
| clustering) (range 0.0-1.0) [0.000]                                            |                                                 |
| <code>--min-aln-len INT</code>                                                 | Minimum alignment length (range 0-INT_MAX) [0]  |
| <code>--seq-id-mode INT</code>                                                 | 0: alignment length 1: shorter, 2: longer       |
| sequence [0]                                                                   |                                                 |
| <code>--alt-ali INT</code>                                                     | Show up to this many alternative alignments [0] |
| <code>--max-rejected INT</code>                                                | Maximum rejected alignments before alignment    |

calculation for a query is stopped [2147483647]

--max-accept INT Maximum accepted alignments before alignment calculation for a query is stopped [2147483647]

--score-bias FLOAT Score bias when computing SW alignment (in bits) [0.000]

--realign BOOL Compute more conservative, shorter alignments (scores and E-values not changed) [0]

--realign-score-bias FLOAT Additional bias when computing realignment [-0.200]

--realign-max-seqs INT Maximum number of results to return in realignment [2147483647]

--corr-score-weight FLOAT Weight of backtrace correlation score that is added to the alignment score [0.000]

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

--zdrop INT Maximal allowed difference between score values before alignment is truncated (nucleotide alignment only) [40]

clust:

--cluster-mode INT 0: Set-Cover (greedy)  
1: Connected component (BLASTclust)  
2,3: Greedy clustering by sequence length

(CDHIT) [0]

--max-iterations INT Maximum depth of breadth first search in connected component clustering [1000]

--similarity-type INT Type of score used for clustering. 1: alignment score 2: sequence identity [2]

--single-step-clustering BOOL Switch from cascaded to simple clustering workflow [0]

--cluster-steps INT Cascaded clustering steps from 1 to -s [3]

--cluster-reassign BOOL Cascaded clustering can cluster sequence that do not fulfill the clustering criteria.  
Cluster reassignment corrects these errors [0]

kmermatcher:

--weights STR Weights used for cluster prioritization []

--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization [0.900]

--kmer-per-seq INT k-mers per sequence [21]

--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]

--adjust-kmer-len BOOL Adjust k-mer length based on specificity (only for nucleotides) [0]

--hash-shift INT Shift k-mer hash initialization [67]

--include-only-extendable BOOL Include only extendable [0]

--ignore-multi-kmer BOOL Skip k-mers occurring multiple times (>=2) [0]

profile:

--pca Pseudo count admixture strength []

--pcb Pseudo counts: Neff at half of maximum

admixture (range 0.0-inf) []

misc:

```

--taxon-list STR          Taxonomy ID, possibly multiple values separated
by ',' []
--rescore-mode INT        Rescore diagonals with:
                           0: Hamming distance
                           1: local alignment (score only)
                           2: local alignment
                           3: global alignment
                           4: longest alignment fulfilling window quality

criterion [0]
--dbtype INT              Database type 0: auto, 1: amino acid 2:
nucleotides [0]
--shuffle BOOL            Shuffle input database [1]
--createdb-mode INT       Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
--id-offset INT           Numeric ids in index file are offset by this
value [0]
common:
--sub-mat TWIN            Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT         Maximum sequence length [65535]
--db-load-mode INT        Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT             Number of CPU-cores used (all by default) [10]
--compressed INT          Write compressed output [0]
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--remove-tmp-files BOOL   Delete temporary files [1]
--force-reuse BOOL        Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--write-lookup INT        write .lookup file containing mapping from
internal id, fasta id and file number [0]

examples:
mmseqs easy-cluster examples/DB.fasta result tmp
# Cluster output
# - result_rep_seq.fasta: Representatives
# - result_all_seqs.fasta: FASTA-like per cluster
# - result_cluster.tsv: Adjacency list

# Important parameter: --min-seq-id, --cov-mode and -c
# --cov-mode
# 0 1 2
# Q: MAVGTACRPA 60% IGN 60%

```

```
# T: -AVGTAC--- 60% 100% IGN
#      -c 0.7    -    +    -
#      -c 0.6    +    +    +

# Cascaded clustering with reassignment
# - Corrects criteria-violations of cascaded merging
# - Produces more clusters and is a bit slower
mmseqs easy-cluster examples/DB.fasta result tmp --cluster-reassign
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

### 6.2.40. easy-linclust

Fast linear time cluster, less sensitive clustering.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy\_workflows.

### Usage

```
usage: mmseqs easy-linclust <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]
```

### Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --alph-size            | Alphabet size (range 2-21)                                           |

| Option                | Purpose                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                                                          |
| --mask                | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob           | Mask sequences is probablity is above threshold                                                              |
| --mask-lower-case     | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat       | Repeat letters that occure > threshold in a rwo                                                              |
| -k                    | k-mer length (0: automatically set to optimum)                                                               |
| --split-memory-limit  | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |

### Full CLI Help Snapshot

usage: mmseqs easy-linclust <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>  
<o:clusterPrefix> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

|                                                                          |                                                 |
|--------------------------------------------------------------------------|-------------------------------------------------|
| --comp-bias-corr INT<br>composition (range 0-1) [1]                      | Correct for locally biased amino acid           |
| --comp-bias-corr-scale FLOAT<br>composition (range 0-1) [1.000]          | Correct for locally biased amino acid           |
| --add-self-matches BOOL<br>themselves (for clustering) [0]               | Artificially add entries of queries with        |
| --alph-size TWIN                                                         | Alphabet size (range 2-21) [aa:21,nucl:5]       |
| --spaced-kmer-mode INT<br>spaced k-mers [0]                              | 0: use consecutive positions in k-mers; 1: use  |
| --spaced-kmer-pattern STR                                                | User-specified spaced k-mer pattern []          |
| --mask INT                                                               | Mask sequences in prefilter stage with tantan:  |
| 0: w/o low complexity masking, 1: with low complexity masking [1]        |                                                 |
| --mask-prob FLOAT<br>[0.900]                                             | Mask sequences is probablity is above threshold |
| --mask-lower-case INT<br>search 0: include region, 1: exclude region [0] | Lowercase letters will be excluded from k-mer   |
| --mask-n-repeat INT<br>[0]                                               | Repeat letters that occure > threshold in a rwo |
| -k INT<br>[0]                                                            | k-mer length (0: automatically set to optimum)  |
| --split-memory-limit BYTE                                                | Set max memory per split. E.g. 800B, 5K, 10M,   |

1G. Default (0) to all available system memory [0]

align:

- a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]
- alignment-mode INT How to compute the alignment:
  - 0: automatic
  - 1: only score and end\_pos
  - 2: also start\_pos and cov
  - 3: also seq.id
  - 4: only ungapped alignment [0]
- alignment-output-mode INT How to compute the alignment:
  - 0: automatic
  - 1: only score and end\_pos
  - 2: also start\_pos and cov
  - 3: also seq.id
  - 4: only ungapped alignment
  - 5: score only (output) cluster format [0]
- wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]
- e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]
- min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]
- min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]
- seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]
- alt-ali INT Show up to this many alternative alignments [0]
- c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.800]
- cov-mode INT
  - 0: coverage of query and target
  - 1: coverage of target
  - 2: coverage of query
  - 3: target seq. length has to be at least x% of query length
  - 4: query seq. length has to be at least x% of target length
  - 5: short seq. needs to be at least x% of the other seq. length [0]
- max-rejected INT Maximum rejected alignments before alignment calculation for a query is stopped [2147483647]
- max-accept INT Maximum accepted alignments before alignment calculation for a query is stopped [2147483647]
- score-bias FLOAT Score bias when computing SW alignment (in bits) [0.000]
- realign BOOL Compute more conservative, shorter alignments (scores and E-values not changed) [0]
- realign-score-bias FLOAT Additional bias when computing realignment [-0.200]
- realign-max-seqs INT Maximum number of results to return in



```

realignment [2147483647]
  --corr-score-weight FLOAT      Weight of backtrace correlation score that is
added to the alignment score [0.000]
  --gap-open TWIN                Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN              Gap extension cost [aa:1,nucl:2]
  --zdrop INT                    Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
clust:
  --cluster-mode INT             0: Set-Cover (greedy)
                                1: Connected component (BLASTclust)
                                2,3: Greedy clustering by sequence length

(CDHIT) [0]
  --max-iterations INT           Maximum depth of breadth first search in
connected component clustering [1000]
  --similarity-type INT           Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
kmermatcher:
  --weights STR                  Weights used for cluster prioritization []
  --cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
  --kmer-per-seq INT             k-mers per sequence [21]
  --kmer-per-seq-scale TWIN       Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
  --adjust-kmer-len BOOL          Adjust k-mer length based on specificity (only
for nucleotides) [0]
  --hash-shift INT               Shift k-mer hash initialization [67]
  --include-only-extendable BOOL  Include only extendable [0]
  --ignore-multi-kmer BOOL        Skip k-mers occurring multiple times (>=2) [0]
profile:
  --pca                          Pseudo count admixture strength []
  --pcb                          Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
misc:
  --rescore-mode INT             Rescore diagonals with:
                                0: Hamming distance
                                1: local alignment (score only)
                                2: local alignment
                                3: global alignment
                                4: longest alignment fulfilling window quality

criterion [0]
  --dbtype INT                   Database type 0: auto, 1: amino acid 2:
nucleotides [0]
  --shuffle BOOL                 Shuffle input database [1]
  --createdb-mode INT            Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
  --id-offset INT                Numeric ids in index file are offset by this
value [0]
common:
  --threads INT                  Number of CPU-cores used (all by default) [10]

```

```

--compressed INT          Write compressed output [0]
-v INT                    Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--sub-mat TWIN            Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT         Maximum sequence length [65535]
--db-load-mode INT        Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--remove-tmp-files BOOL   Delete temporary files [1]
--force-reuse BOOL        Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--write-lookup INT        write .lookup file containing mapping from
internal id, fasta id and file number [0]

```

examples:

```
mmseqs easy-linclust examples/DB.fasta result tmp
```

# Linclust output

```

# - result_rep_seq.fasta: Representatives
# - result_all_seqs.fasta: FASTA-like per cluster
# - result_cluster.tsv: Adjecency list

```

# Important parameter: --min-seq-id, --cov-mode and -c

```

#           --cov-mode
#           0     1     2
# Q: MAVGTACRPA 60%  IGN  60%
# T: -AVGTAC--- 60% 100% IGN
#       -c 0.7   -    +    -
#       -c 0.6   +    +    +

```

# Cluster nucleotide sequences

```
mmseqs easy-linclust examples/DB.fasta result tmp --kmer-per-seq-scale 0.3
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

#### 6.2.41. easy-linsearch

Fast, less sensitive homology search.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `easy_workflows`.

## Usage

```
usage:                                     mmseqs
easy-linsearch <i:queryFastaFile1[.gz|.bz2]> ... <i:queryFastaFileN[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
```

## Key Options

| Option                               | Purpose                                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--comp-bias-corr</code>        | Correct for locally biased amino acid composition (range 0-1)                                                |
| <code>--comp-bias-corr-scale</code>  | Correct for locally biased amino acid composition (range 0-1)                                                |
| <code>--add-self-matches</code>      | Artificially add entries of queries with themselves (for clustering)                                         |
| <code>--seed-sub-mat</code>          | Substitution matrix file for k-mer generation                                                                |
| <code>--mask</code>                  | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| <code>--mask-prob</code>             | Mask sequences is probability is above threshold                                                             |
| <code>--mask-lower-case</code>       | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| <code>--mask-n-repeat</code>         | Repeat letters that occur > threshold in a row                                                               |
| <code>--split-memory-limit</code>    | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| <code>-a</code>                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                  |
| <code>--alignment-mode</code>        | How to compute the alignment:                                                                                |
| <code>--alignment-output-mode</code> | How to compute the alignment:                                                                                |

## Full CLI Help Snapshot

```
usage: mmseqs easy-linsearch <i:queryFastaFile1[.gz|.bz2]> ...
<i:queryFastaFileN[.gz|.bz2]> <i:targetFastaFile[.gz|.bz2]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  --mask INT                   Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT            Mask sequences is probability is above threshold
[0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a rwo
[0]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [3]
  --alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
  --wrapped-scoring BOOL        Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                     List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --min-seq-id FLOAT            List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT             Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT             0: alignment length 1: shorter, 2: longer sequence
```

```

[0]
--alt-ali INT          Show up to this many alternative alignments [0]
-c FLOAT              List matches above this fraction of aligned
(cov) residues (see --cov-mode) [0.000]
--cov-mode INT        0: coverage of query and target
                      1: coverage of target
                      2: coverage of query
                      3: target seq. length has to be at least x% of
query length
                      4: query seq. length has to be at least x% of
target length
                      5: short seq. needs to be at least x% of the other
seq. length [0]
--max-rejected INT     Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
--max-accept INT       Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT     Score bias when computing SW alignment (in bits)
[0.000]
--realign BOOL         Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT Maximum number of results to return in realignment
[2147483647]
--corr-score-weight FLOAT Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN        Gap open cost [aa:11,nucl:5]
--gap-extend TWIN      Gap extension cost [aa:1,nucl:2]
--zdrop INT            Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
kmermatcher:
--kmer-per-seq INT     k-mers per sequence [21]
--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length
as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--pick-n-sim-kmer INT  Add N similar k-mers to search [1]
--result-direction INT result is 0: query, 1: target centric [1]
profile:
--pca                  Pseudo count admixture strength []
--pcb                  Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
--min-length INT       Minimum codon number in open reading frames [30]
--max-length INT       Maximum codon number in open reading frames
[32734]
--max-gaps INT         Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT Contig start can be 0: incomplete, 1: complete, 2:
both [2]

```

```

--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT          Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR          Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR          Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT        1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this value
[0]
--search-type INT              Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
--format-mode INT              Output format:
0: BLAST-TAB
1: SAM
2: BLAST-TAB + query/db length
3: Pretty HTML
4: BLAST-TAB + column headers
BLAST-TAB (0) and BLAST-TAB + column headers (4)
support custom output formats (--format-output) [0]
--format-output STR            Choose comma separated list of output columns
from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen
tstart,tend,tlen,alrlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos
[query,target,fident,alrlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
common:
--sub-mat TWIN                 Substitution matrix file
[aa:blosum62.out,nuc1:nucleotide.out]
--max-seq-len INT              Maximum sequence length [65535]
--db-load-mode INT             Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT               Write compressed output [0]

```

```

-v INT                      Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR            Use MPI on compute cluster with this MPI command
(e.g. "mpirun -np 42") []
--force-reuse B00L         Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files B00L    Delete temporary files [1]
expert:
--create-lookup INT        Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT          Combine ORFs/split sequences to a single entry [1]
--db-output B00L           Return a result DB instead of a text file [0]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. Nature Communications, 9(1), 2542 (2018)

#### 6.2.42. easy-rbh

Find reciprocal best hit.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy\_workflows.

#### Usage

```
usage:          mmseqs          easy-rbh          <i:queryFastaFile1[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
```

#### Key Options

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| --add-self-matches   | Artificially add entries of queries with themselves (for clustering)                   |
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                          |
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                   | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically     |
| --split-mode         | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

```
usage: mmseqs easy-rbh <i:queryFastaFile1[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
  By Eli Levy Karin & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [5.700]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
```



--max-seqs INT Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]  
 --split INT Split input into N equally distributed chunks.  
 0: set the best split automatically [0]  
 --split-mode INT 0: split target db; 1: split query db; 2: auto, depending on main memory [2]  
 --split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory [0]  
 --diag-score BOOL Use ungapped diagonal scoring during prefilter [1]  
 --exact-kmer-matching INT Extract only exact k-mers for matching (range 0-1) [0]  
 --mask INT Mask sequences in prefilter stage with tantan:  
 0: w/o low complexity masking, 1: with low complexity masking [1]  
 --mask-prob FLOAT Mask sequences is probability is above threshold [0.900]  
 --mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]  
 --mask-n-repeat INT Repeat letters that occur > threshold in a row [0]  
 --min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]  
 --spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use spaced k-mers [1]  
 --spaced-kmer-pattern STR User-specified spaced k-mer pattern []  
 --local-tmp STR Path where some of the temporary files will be created []  
 --disk-space-limit BYTE Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]  
 align:  
 -a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]  
 --alignment-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment [3]  
 --alignment-output-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment  
 5: score only (output) cluster format [0]  
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]  
 -e DOUBLE List matches below this E-value (range 0.0-inf)

```

[1.000E-03]
--min-seq-id FLOAT          List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT           Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT           0: alignment length 1: shorter, 2: longer
sequence [0]
--alt-ali INT               Show up to this many alternative alignments [0]
-c FLOAT                    List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
--cov-mode INT              0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of
query length
                           4: query seq. length has to be at least x% of
target length
                           5: short seq. needs to be at least x% of the
other seq. length [0]
--max-rejected INT          Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
--max-accept INT            Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT          Score bias when computing SW alignment (in
bits) [0.000]
--realign BOOL              Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT  Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT      Maximum number of results to return in
realignment [2147483647]
--corr-score-weight FLOAT   Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN             Gap open cost [aa:11,nucl:5]
--gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
--zdrop INT                 Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
--exhaustive-search-filter INT Filter result during search: 0: do not filter,
1: filter [0]
profile:
--pca                       Pseudo count admixture strength []
--pcb                       Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT           Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE           Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg BOOL                    Use global sequence weighting for profile
calculation [0]
--filter-msa INT             Filter msa: 0: do not filter, 1: filter [1]

```

--filter-min-enable INT                      Only filter MSAs with more than N sequences, 0  
 always filters [0]  
 --max-seq-id FLOAT                            Reduce redundancy of output MSA using max.  
 pairwise sequence identity [0.0,1.0] [0.900]  
 --qid STR                                      Reduce diversity of output MSAs using min.seq.  
 identity with query sequences [0.0,1.0]  
                                             Alternatively, can be a list of multiple  
 thresholds:  
                                             E.g.: 0.15,0.30,0.50 to defines filter buckets  
 of ]0.15-0.30] and ]0.30-0.50] [0.0]  
 --qsc FLOAT                                    Reduce diversity of output MSAs using min.  
 score per aligned residue with query sequences [-50.0,100.0] [-20.000]  
 --cov FLOAT                                    Filter output MSAs using min. fraction of query  
 residues covered by matched sequences [0.0,1.0] [0.000]  
 --diff INT                                     Filter MSAs by selecting most diverse set of  
 sequences, keeping at least this many seqs in each MSA block of length 50 [1000]  
 --pseudo-cnt-mode INT                        use 0: substitution-matrix or 1: context-  
 specific pseudocounts [0]  
 --profile-output-mode INT                    Profile output mode: 0: binary log-odds 1:  
 human-readable frequencies [0]  
 --num-iterations INT                         Number of iterative profile search iterations  
 [1]  
 --exhaustive-search BOOL                    For bigger profile DB, run iteratively the  
 search by greedily swapping the search results [0]  
 --lca-search BOOL                            Efficient search for LCA candidates [0]  
 misc:  
 --taxon-list STR                            Taxonomy ID, possibly multiple values separated  
 by ',' []  
 --prefilter-mode INT                        prefilter mode: 0: kmer/ungapped 1: ungapped,  
 2: nofilter, 3: ungapped&gapped [0]  
 --rescore-mode INT                          Rescore diagonals with:  
                                             0: Hamming distance  
                                             1: local alignment (score only)  
                                             2: local alignment  
                                             3: global alignment  
                                             4: longest alignment fulfilling window quality  
 criterion [0]  
 --allow-deletion BOOL                        Allow deletions in a MSA [0]  
 --min-length INT                            Minimum codon number in open reading frames  
 [30]  
 --max-length INT                            Maximum codon number in open reading frames  
 [32734]  
 --max-gaps INT                               Maximum number of codons with gaps or unknown  
 residues before an open reading frame is rejected [2147483647]  
 --contig-start-mode INT                      Contig start can be 0: incomplete, 1: complete,  
 2: both [2]  
 --contig-end-mode INT                       Contig end can be 0: incomplete, 1: complete,  
 2: both [2]  
 --orf-start-mode INT                        Orf fragment can be 0: from start to stop, 1:

from any to stop, 2: from last encountered start to stop (no start in the middle)  
[1]

--forward-frames STR Comma-separated list of frames on the forward  
strand to be extracted [1,2,3]

--reverse-frames STR Comma-separated list of frames on the reverse  
strand to be extracted [1,2,3]

--translation-table INT 1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3)  
YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE  
9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11)  
PROKARYOTE, 12) ALT\_YEAST, 13) ASCIDIAN\_MITOCHONDRIAL, 14)  
ALT\_FLATWORM\_MITOCHONDRIAL

15) BLEPHARISMA, 16)  
CHLOROPHYCEAN\_MITOCHONDRIAL, 21) TREMATODE\_MITOCHONDRIAL, 22)  
SCENEDESMUS\_MITOCHONDRIAL

23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24)  
PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)  
KARYORELICT, 28) CONDYLOSTOMA

29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]

--translate INT Translate ORF to amino acid [0]  
--use-all-table-starts BOOL Use all alternatives for a start codon in the  
genetic table, if false - only ATG (AUG) [0]

--id-offset INT Numeric ids in index file are offset by this  
value [0]

--sequence-overlap INT Overlap between sequences [0]  
--sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link

data and write new index, [1]  
--headers-split-mode INT Header split mode: 0: split position, 1:  
original header [0]

--search-type INT Search type 0: auto 1: amino acid, 2:  
translated, 3: nucleotide, 4: translated nucleotide alignment [0]

--start-sens FLOAT Start sensitivity [4.000]  
--sens-steps INT Number of search steps performed from --start-  
sens to -s [1]

--translation-mode INT Translation AA seq from nucleotide by 0: ORFs,  
1: full reading frames [0]

--format-mode INT Output format:  
0: BLAST-TAB  
1: SAM  
2: BLAST-TAB + query/db length  
3: Pretty HTML  
4: BLAST-TAB + column headers  
BLAST-TAB (0) and BLAST-TAB + column headers

(4) support custom output formats (--format-output) [0]

--format-output STR Choose comma separated list of output columns  
from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen

tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c  
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos

```

[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
--overlap FLOAT                Maximum overlap of covered regions [0.000]
--dbtype INT                    Database type 0: auto, 1: amino acid 2:
nucleotides [0]
--shuffle BOOL                  Shuffle input database [1]
--createdb-mode INT             Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
--greedy-best-hits BOOL         Choose the best hits greedily to cover the
query [0]
common:
--sub-mat TWIN                  Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]
--max-seq-len INT               Maximum sequence length [65535]
--db-load-mode INT              Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT                   Number of CPU-cores used (all by default) [10]
--compressed INT                Write compressed output [0]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--gpu INT                       Use GPU (CUDA) if possible [0]
--gpu-server INT                Use GPU server [0]
--gpu-server-wait-timeout INT    Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR                Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL              Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL         Delete temporary files [1]
expert:
--filter-hits BOOL              Filter hits by seq.id. and coverage [0]
--sort-results INT              Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT             Create database lookup file (can be very large)
[0]
--chain-alignments INT          Chain overlapping alignments [0]
--merge-query INT               Combine ORFs/split sequences to a single entry
[1]
--strand INT                    Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output BOOL                Return a result DB instead of a text file [0]
--write-lookup INT              write .lookup file containing mapping from
internal id, fasta id and file number [0]

examples:
# Assign reciprocal best hit
mmseqs easy-rbh examples/QUERY.fasta examples/DB.fasta result tmp

references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.43. easy-search

Sensitive homology search.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy\_workflows.

#### Usage

```
usage:      mmseqs      easy-search      <i:queryFastaFile1[.gz|.bz2]>      ...
<i:queryFastaFileN[.gz|.bz2]>|<i:stdin>      <i:targetFastaFile[.gz]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]
```

#### Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |
| --k-score              | k-mer threshold for generating similar k-mer lists                   |
| --alph-size            | Alphabet size (range 2-21)                                           |

| Option       | Purpose                                                                                |
|--------------|----------------------------------------------------------------------------------------|
| --max-seqs   | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split      | Split input into N equally distributed chunks.<br>0: set the best split automatically  |
| --split-mode | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

```
usage: mmseqs easy-search <i:queryFastaFile1[.gz|.bz2]> ...
<i:queryFastaFileN[.gz|.bz2]>|<i:stdin> <i:targetFastaFile[.gz]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [5.700]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT               Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
  --split INT                  Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --diag-score BOOL            Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT     Extract only exact k-mers for matching (range
0-1) [0]
  --mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
```

--mask-prob FLOAT Mask sequences is probablity is above threshold [0.900]  
 --mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]  
 --mask-n-repeat INT Repeat letters that occure > threshold in a rwo [0]  
 --min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]  
 --spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use spaced k-mers [1]  
 --spaced-kmer-pattern STR User-specified spaced k-mer pattern []  
 --local-tmp STR Path where some of the temporary files will be created []  
 --disk-space-limit BYTE Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]  
 align:  
 -a B00L Add backtrace string (convert to alignments with mmseqs convertalis module) [0]  
 --alignment-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment [3]  
 --alignment-output-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment  
 5: score only (output) cluster format [0]  
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]  
 -e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]  
 --min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]  
 --min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]  
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]  
 --alt-ali INT Show up to this many alternative alignments [0]  
 -c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]  
 --cov-mode INT 0: coverage of query and target  
 1: coverage of target  
 2: coverage of query  
 3: target seq. length has to be at least x% of query length



4: query seq. length has to be at least x% of  
 target length  
 5: short seq. needs to be at least x% of the  
 other seq. length [0]  
 --max-rejected INT Maximum rejected alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --max-accept INT Maximum accepted alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --score-bias FLOAT Score bias when computing SW alignment (in  
 bits) [0.000]  
 --realign BOOL Compute more conservative, shorter alignments  
 (scores and E-values not changed) [0]  
 --realign-score-bias FLOAT Additional bias when computing realignment  
 [-0.200]  
 --realign-max-seqs INT Maximum number of results to return in  
 realignment [2147483647]  
 --corr-score-weight FLOAT Weight of backtrace correlation score that is  
 added to the alignment score [0.000]  
 --gap-open TWIN Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]  
 --zdrop INT Maximal allowed difference between score values  
 before alignment is truncated (nucleotide alignment only) [40]  
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,  
 1: filter [0]  
 profile:  
 --pca Pseudo count admixture strength []  
 --pcb Pseudo counts: Neff at half of maximum  
 admixture (range 0.0-inf) []  
 --mask-profile INT Mask query sequence of profile using tantan  
 [0,1] [1]  
 --e-profile DOUBLE Include sequences matches with < E-value thr.  
 into the profile (>=0.0) [1.000E-03]  
 --wg BOOL Use global sequence weighting for profile  
 calculation [0]  
 --filter-msa INT Filter msa: 0: do not filter, 1: filter [1]  
 --filter-min-enable INT Only filter MSAs with more than N sequences, 0  
 always filters [0]  
 --max-seq-id FLOAT Reduce redundancy of output MSA using max.  
 pairwise sequence identity [0.0,1.0] [0.900]  
 --qid STR Reduce diversity of output MSAs using min.seq.  
 identity with query sequences [0.0,1.0]  
 Alternatively, can be a list of multiple  
 thresholds:  
 E.g.: 0.15,0.30,0.50 to defines filter buckets  
 of ]0.15-0.30] and ]0.30-0.50] [0.0]  
 --qsc FLOAT Reduce diversity of output MSAs using min.  
 score per aligned residue with query sequences [-50.0,100.0] [-20.000]  
 --cov FLOAT Filter output MSAs using min. fraction of query  
 residues covered by matched sequences [0.0,1.0] [0.000]

```

--diff INT                                Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT                      use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT                  Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--num-iterations INT                      Number of iterative profile search iterations
[1]
--exhaustive-search BOOL                   For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL                         Efficient search for LCA candidates [0]
misc:
--taxon-list STR                          Taxonomy ID, possibly multiple values separated
by ',' [ ]
--prefilter-mode INT                      prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT                        Rescore diagonals with:
0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality
criterion [0]
--allow-deletion BOOL                     Allow deletions in a MSA [0]
--min-length INT                          Minimum codon number in open reading frames
[30]
--max-length INT                          Maximum codon number in open reading frames
[32734]
--max-gaps INT                            Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT                   Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT                     Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT                      Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR                      Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR                      Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT                   1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL

```

23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24) PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27) KARYORELICT, 28) CONDYLOSTOMA  
 29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]

--translate INT Translate ORF to amino acid [0]  
 --use-all-table-starts BOOL Use all alternatives for a start codon in the genetic table, if false - only ATG (AUG) [0]  
 --id-offset INT Numeric ids in index file are offset by this value [0]  
 --sequence-overlap INT Overlap between sequences [0]  
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link data and write new index, [1]  
 --headers-split-mode INT Header split mode: 0: split position, 1: original header [0]  
 --search-type INT Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment [0]  
 --start-sens FLOAT Start sensitivity [4.000]  
 --sens-steps INT Number of search steps performed from --start-sens to -s [1]  
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs, 1: full reading frames [0]  
 --format-mode INT Output format:  
 0: BLAST-TAB  
 1: SAM  
 2: BLAST-TAB + query/db length  
 3: Pretty HTML  
 4: BLAST-TAB + column headers  
 BLAST-TAB (0) and BLAST-TAB + column headers  
 (4) support custom output formats (--format-output) [0]  
 --format-output STR Choose comma separated list of output columns from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen,tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos [query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]  
 --overlap FLOAT Maximum overlap of covered regions [0.000]  
 --dbtype INT Database type 0: auto, 1: amino acid 2: nucleotides [0]  
 --shuffle BOOL Shuffle input database [1]  
 --createdb-mode INT Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) [0]  
 --greedy-best-hits BOOL Choose the best hits greedily to cover the query [0]  
 common:  
 --sub-mat TWIN Substitution matrix file [aa:blosum62.out,nuc:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:

```

mmap, 3: mmap+touch [0]
--threads INT                Number of CPU-cores used (all by default) [10]
--compressed INT             Write compressed output [0]
-v INT                        Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--gpu INT                     Use GPU (CUDA) if possible [0]
--gpu-server INT              Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR              Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL            Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL       Delete temporary files [1]
expert:
--filter-hits BOOL            Filter hits by seq.id. and coverage [0]
--sort-results INT            Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT           Create database lookup file (can be very large)
[0]
--chain-alignments INT         Chain overlapping alignments [0]
--merge-query INT              Combine ORFs/split sequences to a single entry
[1]
--strand INT                  Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output BOOL              Return a result DB instead of a text file [0]
--write-lookup INT             write .lookup file containing mapping from
internal id, fasta id and file number [0]

examples:
# Search multiple FASTA against FASTA (like BLASTP, TBLASTN, BLASTX, BLASTN --
search-type 3, TBLASTX --search-type 2)
mmseqs easy-search examples/QUERY.fasta examples/QUERY.fasta examples/DB.fasta
result.m8 tmp

# Iterative profile search from stdin (like PSI-BLAST)
cat examples/QUERY.fasta | mmseqs easy-search stdin examples/DB.fasta result.m8
tmp --num-iterations 2

# Profile search against small databases (e.g. PFAM, eggNOG)
mmseqs databases PFAM pfam_db tmp
mmseqs easy-search examples/QUERY.fasta pfam_db res.m8 tmp

# Exhaustive search against sequences or profiles (works for large DBs)
mmseqs easy-search examples/QUERY.fasta targetProfiles res.m8 tmp --exhaustive-
search

# Increasing sensitivity search (from 2 to 7 in 3 steps)
mmseqs easy-search examples/QUERY.fasta examples/DB.fasta result.m8 tmp --start-

```

```
sens 2 -s 7 --sens-steps 3
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. *bioRxiv*, 2024.11.13.623350 (2024)

#### 6.2.44. easy-taxonomy

Taxonomic classification.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 11 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy\_workflows.

#### Usage

```
usage: mmseqs easy-taxonomy <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<i:targetDB> <o:taxReports> <tmpDir> [options]
```

#### Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score            | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size          | Alphabet size (range 2-21)                                                             |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split              | Split input into N equally distributed chunks.<br>0: set the best split automatically  |
| --split-mode         | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

usage: mmseqs easy-taxonomy <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>  
<i:targetDB> <o:taxReports> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

|                              |                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------|
| --comp-bias-corr INT         | Correct for locally biased amino acid composition (range 0-1) [1]                                |
| --comp-bias-corr-scale FLOAT | Correct for locally biased amino acid composition (range 0-1) [1.000]                            |
| --add-self-matches BOOL      | Artificially add entries of queries with themselves (for clustering) [0]                         |
| --seed-sub-mat TWIN          | Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]                |
| -s FLOAT                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [4.000]                                         |
| -k INT                       | k-mer length (0: automatically set to optimum) [0]                                               |
| --target-search-mode INT     | target search mode (0: regular k-mer, 1: similar k-mer) [0]                                      |
| --k-score TWIN               | k-mer threshold for generating similar k-mer lists [seq:2147483647,prof:2147483647]              |
| --alph-size TWIN             | Alphabet size (range 2-21) [aa:21,nucl:5]                                                        |
| --max-seqs INT               | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]     |
| --split INT                  | Split input into N equally distributed chunks. 0: set the best split automatically [0]           |
| --split-mode INT             | 0: split target db; 1: split query db; 2: auto, depending on main memory [2]                     |
| --split-memory-limit BYTE    | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory [0] |
| --diag-score BOOL            | Use ungapped diagonal scoring during prefilter                                                   |

```

[1]
--exact-kmer-matching INT      Extract only exact k-mers for matching (range
0-1) [0]
--mask INT                     Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT             Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT         Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT           Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT      Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT        0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR     User-specified spaced k-mer pattern []
--local-tmp STR               Path where some of the temporary files will be
created []
--disk-space-limit BYTE       Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                       Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT          How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [0]
--alignment-output-mode INT   How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment
5: score only (output) cluster format [0]
--wrapped-scoring BOOL       Double the (nucleotide) query sequence during
the scoring process to allow wrapped diagonal scoring around end and start [0]
-e DOUBLE                     List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT           List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT            0: alignment length 1: shorter, 2: longer
sequence [0]
--alt-ali INT                Show up to this many alternative alignments [0]
-c FLOAT                     List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]

```

```

--cov-mode INT          0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
--max-rejected INT      Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
--max-accept INT        Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT      Score bias when computing SW alignment (in
bits) [0.000]
--realign BOOL          Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT  Maximum number of results to return in
realignment [2147483647]
--corr-score-weight FLOAT Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN         Gap open cost [aa:11,nucl:5]
--gap-extend TWIN       Gap extension cost [aa:1,nucl:2]
--zdrop INT             Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
--exhaustive-search-filter INT Filter result during search: 0: do not filter,
1: filter [0]
profile:
--pca                  Pseudo count admixture strength []
--pcb                  Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT     Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE     Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg BOOL              Use global sequence weighting for profile
calculation [0]
--filter-msa INT       Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT     Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR              Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                        Alternatively, can be a list of multiple
thresholds:
                        E.g.: 0.15,0.30,0.50 to defines filter buckets

```



of ]0.15-0.30] and ]0.30-0.50] [0.0]

--qsc FLOAT                      Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT                      Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT                      Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT              use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT          Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--exhaustive-search BOOL          For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL                  Efficient search for LCA candidates [0]

misc:

--orf-filter INT                  Prefilter query ORFs with non-selective search  
Only used during nucleotide-vs-protein

classification

NOTE: Consider disabling when classifying short reads [0]

--orf-filter-e DOUBLE              E-value threshold used for query ORF prefiltering [1.000E+02]

--orf-filter-s FLOAT              Sensitivity used for query ORF prefiltering [2.000]

--lca-mode INT                    LCA Mode 1: single search LCA , 2/3: approximate 2bLCA, 4: top hit [3]

--majority FLOAT                  minimal fraction of agreement among taxonomically assigned sequences of a set [0.500]

--vote-mode INT                  Mode of assigning weights to compute majority. 0: uniform, 1: minus log E-value, 2: score [1]

--lca-ranks STR                  Add column with specified ranks (',' separated) [ ]

--tax-lineage INT                  0: don't show, 1: add all lineage names, 2: add all lineage taxids [0]

--blacklist STR                  Comma separated list of ignored taxa in LCA computation [12908:unclassified sequences,28384:other sequences]

--taxon-list STR                  Taxonomy ID, possibly multiple values separated by ',' [ ]

--prefilter-mode INT              prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT              Rescore diagonals with:  
0: Hamming distance  
1: local alignment (score only)  
2: local alignment  
3: global alignment  
4: longest alignment fulfilling window quality

criterion [0]

--allow-deletion BOOL              Allow deletions in a MSA [0]

--min-length INT                  Minimum codon number in open reading frames

```

[30]
--max-length INT                Maximum codon number in open reading frames
[32734]
--max-gaps INT                  Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT         Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT           Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT            Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR            Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR            Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT         1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)
BLASTOCRITHIDIA [1]
--translate INT                 Translate ORF to amino acid [0]
--use-all-table-starts BOOL     Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                 Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT           Overlap between sequences [0]
--sequence-split-mode INT        Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT         Header split mode: 0: split position, 1:
original header [0]
--search-type INT                Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--translation-mode INT           Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
--report-mode INT                Taxonomy report mode 0: Kraken 1: Krona [0]
--format-mode INT                Output format:
0: BLAST-TAB
1: SAM
2: BLAST-TAB + query/db length
3: Pretty HTML

```

4: BLAST-TAB + column headers  
 BLAST-TAB (0) and BLAST-TAB + column headers

(4) support custom output formats (--format-output) [0]  
 --format-output STR Choose comma separated list of output columns  
 from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen  
 tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c  
 qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos  
 [query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]  
 --first-seq-as-repr BOOL Use the first sequence of the clustering result  
 as representative sequence [0]  
 --target-column INT Select a target column (default 1), 0 if no  
 target id exists [1]  
 --full-header BOOL Replace DB ID by its corresponding Full Header  
 [0]  
 --idx-seq-src INT 0: auto, 1: split/translated sequences, 2:  
 input sequences [0]  
 --dbtype INT Database type 0: auto, 1: amino acid 2:  
 nucleotides [0]  
 --shuffle BOOL Shuffle input database [1]  
 --createdb-mode INT Createdb mode 0: copy data, 1: soft link data  
 and write new index (works only with single line fasta/q) [1]  
 common:  
 --compressed INT Write compressed output [0]  
 --threads INT Number of CPU-cores used (all by default) [10]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:  
 +warnings, 3: +info [3]  
 --sub-mat TWIN Substitution matrix file  
 [aa:blosum62.out,nucl:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:  
 mmap, 3: mmap+touch [0]  
 --gpu INT Use GPU (CUDA) if possible [0]  
 --gpu-server INT Use GPU server [0]  
 --gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no  
 wait limit: >0 this many seconds [600]  
 --mpi-runner STR Use MPI on compute cluster with this MPI  
 command (e.g. "mpirun -np 42") []  
 --force-reuse BOOL Reuse tmp filse in tmp/latest folder ignoring  
 parameters and version changes [0]  
 --remove-tmp-files BOOL Delete temporary files [1]  
 expert:  
 --filter-hits BOOL Filter hits by seq.id. and coverage [0]  
 --sort-results INT Sort results: 0: no sorting, 1: sort by E-value  
 (Alignment) or seq.id. (Hamming) [0]  
 --create-lookup INT Create database lookup file (can be very large)  
 [0]  
 --chain-alignments INT Chain overlapping alignments [0]  
 --merge-query INT Combine ORFs/split sequences to a single entry

```
[1]
--strand INT                Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output B00L            Return a result DB instead of a text file [0]
--write-lookup INT          write .lookup file containing mapping from
internal id, fasta id and file number [0]
```

examples:

```
# Assign taxonomic labels to FASTA sequences
- result_tophit_aln: top hits
- result_tophit_report: coverage profiles per database entry
- result_report: kraken style report
# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Assign taxonomy based on 2bLCA hit
mmseqs easy-taxonomy examples/DB.fasta swissprotDB result tmp

# Assign taxonomy based on top hit
mmseqs easy-taxonomy examples/DB.fasta swissprotDB result tmp --lca-mode 4

# Assign taxonomy without ORF prefilter
# Classifies higher percentage for short nucleotide input (e.g. short reads) at
the cost of speed
mmseqs easy-taxonomy queryNuclDB swissprotDB result tmp --orf-filter 0
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics*, btab184 (2021)

#### 6.2.45. expand2profile

Expand an alignment result based on another and create a profile.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

usage: mmseqs expand2profile <inputDB(s)> <outputDB> [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs expand2profile locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.46. expandaln

Expand an alignment result based on another.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

### Usage

usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB|ca3mDB> <o:alignmentDB> [options]

### Key Options

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |
| --gap-open             | Gap open cost                                                 |
| --gap-extend           | Gap extension cost                                            |

| Option            | Purpose                                                                         |
|-------------------|---------------------------------------------------------------------------------|
| --score-bias      | Score bias when computing SW alignment (in bits)                                |
| -e                | List matches below this E-value (range 0.0-inf)                                 |
| --min-seq-id      | List matches above this sequence identity (for clustering) (range 0.0-1.0)      |
| -c                | List matches above this fraction of aligned (covered) residues (see --cov-mode) |
| --cov-mode        | 0: coverage of query and target                                                 |
| --pseudo-cnt-mode | use 0: substitution-matrix or 1: context-specific pseudocounts                  |
| --pca             | Pseudo count admixture strength                                                 |
| --pcb             | Pseudo counts: Neff at half of maximum admixture (range 0.0-inf)                |

### Full CLI Help Snapshot

usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB|ca3mDB> <o:alignmentDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT           Correct for locally biased amino acid composition (range 0-1) [1]

--comp-bias-corr-scale FLOAT   Correct for locally biased amino acid composition (range 0-1) [1.000]

align:

--gap-open TWIN               Gap open cost [aa:11,nucl:5]

--gap-extend TWIN             Gap extension cost [aa:1,nucl:2]

--score-bias FLOAT            Score bias when computing SW alignment (in bits) [0.000]

-e DOUBLE                      List matches below this E-value (range 0.0-inf) [1.000E-03]

--min-seq-id FLOAT            List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]

-c FLOAT                       List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]

--cov-mode INT                0: coverage of query and target  
                               1: coverage of target  
                               2: coverage of query  
                               3: target seq. length has to be at least x% of query length  
                               4: query seq. length has to be at least x% of target length  
                               5: short seq. needs to be at least x% of the

```

other seq. length [0]
profile:
  --pseudo-cnt-mode INT          use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
  --pca                          Pseudo count admixture strength []
  --pcb                          Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
  --filter-min-enable INT        Only filter MSAs with more than N sequences, 0
always filters [0]
  --max-seq-id FLOAT            Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
  --qid STR                      Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                                Alternatively, can be a list of multiple
thresholds:
                                E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
  --qsc FLOAT                   Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
  --cov FLOAT                   Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
  --diff INT                    Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
  --expansion-mode INT          Update score, E-value, and sequence identity by
0: input alignment 1: rescoring the inferred backtrace [0]
  --expand-filter-clusters INT  Filter each target cluster during expansion 0: no
filter 1: filter [0]
common:
  --sub-mat TWIN                Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT             Maximum sequence length [65535]
  --db-load-mode INT            Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
  --compressed INT             Write compressed output [0]
  --threads INT                Number of CPU-cores used (all by default) [10]
  -v INT                       Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.47. extractalignedregion

Extract aligned sequence region from query.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

## Usage

```
usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB>
<o:sequenceDB> [options]
```

## Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --extract-mode | Extract from 1: Query, 2: Target                                |
| --compressed   | Write compressed output                                         |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads      | Number of CPU-cores used (all by default)                       |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

## Full CLI Help Snapshot

```
usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB>
<o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --extract-mode INT    Extract from 1: Query, 2: Target [2]
common:
  --compressed INT      Write compressed output [0]
  --db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT         Number of CPU-cores used (all by default) [10]
  -v INT                Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.48. extractdomains

Extract highest scoring alignment regions for each sequence from BLAST-tab file.



Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

### Usage

```
usage: mmseqs extractdomains <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs extractdomains` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.49. **extractframes**

Extract frames from a nucleotide sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `sequence_manipulation`.

### Usage

```
usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]
```

### Key Options

| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --forward-frames    | Comma-separated list of frames on the forward strand to be extracted                                                                |
| --reverse-frames    | Comma-separated list of frames on the reverse strand to be extracted                                                                |
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --translate         | Translate ORF to amino acid                                                                                                         |
| --threads           | Number of CPU-cores used (all by default)                                                                                           |
| --compressed        | Write compressed output                                                                                                             |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                       |
| --create-lookup     | Create database lookup file (can be very large)                                                                                     |

### Full CLI Help Snapshot

```
usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --forward-frames STR      Comma-separated list of frames on the forward strand
to be extracted [1,2,3]
  --reverse-frames STR      Comma-separated list of frames on the reverse strand
to be extracted [1,2,3]
  --translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL, 21)
TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA [1]
  --translate INT           Translate ORF to amino acid [0]
common:
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT           Write compressed output [0]
  -v INT                     Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
```

--create-lookup INT            Create database lookup file (can be very large) [0]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.50. extractorfs

Six-frame extraction of open reading frames.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 6 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

#### Usage

usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]

#### Key Options

| Option              | Purpose                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| --min-length        | Minimum codon number in open reading frames                                                                                     |
| --max-length        | Maximum codon number in open reading frames                                                                                     |
| --max-gaps          | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected                                 |
| --contig-start-mode | Contig start can be 0: incomplete, 1: complete, 2: both                                                                         |
| --contig-end-mode   | Contig end can be 0: incomplete, 1: complete, 2: both                                                                           |
| --orf-start-mode    | Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle) |
| --forward-frames    | Comma-separated list of frames on the forward strand to be extracted                                                            |

| Option                 | Purpose                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --reverse-frames       | Comma-separated list of frames on the reverse strand to be extracted                                                                |
| --translation-table    | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --translate            | Translate ORF to amino acid                                                                                                         |
| --use-all-table-starts | Use all alternatives for a start codon in the genetic table, if false - only ATG (AUG)                                              |
| --id-offset            | Numeric ids in index file are offset by this value                                                                                  |

### Full CLI Help Snapshot

```
usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --min-length INT           Minimum codon number in open reading frames [30]
  --max-length INT           Maximum codon number in open reading frames [32734]
  --max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
  --contig-start-mode INT    Contig start can be 0: incomplete, 1: complete, 2:
both [2]
  --contig-end-mode INT     Contig end can be 0: incomplete, 1: complete, 2:
both [2]
  --orf-start-mode INT      Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
  --forward-frames STR      Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
  --reverse-frames STR      Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
  --translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
```

```

--translate INT          Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT          Numeric ids in index file are offset by this value
[0]
common:
--threads INT            Number of CPU-cores used (all by default) [10]
--compressed INT        Write compressed output [0]
-v INT                  Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--create-lookup INT      Create database lookup file (can be very large) [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

### 6.2.51. filterdb

DB filtering by given conditions.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 8 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

#### Usage

usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]

#### Key Options

| Option              | Purpose                                                              |
|---------------------|----------------------------------------------------------------------|
| --add-self-matches  | Artificially add entries of queries with themselves (for clustering) |
| --filter-expression | Specify a mathematical expression to filter lines                    |
| --filter-column     | column                                                               |
| --column-to-take    | column to take in join mode. If -1, the whole line is taken          |

| Option               | Purpose                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------|
| --filter-regex       | Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9])                      |
| --positive-filter    | Used in conjunction with --filter-file. If true, out = in filter ; if false, out = in - filter |
| --filter-file        | Specify a file that contains the filtering elements                                            |
| --beats-first        | Filter by comparing each entry to the first entry                                              |
| --mapping-file       | Specify a file that translates the keys of a DB to new keys, TSV format                        |
| --weights            | Weights used for cluster prioritization                                                        |
| --trim-to-one-column | Output only the column specified by --filter-column                                            |
| --extract-lines      | Extract n lines of each entry                                                                  |

### Full CLI Help Snapshot

```
usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]
  By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --add-self-matches BOOL      Artificially add entries of queries with themselves
                                (for clustering) [0]
misc:
  --filter-expression STR      Specify a mathematical expression to filter lines []
  --filter-column INT          column [1]
  --column-to-take INT         column to take in join mode. If -1, the whole line
                                is taken [-1]
  --filter-regex STR           Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9]) [^.*$]
  --positive-filter BOOL       Used in conjunction with --filter-file. If true, out
                                = in \intersect filter ; if false, out = in - filter [1]
  --filter-file STR            Specify a file that contains the filtering elements
                                []
  --beats-first BOOL           Filter by comparing each entry to the first entry
                                [0]
  --mapping-file STR           Specify a file that translates the keys of a DB to
                                new keys, TSV format []
  --weights STR                Weights used for cluster prioritization []
  --trim-to-one-column BOOL    Output only the column specified by --filter-column
                                [0]
  --extract-lines INT          Extract n lines of each entry [0]
  --comparison-operator STR    Filter by comparing each entry row numerically by
                                using the le) less-than-equal, ge) greater-than-equal or e) equal operator []
```

```

--comparison-value DOUBLE    Filter by comparing each entry to this value
[0.000E+00]
--sort-entries INT           Sort column set by --filter-column, by 0: no
sorting, 1: increasing, 2: decreasing, 3: random shuffle, 4: priority [0]
--join-db STR                Join another database entry with respect to the
database identifier in the chosen column []
common:
--threads INT                Number of CPU-cores used (all by default) [10]
--compressed INT            Write compressed output [0]
-v INT                       Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]

```

examples:

```

# Retain top alignment for each query (alignment DBs are sorted by E-value)
mmseqs filterdb alignmentDB topHitAlignmentDB --extract-lines 1

# Extract alignments with Seq.id. greater than 90%
mmseqs filterdb alignmentDB scoreGreater35AlignmentDB --comparison-operator ge
--comparison-value 0.9 --filter-column 2

# Retain all hits matching a regular expression
mmseqs filterdb alignmentDB regexFilteredDB --filter-regex '^[1-9].$' --filter-
column 2

# Remove all hits to target keys contained in file db.index
mmseqs filterdb --filter-file db.index --positive-filter false

# Retain all hits matching any boolean expression
mmseqs filterdb --filter-expression '$1 * $2 >= 200'

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.52. filterresult

Pairwise alignment result filter.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

## Usage

usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>  
[options]

## Key Options

| Option                 | Purpose                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                          |
| --add-self-matches     | keep the query (representative) sequence                                                                               |
| --gap-open             | Gap open cost                                                                                                          |
| --gap-extend           | Gap extension cost                                                                                                     |
| --filter-min-enable    | Only filter MSAs with more than N sequences, 0 always filters                                                          |
| --max-seq-id           | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]                                        |
| --qid                  | Reduce diversity of output MSAs using min.seq. identity with query sequences                                           |
| --qsc                  | Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0]                |
| --cov                  | Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0]                        |
| --diff                 | Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 |
| --allow-deletion       | Allow deletions in a MSA                                                                                               |

## Full CLI Help Snapshot

usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>  
[options]

By Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT           Correct for locally biased amino acid composition  
(range 0-1) [1]  
--comp-bias-corr-scale FLOAT   Correct for locally biased amino acid composition  
(range 0-1) [1.000]



```

--add-self-matches BOOL      keep the query (representative) sequence [0]
align:
--gap-open TWIN              Gap open cost [aa:11,nucl:5]
--gap-extend TWIN            Gap extension cost [aa:1,nucl:2]
profile:
--filter-min-enable INT      Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT           Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
--qid STR                     Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple
thresholds:
E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
--qsc FLOAT                  Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                  Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                   Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--allow-deletion BOOL        Allow deletions in a MSA [0]
common:
--sub-mat TWIN               Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT           Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT                Number of CPU-cores used (all by default) [10]
--compressed INT             Write compressed output [0]
-v INT                       Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

### 6.2.53. filtertaxdb

Filter taxonomy result database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

## Usage

```
usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]
```

## Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --taxon-list | Taxonomy ID, possibly multiple values separated by ‘,’        |
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## Full CLI Help Snapshot

```
usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

```
--taxon-list STR    Taxonomy ID, possibly multiple values separated by ',' []
```

common:

```
--compressed INT    Write compressed output [0]
```

```
--threads INT        Number of CPU-cores used (all by default) [10]
```

```
-v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
```

[3]

examples:

```
# Download a sequence database with taxonomy information
```

```
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp
```

```
# Annotate a queryDB with taxonomy information
```

```
mmseqs taxonomy queryDB swissprotDB taxDB tmp
```

```
# Retain all unclassified hits
```

```
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list 0
```

```
mmseqs createsubdb <(awk '$3 == 1' filteredTaxDB.index) queryDB
```

```
queryUnclassifiedDB
```

```
# Retain all eukaryotic hits except fungi
```

```
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list '2759&&!4751'
```

```
# Retain all human and chlamydia hits
```

```
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list '9606||810'
```

references:

```
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
```

sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184 (2021)

#### 6.2.54. **filtertaxseqdb**

Filter taxonomy sequence database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### **Usage**

```
usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]
```

#### **Key Options**

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --taxon-list | Taxonomy ID, possibly multiple values separated by ‘;         |
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index     |
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### **Full CLI Help Snapshot**

```
usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]
  By Eli Levy Karin <eli.levy.karin@gmail.com> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: misc:
  --taxon-list STR  Taxonomy ID, possibly multiple values separated by ',' []
  --subdb-mode INT  Subdb mode 0: copy data 1: soft link data and write index [0]
common:
  --compressed INT  Write compressed output [0]
  --threads INT     Number of CPU-cores used (all by default) [10]
  -v INT            Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

examples:

```
# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Retain all bacterial sequences
mmseqs filtertaxseqdb swissprotDB swissprotDB_only_bac --taxon-list 2

# Retain all eukaryotic sequences except fungi
mmseqs filtertaxseqdb swissprotDB swissprotDB_euk_wo_fungi --taxon-list '2759&&!4751'

# Retain all human and chlamydia sequences
mmseqs filtertaxseqdb swissprotDB swissprotDB_human_and_chlamydia --taxon-list '9606||810'
```

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics*, btab184 (2021)

#### 6.2.55. fwbw

Forward Backward Alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

#### Usage

usage: mmseqs fwbw <inputDB(s)> <outputDB> [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs fwbw locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

**Note**

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.56. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

#### Usage

usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]

#### Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --min-ungapped-score   | Accept only matches with ungapped alignment score above threshold                      |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --gap-open             | Gap open cost                                                                          |
| --gap-extend           | Gap extension cost                                                                     |
| -e                     | List matches below this E-value (range 0.0-inf)                                        |
| -c                     | List matches above this fraction of aligned (covered) residues (see --cov-mode)        |
| --cov-mode             | 0: coverage of query and target                                                        |
| --taxon-list           | Taxonomy ID, possibly multiple values separated by ‘;                                  |
| --sub-mat              | Substitution matrix file                                                               |

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |

## Full CLI Help Snapshot

usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]  
 By Milot Mirdita <milot@mirdita.de> & Martin Steinegger  
 <martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT           Correct for locally biased amino acid composition  
 (range 0-1) [1]  
 --comp-bias-corr-scale FLOAT   Correct for locally biased amino acid composition  
 (range 0-1) [1.000]  
 --min-ungapped-score INT       Accept only matches with ungapped alignment score  
 above threshold [15]  
 --max-seqs INT                Maximum results per query sequence allowed to pass  
 the prefilter (affects sensitivity) [300]

align:

--gap-open TWIN               Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN            Gap extension cost [aa:1,nucl:2]  
 -e DOUBLE                    List matches below this E-value (range 0.0-inf)  
 [1.000E-03]  
 -c FLOAT                    List matches above this fraction of aligned  
 (covered) residues (see --cov-mode) [0.000]  
 --cov-mode INT                0: coverage of query and target  
                               1: coverage of target  
                               2: coverage of query  
                               3: target seq. length has to be at least x% of  
 query length  
                               4: query seq. length has to be at least x% of  
 target length  
                               5: short seq. needs to be at least x% of the other  
 seq. length [0]

misc:

--taxon-list STR              Taxonomy ID, possibly multiple values separated by  
 ',' []

common:

--sub-mat TWIN                Substitution matrix file  
 [aa:blosum62.out,nucl:nucleotide.out]  
 --db-load-mode INT           Database preload mode 0: auto, 1: fread, 2: mmap,  
 3: mmap+touch [0]  
 --threads INT                Number of CPU-cores used (all by default) [10]  
 --compressed INT             Write compressed output [0]  
 -v INT                        Verbosity level: 0: quiet, 1: +errors, 2:  
 +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. *bioRxiv*, 2024.11.13.623350 (2024)

#### 6.2.57. **gff2db**

Extract regions from a sequence database based on a GFF3 file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

#### Usage

```
usage: mmseqs gff2db <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs gff2db` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

##### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.58. **gpuserver**

Start a GPU server.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

### Usage

usage: mmseqs gpuserver <i:DB> [options]

### Key Options

| Option           | Purpose                                                                                |
|------------------|----------------------------------------------------------------------------------------|
| --max-seqs       | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --prefilter-mode | prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped          |
| --gpu            | Use GPU (CUDA) if possible                                                             |
| --db-load-mode   | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                        |

### Full CLI Help Snapshot

usage: mmseqs gpuserver <i:DB> [options]

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger

<martin.steinegger@snu.ac.kr>

options: prefilter:

--max-seqs INT Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]

misc:

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

common:

--gpu INT Use GPU (CUDA) if possible [0]

--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]

references:

- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv, 2024.11.13.623350 (2024)

### 6.2.59. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.



This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

## Usage

```
usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]
```

## Key Options

| Option                | Purpose                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| --alph-size           | Alphabet size (range 2-21)                                                                                   |
| --spaced-kmer-mode    | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern | User-specified spaced k-mer pattern                                                                          |
| --mask                | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob           | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case     | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat       | Repeat letters that occur > threshold in a two                                                               |
| -k                    | k-mer length (0: automatically set to optimum)                                                               |
| --split-memory-limit  | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| --min-seq-id          | List matches above this sequence identity (for clustering) (range 0.0-1.0)                                   |
| --cov-mode            | 0: coverage of query and target                                                                              |
| -c                    | List matches above this fraction of aligned (covered) residues (see --cov-mode)                              |

## Full CLI Help Snapshot

```
usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

```

--alph-size TWIN           Alphabet size (range 2-21) [aa:13,nucl:5]
--spaced-kmer-mode INT     0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
--spaced-kmer-pattern STR  User-specified spaced k-mer pattern []
--mask INT                Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
--mask-prob FLOAT         Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT     Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT       Repeat letters that occure > threshold in a rwo
[0]
-k INT                    k-mer length (0: automatically set to optimum)
[0]
--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
align:
--min-seq-id FLOAT        List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--cov-mode INT            0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
-c FLOAT                 List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
kmermatcher:
--kmer-per-seq INT        k-mers per sequence [0]
--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL    Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT          Shift k-mer hash initialization [67]
--include-only-extendable BOOL Include only extendable [0]
--ignore-multi-kmer BOOL  Skip k-mers occurring multiple times (>=2) [0]
--weights STR             Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
common:
--sub-mat TWIN            Substitution matrix file
[aa:blos62.out,nucl:nucleotide.out]
--max-seq-len INT         Maximum sequence length [65535]
--threads INT             Number of CPU-cores used (all by default) [10]
--compressed INT          Write compressed output [0]
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2:

```

+warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.60. kmersearch

Find bottom-m-hashed k-mer matches between target and query DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

### Usage

usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]

### Key Options

| Option               | Purpose                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                                                |
| --mask               | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob          | Mask sequences is probablity is above threshold                                                              |
| --mask-lower-case    | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat      | Repeat letters that occure > threshold in a rwo                                                              |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| --cov-mode           | 0: coverage of query and target                                                                              |
| -c                   | List matches above this fraction of aligned (covered) residues (see --cov-mode)                              |
| --kmer-per-seq       | k-mers per sequence                                                                                          |

| Option               | Purpose                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| --kmer-per-seq-scale | Scale k-mer per sequence based on sequence length as kmer-per-seq val + scale x seqlen |
| --pick-n-sim-kmer    | Add N similar k-mers to search                                                         |
| --result-direction   | result is 0: query, 1: target centric                                                  |

### Full CLI Help Snapshot

```
usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
                                [aa:VTML80.out,nucl:nucleotide.out]
  --mask INT                   Mask sequences in prefilter stage with tantan: 0: w/o
                                low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT            Mask sequences is probablity is above threshold
                                [0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer search
                                0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a rwo [0]
  --split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M, 1G.
                                Default (0) to all available system memory [0]
align:
  --cov-mode INT               0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of query
                                length
                                4: query seq. length has to be at least x% of target
                                length
                                5: short seq. needs to be at least x% of the other
                                seq. length [0]
  -c FLOAT                    List matches above this fraction of aligned (covered)
                                residues (see --cov-mode) [0.800]
kmermatcher:
  --kmer-per-seq INT           k-mers per sequence [0]
  --kmer-per-seq-scale TWIN    Scale k-mer per sequence based on sequence length as
                                kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
  --pick-n-sim-kmer INT        Add N similar k-mers to search [1]
  --result-direction INT       result is 0: query, 1: target centric [1]
common:
  --max-seq-len INT            Maximum sequence length [65535]
  --threads INT                Number of CPU-cores used (all by default) [10]
  --compressed INT             Write compressed output [0]
  -v INT                       Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
                                3: +info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.61. lcaalign

Efficient gapped alignment for lca computation.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB>
[options]
```

### Key Options

| Option                  | Purpose                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr        | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --comp-bias-corr-scale  | Correct for locally biased amino acid composition (range 0-1)                                                            |
| --add-self-matches      | Artificially add entries of queries with themselves (for clustering)                                                     |
| -a                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |
| --alignment-mode        | How to compute the alignment:                                                                                            |
| --alignment-output-mode | How to compute the alignment:                                                                                            |
| --wrapped-scoring       | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| -e                      | List matches below this E-value (range 0.0-inf)                                                                          |
| --min-seq-id            | List matches above this sequence identity (for clustering) (range 0.0-1.0)                                               |

| Option        | Purpose                                            |
|---------------|----------------------------------------------------|
| --min-aln-len | Minimum alignment length (range 0-INT_MAX)         |
| --seq-id-mode | 0: alignment length 1: shorter, 2: longer sequence |
| --alt-ali     | Show up to this many alternative alignments        |

### Full CLI Help Snapshot

```
usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB>
[options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
                                (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
                                (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
                                themselves (for clustering) [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
                                mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
  --alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
  --wrapped-scoring BOOL       Double the (nucleotide) query sequence during the
                                scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
                                [1.000E-03]
  --min-seq-id FLOAT           List matches above this sequence identity (for
                                clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT            0: alignment length 1: shorter, 2: longer sequence
                                [0]
  --alt-ali INT                Show up to this many alternative alignments [0]
  -c FLOAT                     List matches above this fraction of aligned
                                (covered) residues (see --cov-mode) [0.000]
  --cov-mode INT               0: coverage of query and target
```

1: coverage of target  
 2: coverage of query  
 3: target seq. length has to be at least x% of  
 query length  
 4: query seq. length has to be at least x% of  
 target length  
 5: short seq. needs to be at least x% of the other  
 seq. length [0]  
 --max-rejected INT Maximum rejected alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --max-accept INT Maximum accepted alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --score-bias FLOAT Score bias when computing SW alignment (in bits)  
 [0.000]  
 --realign BOOL Compute more conservative, shorter alignments  
 (scores and E-values not changed) [0]  
 --realign-score-bias FLOAT Additional bias when computing realignment  
 [-0.200]  
 --realign-max-seqs INT Maximum number of results to return in realignment  
 [2147483647]  
 --corr-score-weight FLOAT Weight of backtrace correlation score that is  
 added to the alignment score [0.000]  
 --gap-open TWIN Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]  
 --zdrop INT Maximal allowed difference between score values  
 before alignment is truncated (nucleotide alignment only) [40]  
 profile:  
 --pca Pseudo count admixture strength []  
 --pcb Pseudo counts: Neff at half of maximum admixture  
 (range 0.0-inf) []  
 common:  
 --sub-mat TWIN Substitution matrix file  
 [aa:blosom62.out,nucl:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap,  
 3: mmap+touch [0]  
 --threads INT Number of CPU-cores used (all by default) [10]  
 --compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:  
 +warnings, 3: +info [3]  
 references:  
 - Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and  
 sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184  
 (2021)

#### 6.2.62. lca

Compute the lowest common ancestor.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

## Usage

```
usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]
```

## Key Options

| Option        | Purpose                                                            |
|---------------|--------------------------------------------------------------------|
| --lca-ranks   | Add column with specified ranks (',' separated)                    |
| --blacklist   | Comma separated list of ignored taxa in LCA computation            |
| --tax-lineage | 0: don't show, 1: add all lineage names, 2: add all lineage taxids |
| --compressed  | Write compressed output                                            |
| --threads     | Number of CPU-cores used (all by default)                          |
| -v            | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info      |

## Full CLI Help Snapshot

```
usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: misc:
```

```
--lca-ranks STR    Add column with specified ranks (',' separated) []
```

```
--blacklist STR    Comma separated list of ignored taxa in LCA computation
```

```
[12908:unclassified sequences,28384:other sequences]
```

```
--tax-lineage INT  0: don't show, 1: add all lineage names, 2: add all lineage taxids [0]
```

```
common:
```

```
--compressed INT  Write compressed output [0]
```

```
--threads INT      Number of CPU-cores used (all by default) [10]
```

```
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
```

```
+info [3]
```

```
references:
```



- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics*, btab184 (2021)

### 6.2.63. **linclust**

Fast, less sensitive clustering.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 8 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

#### Usage

usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]

#### Key Options

| Option                 | Purpose                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                                         |
| --alph-size            | Alphabet size (range 2-21)                                                                                   |
| --spaced-kmer-mode     | 0: use consecutive positions in k-mers; 1: use spaced k-mers                                                 |
| --spaced-kmer-pattern  | User-specified spaced k-mer pattern                                                                          |
| --mask                 | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob            | Mask sequences is probability is above threshold                                                             |

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --mask-lower-case    | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region    |
| --mask-n-repeat      | Repeat letters that occure > threshold in a rwo                                              |
| -k                   | k-mer length (0: automatically set to optimum)                                               |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |

### Full CLI Help Snapshot

```
usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --spaced-kmer-mode INT       0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
  --spaced-kmer-pattern STR     User-specified spaced k-mer pattern []
  --mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT            Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a rwo
[0]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [2]
  --alignment-output-mode INT   How to compute the alignment:
```

|                                                                                |                                                 |
|--------------------------------------------------------------------------------|-------------------------------------------------|
|                                                                                | 0: automatic                                    |
|                                                                                | 1: only score and end_pos                       |
|                                                                                | 2: also start_pos and cov                       |
|                                                                                | 3: also seq.id                                  |
|                                                                                | 4: only ungapped alignment                      |
|                                                                                | 5: score only (output) cluster format [0]       |
| --wrapped-scoring BOOL                                                         | Double the (nucleotide) query sequence during   |
| the scoring process to allow wrapped diagonal scoring around end and start [0] |                                                 |
| -e DOUBLE                                                                      | List matches below this E-value (range 0.0-inf) |
| [1.000E-03]                                                                    |                                                 |
| --min-seq-id FLOAT                                                             | List matches above this sequence identity (for  |
| clustering) (range 0.0-1.0) [0.900]                                            |                                                 |
| --min-aln-len INT                                                              | Minimum alignment length (range 0-INT_MAX) [0]  |
| --seq-id-mode INT                                                              | 0: alignment length 1: shorter, 2: longer       |
| sequence [0]                                                                   |                                                 |
| --alt-ali INT                                                                  | Show up to this many alternative alignments [0] |
| -c FLOAT                                                                       | List matches above this fraction of aligned     |
| (covered) residues (see --cov-mode) [0.800]                                    |                                                 |
| --cov-mode INT                                                                 | 0: coverage of query and target                 |
|                                                                                | 1: coverage of target                           |
|                                                                                | 2: coverage of query                            |
|                                                                                | 3: target seq. length has to be at least x% of  |
| query length                                                                   |                                                 |
|                                                                                | 4: query seq. length has to be at least x% of   |
| target length                                                                  |                                                 |
|                                                                                | 5: short seq. needs to be at least x% of the    |
| other seq. length [0]                                                          |                                                 |
| --max-rejected INT                                                             | Maximum rejected alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --max-accept INT                                                               | Maximum accepted alignments before alignment    |
| calculation for a query is stopped [2147483647]                                |                                                 |
| --score-bias FLOAT                                                             | Score bias when computing SW alignment (in      |
| bits) [0.000]                                                                  |                                                 |
| --realign BOOL                                                                 | Compute more conservative, shorter alignments   |
| (scores and E-values not changed) [0]                                          |                                                 |
| --realign-score-bias FLOAT                                                     | Additional bias when computing realignment      |
| [-0.200]                                                                       |                                                 |
| --realign-max-seqs INT                                                         | Maximum number of results to return in          |
| realignment [2147483647]                                                       |                                                 |
| --corr-score-weight FLOAT                                                      | Weight of backtrace correlation score that is   |
| added to the alignment score [0.000]                                           |                                                 |
| --gap-open TWIN                                                                | Gap open cost [aa:11,nucl:5]                    |
| --gap-extend TWIN                                                              | Gap extension cost [aa:1,nucl:2]                |
| --zdrop INT                                                                    | Maximal allowed difference between score values |
| before alignment is truncated (nucleotide alignment only) [40]                 |                                                 |
| clust:                                                                         |                                                 |
| --cluster-mode INT                                                             | 0: Set-Cover (greedy)                           |
|                                                                                | 1: Connected component (BLASTclust)             |
|                                                                                | 2,3: Greedy clustering by sequence length       |

```

(CDHIT) [0]
--max-iterations INT          Maximum depth of breadth first search in
connected component clustering [1000]
--similarity-type INT          Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
kmermatcher:
--weights STR                  Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
--kmer-per-seq INT             k-mers per sequence [21]
--kmer-per-seq-scale TWIN       Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL         Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT               Shift k-mer hash initialization [67]
--include-only-extendable BOOL Include only extendable [0]
--ignore-multi-kmer BOOL       Skip k-mers occurring multiple times (>=2) [0]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
misc:
--rescore-mode INT             Rescore diagonals with:
                                0: Hamming distance
                                1: local alignment (score only)
                                2: local alignment
                                3: global alignment
                                4: longest alignment fulfilling window quality

criterion [0]
common:
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT               Write compressed output [0]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--sub-mat TWIN                  Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT               Maximum sequence length [65535]
--db-load-mode INT              Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--remove-tmp-files BOOL         Delete temporary files [0]
--force-reuse BOOL              Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--mpi-runner STR                Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL              Filter hits by seq.id. and coverage [0]
--sort-results INT              Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]

```

examples:

```
# Linear-time clustering of FASTA file
mmseqs linclust sequenceDB clusterDB tmp

--cov-mode
# Sequence      0    1    2
# Q: MAVGTACRPA 60%  IGN 60%
# T: -AVGTAC-- 60% 100% IGN
# Cutoff -c 0.7  -    +   -
#          -c 0.6  +    +   +

# Cluster nucleotide sequences
mmseqs easy-linclust nucl.fasta result tmp --kmer-per-seq-scale 0.3
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

#### 6.2.64. linsearch

Fast, less sensitive homology search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 2 upstream caller(s) and 9 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: search\_workflows.

#### Usage

usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

#### Key Options

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1) |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

| Option                  | Purpose                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| --add-self-matches      | Artificially add entries of queries with themselves (for clustering)                                         |
| --seed-sub-mat          | Substitution matrix file for k-mer generation                                                                |
| --mask                  | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob             | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case       | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat         | Repeat letters that occur > threshold in a row                                                               |
| --split-memory-limit    | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory                 |
| -a                      | Add backtrace string (convert to alignments with mmseqs convertalis module)                                  |
| --alignment-mode        | How to compute the alignment:                                                                                |
| --alignment-output-mode | How to compute the alignment:                                                                                |

### Full CLI Help Snapshot

```
usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:blosum62.out,nucl:nucleotide.out]
  --mask INT                   Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT           Mask sequences is probability is above threshold
[0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT         Repeat letters that occur > threshold in a row
[0]
  --split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M, 1G.
```

Default (0) to all available system memory [0]

align:

- a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]
- alignment-mode INT How to compute the alignment:
  - 0: automatic
  - 1: only score and end\_pos
  - 2: also start\_pos and cov
  - 3: also seq.id
  - 4: only ungapped alignment [2]
- alignment-output-mode INT How to compute the alignment:
  - 0: automatic
  - 1: only score and end\_pos
  - 2: also start\_pos and cov
  - 3: also seq.id
  - 4: only ungapped alignment
  - 5: score only (output) cluster format [0]
- wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]
- e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]
- min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]
- min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]
- seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]
- alt-ali INT Show up to this many alternative alignments [0]
- c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]
- cov-mode INT
  - 0: coverage of query and target
  - 1: coverage of target
  - 2: coverage of query
  - 3: target seq. length has to be at least x% of query length
  - 4: query seq. length has to be at least x% of target length
  - 5: short seq. needs to be at least x% of the other seq. length [0]
- max-rejected INT Maximum rejected alignments before alignment calculation for a query is stopped [2147483647]
- max-accept INT Maximum accepted alignments before alignment calculation for a query is stopped [2147483647]
- score-bias FLOAT Score bias when computing SW alignment (in bits) [0.000]
- realign BOOL Compute more conservative, shorter alignments (scores and E-values not changed) [0]
- realign-score-bias FLOAT Additional bias when computing realignment [-0.200]
- realign-max-seqs INT Maximum number of results to return in realignment

```

[2147483647]
--corr-score-weight FLOAT      Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN                Gap open cost [aa:11,nucl:5]
--gap-extend TWIN              Gap extension cost [aa:1,nucl:2]
--zdrop INT                    Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
kmermatcher:
--kmer-per-seq INT             k-mers per sequence [21]
--kmer-per-seq-scale TWIN      Scale k-mer per sequence based on sequence length
as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--pick-n-sim-kmer INT          Add N similar k-mers to search [1]
--result-direction INT         result is 0: query, 1: target centric [1]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
--min-length INT               Minimum codon number in open reading frames [30]
--max-length INT               Maximum codon number in open reading frames
[32734]
--max-gaps INT                 Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT        Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR           Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR           Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT        1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this value

```



```

[0]
--search-type INT          Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nuc1:nucleotide.out]
--max-seq-len INT          Maximum sequence length [65535]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR           Use MPI on compute cluster with this MPI command
(e.g. "mpirun -np 42") []
--force-reuse BOOL         Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--create-lookup INT        Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT          Combine ORFs/split sequences to a single entry [1]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.65. lndb

Symlink a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]

#### Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.66. majoritylca

Compute the lowest common ancestor using majority voting.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### Usage

usage: mmseqs majoritylca <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs majoritylca locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.67. map

Map nearly identical sequences.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 0 upstream caller(s) and 1 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: search\_workflows.

#### Usage

usage: mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

#### Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                                |
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                             |
| -k                   | k-mer length (0: automatically set to optimum)                                               |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                      |
| --k-score            | k-mer threshold for generating similar k-mer lists                                           |
| --alph-size          | Alphabet size (range 2-21)                                                                   |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)       |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically           |
| --split-mode         | 0: split target db; 1: split query db; 2: auto, depending on main memory                     |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --comp-bias-corr     | Correct for locally biased amino acid composition (range 0-1)                                |

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

### Full CLI Help Snapshot

```
usage: mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                    Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
[2.000]
  -k INT                      k-mer length (0: automatically set to optimum) [0]
  --target-search-mode INT    target search mode (0: regular k-mer, 1: similar k-
mer) [0]
  --k-score TWIN              k-mer threshold for generating similar k-mer lists
[seq:2147483647,prof:2147483647]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT              Maximum results per query sequence allowed to pass
the prefilter (affects sensitivity) [300]
  --split INT                 Split input into N equally distributed chunks. 0:
set the best split automatically [0]
  --split-mode INT            0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
  --comp-bias-corr INT        Correct for locally biased amino acid composition
(range 0-1) [0]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --diag-score BOOL           Use ungapped diagonal scoring during prefilter [1]
  --exact-kmer-matching INT   Extract only exact k-mers for matching (range 0-1)
[0]
  --mask INT                  Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT         Repeat letters that occure > threshold in a rwo [0]
  --min-ungapped-score INT    Accept only matches with ungapped alignment score
above threshold [15]
  --add-self-matches BOOL     Artificially add entries of queries with themselves
(for clustering) [0]
  --spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
```

```

--spaced-kmer-pattern STR    User-specified spaced k-mer pattern []
--local-tmp STR              Path where some of the temporary files will be
created []
align:
  -c FLOAT                    List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.950]
  --cov-mode INT              0: coverage of query and target
                             1: coverage of target
                             2: coverage of query
                             3: target seq. length has to be at least x% of
query length
                             4: query seq. length has to be at least x% of
target length
                             5: short seq. needs to be at least x% of the other
seq. length [2]
  --wrapped-scoring BOOL      Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                   List matches below this E-value (range 0.0-inf)
[1.000E-03]
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --min-seq-id FLOAT           List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.900]
  --min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT            0: alignment length 1: shorter, 2: longer sequence
[0]
profile:
  --pca                        Pseudo count admixture strength []
  --pcb                        Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
  --taxon-list STR             Taxonomy ID, possibly multiple values separated by
',' []
  --rescore-mode INT           Rescore diagonals with:
                             0: Hamming distance
                             1: local alignment (score only)
                             2: local alignment
                             3: global alignment
                             4: longest alignment fulfilling window quality

criterion [2]
  --min-length INT             Minimum codon number in open reading frames [10]
  --max-length INT             Maximum codon number in open reading frames [32734]
  --max-gaps INT               Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
  --contig-start-mode INT      Contig start can be 0: incomplete, 1: complete, 2:
both [2]
  --contig-end-mode INT        Contig end can be 0: incomplete, 1: complete, 2:
both [2]
  --orf-start-mode INT         Orf fragment can be 0: from start to stop, 1: from

```

any to stop, 2: from last encountered start to stop (no start in the middle) [1]

--forward-frames STR Comma-separated list of frames on the forward strand to be extracted [1,2,3]

--reverse-frames STR Comma-separated list of frames on the reverse strand to be extracted [1,2,3]

--translation-table INT 1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3) YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE 9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11) PROKARYOTE, 12) ALT\_YEAST, 13) ASCIDIAN\_MITOCHONDRIAL, 14) ALT\_FLATWORM\_MITOCHONDRIAL 15) BLEPHARISMA, 16) CHLOROPHYCEAN\_MITOCHONDRIAL, 21) TREMATODE\_MITOCHONDRIAL, 22) SCENEDESMUS\_MITOCHONDRIAL 23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24) PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27) KARYORELICT, 28) CONDYLOSTOMA 29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA

[1]

--translate INT Translate ORF to amino acid [0]

--use-all-table-starts BOOL Use all alternatives for a start codon in the genetic table, if false - only ATG (AUG) [0]

--id-offset INT Numeric ids in index file are offset by this value [0]

--start-sens FLOAT Start sensitivity [4.000]

--sens-steps INT Number of search steps performed from --start-sens to -s [1]

common:

--sub-mat TWIN Substitution matrix file [aa:blosum62.out,nucl:nucleotide.out]

--max-seq-len INT Maximum sequence length [65535]

--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

--mpi-runner STR Use MPI on compute cluster with this MPI command (e.g. "mpirun -np 42") []

--force-reuse BOOL Reuse tmp filse in tmp/latest folder ignoring parameters and version changes [0]

--remove-tmp-files BOOL Delete temporary files [0]

expert:

--filter-hits BOOL Filter hits by seq.id. and coverage [0]

--sort-results INT Sort results: 0: no sorting, 1: sort by E-value (Alignment) or seq.id. (Hamming) [1]

--create-lookup INT Create database lookup file (can be very large) [0]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028

(2017)

- Steinegger M, Mirdita M, Soding J: Protein-level assembly increases protein sequence recovery from metagenomic samples manifold. *Nature Methods*, 16(7), 603-606 (2019)

#### **6.2.68. maskbygff**

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

#### **Usage**

```
usage: mmseqs maskbygff <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs maskbygff` locally for exact positional arguments in your build.

#### **Key Options**

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### **Full CLI Help Snapshot**

##### **Note**

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### **6.2.69. masksequence**

Soft mask sequence DB using tantan.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

## Usage

usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]

## Key Options

| Option            | Purpose                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| --mask            | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking |
| --mask-prob       | Mask sequences is probability is above threshold                                                             |
| --mask-lower-case | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region                    |
| --mask-n-repeat   | Repeat letters that occur > threshold in a row                                                               |
| --threads         | Number of CPU-cores used (all by default)                                                                    |
| --compressed      | Write compressed output                                                                                      |
| -v                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                |

## Full CLI Help Snapshot

usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--mask INT Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking [1]  
--mask-prob FLOAT Mask sequences is probability is above threshold [0.900]  
--mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]  
--mask-n-repeat INT Repeat letters that occur > threshold in a row [0]

common:

--threads INT Number of CPU-cores used (all by default) [10]  
--compressed INT Write compressed output [0]  
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.70. mergeclusters

Merge multiple cascaded clustering steps.



Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

### Usage

```
usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ...
<i:clusterDBn> [options]
```

### Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ...
<i:clusterDBn> [options]
  By Maria Hauser & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
```

#### 6.2.71. mergedbs

Merge entries from multiple DBs.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options]

## Key Options

| Option             | Purpose                                                       |
|--------------------|---------------------------------------------------------------|
| --compressed       | Write compressed output                                       |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --prefixes         | Comma separated list of prefixes for each entry               |
| --merge-stop-empty | Don't continue merging entries after an empty entry           |

## Full CLI Help Snapshot

usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

--compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

expert:

--prefixes STR Comma separated list of prefixes for each entry []  
 --merge-stop-empty BOOL Don't continue merging entries after an empty entry [0]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.72. mergeresultsbyset

Merge results from multiple ORFs back to their respective contig.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi\_hit.

### Usage

```
usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]
```

### Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --threads      | Number of CPU-cores used (all by default)                       |
| --compressed   | Write compressed output                                         |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

### Full CLI Help Snapshot

```
usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]
  By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>
options: common:
  --db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT         Number of CPU-cores used (all by default) [10]
  --compressed INT      Write compressed output [0]
  -v INT                Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

#### 6.2.73. msa2profile

Convert a MSA DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

## Usage

usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options]

## Key Options

| Option                 | Purpose                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --gap-open             | Gap open cost                                                                                                                            |
| --gap-extend           | Gap extension cost                                                                                                                       |
| --match-mode           | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept |
| --match-ratio          | Columns that have a residue in this ratio of all sequences are kept                                                                      |
| --pseudo-cnt-mode      | use 0: substitution-matrix or 1: context-specific pseudocounts                                                                           |
| --pca                  | Pseudo count admixture strength                                                                                                          |
| --pcb                  | Pseudo counts: Neff at half of maximum admixture (range 0.0-inf)                                                                         |
| --wg                   | Use global sequence weighting for profile calculation                                                                                    |
| --filter-msa           | Filter msa: 0: do not filter, 1: filter                                                                                                  |
| --filter-min-enable    | Only filter MSAs with more than N sequences, 0 always filters                                                                            |

## Full CLI Help Snapshot

usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT           Correct for locally biased amino acid composition (range 0-1) [1]

--comp-bias-corr-scale FLOAT   Correct for locally biased amino acid composition (range 0-1) [1.000]

align:

```

--gap-open TWIN          Gap open cost [aa:11,nucl:5]
--gap-extend TWIN        Gap extension cost [aa:1,nucl:2]
profile:
--match-mode INT          0: Columns that have a residue in the first
sequence are kept, 1: columns that have a residue in --match-ratio of all
sequences are kept [0]
--match-ratio FLOAT        Columns that have a residue in this ratio of all
sequences are kept [0.500]
--pseudo-cnt-mode INT      use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
--pca                      Pseudo count admixture strength []
--pcb                      Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
--wg B00L                 Use global sequence weighting for profile
calculation [0]
--filter-msa INT           Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT    Only filter MSAs with more than N sequences, 0
always filters [0]
--cov FLOAT               Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--qid STR                 Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple
thresholds:
E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT               Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--max-seq-id FLOAT         Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
--diff INT                 Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--msa-type INT             MSA Type: 0: cA3M, 1: A3M, 2: FASTA [2]
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--skip-query B00L          Skip the query sequence [0]

examples:
# Convert globally aligned MSAs to profiles
# Defines columns as match columns if more than 50% of residues are not gaps
# Non-match columns are discarded
mmseqs msa2profile msaDB profileDB --match-mode 1 --match-ratio 0.5

```

```
# Assign match-columns through the first sequence
# Gaps in query sequence define non-match columns and are discarded
mmseqs msa2profile msaDB profileDB --match-mode 0
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

#### 6.2.74. msa2result

Convert a MSA DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]

#### Key Options

| Option                 | Purpose                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                                            |
| --gap-open             | Gap open cost                                                                                                                            |
| --gap-extend           | Gap extension cost                                                                                                                       |
| --match-mode           | 0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept |
| --match-ratio          | Columns that have a residue in this ratio of all sequences are kept                                                                      |

| Option              | Purpose                                                          |
|---------------------|------------------------------------------------------------------|
| --pseudo-cnt-mode   | use 0: substitution-matrix or 1: context-specific pseudocounts   |
| --pca               | Pseudo count admixture strength                                  |
| --pcb               | Pseudo counts: Neff at half of maximum admixture (range 0.0-inf) |
| --wg                | Use global sequence weighting for profile calculation            |
| --filter-msa        | Filter msa: 0: do not filter, 1: filter                          |
| --filter-min-enable | Only filter MSAs with more than N sequences, 0 always filters    |

### Full CLI Help Snapshot

```
usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
align:
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --match-mode INT             0: Columns that have a residue in the first
sequence are kept, 1: columns that have a residue in --match-ratio of all
sequences are kept [0]
  --match-ratio FLOAT          Columns that have a residue in this ratio of all
sequences are kept [0.500]
  --pseudo-cnt-mode INT        use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
  --pca                        Pseudo count admixture strength []
  --pcb                        Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
  --wg BOOL                    Use global sequence weighting for profile
calculation [0]
  --filter-msa INT             Filter msa: 0: do not filter, 1: filter [1]
  --filter-min-enable INT      Only filter MSAs with more than N sequences, 0
always filters [0]
  --cov FLOAT                  Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
  --qid STR                    Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                              Alternatively, can be a list of multiple
```





| Option              | Purpose                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| --dbtype            | Database type 0: auto, 1: amino acid 2: nucleotides                                                                             |
| --shuffle           | Shuffle input database                                                                                                          |
| --createdb-mode     | Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q)                         |
| --id-offset         | Numeric ids in index file are offset by this value                                                                              |
| --min-length        | Minimum codon number in open reading frames                                                                                     |
| --max-length        | Maximum codon number in open reading frames                                                                                     |
| --max-gaps          | Maximum number of codons with gaps or unknown residues before an open reading frame is rejected                                 |
| --contig-start-mode | Contig start can be 0: incomplete, 1: complete, 2: both                                                                         |
| --contig-end-mode   | Contig end can be 0: incomplete, 1: complete, 2: both                                                                           |
| --orf-start-mode    | Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle) |
| --forward-frames    | Comma-separated list of frames on the forward strand to be extracted                                                            |
| --reverse-frames    | Comma-separated list of frames on the reverse strand to be extracted                                                            |

### Full CLI Help Snapshot

```
usage: mmseqs multihitdb <i:fastaFile1[.gz|bz2]> ... <i:fastaFileN[.gz|bz2]>
<o:setDB> <tmpDir> [options]
  By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>
options: misc:
  --dbtype INT          Database type 0: auto, 1: amino acid 2: nucleotides
  [0]
  --shuffle BOOL        Shuffle input database [1]
  --createdb-mode INT   Createdb mode 0: copy data, 1: soft link data and
  write new index (works only with single line fasta/q) [0]
  --id-offset INT       Numeric ids in index file are offset by this value
```

```

[0]
--min-length INT           Minimum codon number in open reading frames [30]
--max-length INT           Maximum codon number in open reading frames [32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT    Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT      Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT       Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR       Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR       Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA

[1]
--translate INT            Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--add-orf-stop BOOL        Add stop codon '*' at complete start and end [0]
--stat STR                 One of: linecount, mean, min, max, doolittle,
charges, seqlen, firstline []
--tsv BOOL                 Return output in TSV format [0]
common:
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--threads INT              Number of CPU-cores used (all by default) [10]
expert:
--write-lookup INT         write .lookup file containing mapping from internal
id, fasta id and file number [1]
--create-lookup INT        Create database lookup file (can be very large) [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

### 6.2.76. multihitsearch

Search with a grouped set of sequences against another grouped set.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi\_hit.

#### Usage

```
usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir>
[options]
```

#### Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                   |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                          |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                     | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score              | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size            | Alphabet size (range 2-21)                                                             |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split                | Split input into N equally distributed chunks.<br>0: set the best split automatically  |

| Option       | Purpose                                                                  |
|--------------|--------------------------------------------------------------------------|
| --split-mode | 0: split target db; 1: split query db; 2: auto, depending on main memory |

### Full CLI Help Snapshot

usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir> [options]

By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>

options: prefilter:

|                              |                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr INT         | Correct for locally biased amino acid composition (range 0-1) [1]                                                |
| --comp-bias-corr-scale FLOAT | Correct for locally biased amino acid composition (range 0-1) [1.000]                                            |
| --add-self-matches BOOL      | Artificially add entries of queries with themselves (for clustering) [0]                                         |
| --seed-sub-mat TWIN          | Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]                                |
| -s FLOAT                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [5.700]                                                         |
| -k INT                       | k-mer length (0: automatically set to optimum) [0]                                                               |
| --target-search-mode INT     | target search mode (0: regular k-mer, 1: similar k-mer) [0]                                                      |
| --k-score TWIN               | k-mer threshold for generating similar k-mer lists [seq:2147483647,prof:2147483647]                              |
| --alph-size TWIN             | Alphabet size (range 2-21) [aa:21,nucl:5]                                                                        |
| --max-seqs INT               | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]                     |
| --split INT                  | Split input into N equally distributed chunks. 0: set the best split automatically [0]                           |
| --split-mode INT             | 0: split target db; 1: split query db; 2: auto, depending on main memory [2]                                     |
| --split-memory-limit BYTE    | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory [0]                 |
| --diag-score BOOL            | Use ungapped diagonal scoring during prefilter [1]                                                               |
| --exact-kmer-matching INT    | Extract only exact k-mers for matching (range 0-1) [0]                                                           |
| --mask INT                   | Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking [1] |
| --mask-prob FLOAT            | Mask sequences is probablity is above threshold [0.900]                                                          |
| --mask-lower-case INT        | Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]                    |
| --mask-n-repeat INT          | Repeat letters that occure > threshold in a rwo [0]                                                              |
| --min-ungapped-score INT     | Accept only matches with ungapped alignment                                                                      |

score above threshold [15]

--spaced-kmer-mode INT                   0: use consecutive positions in k-mers; 1: use spaced k-mers [1]

--spaced-kmer-pattern STR                User-specified spaced k-mer pattern []

--local-tmp STR                          Path where some of the temporary files will be created []

--disk-space-limit BYTE                 Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]

align:

-a BOOL                                   Add backtrace string (convert to alignments with mmseqs convertalis module) [0]

--alignment-mode INT                    How to compute the alignment:

                                          0: automatic

                                          1: only score and end\_pos

                                          2: also start\_pos and cov

                                          3: also seq.id

                                          4: only ungapped alignment [2]

--alignment-output-mode INT             How to compute the alignment:

                                          0: automatic

                                          1: only score and end\_pos

                                          2: also start\_pos and cov

                                          3: also seq.id

                                          4: only ungapped alignment

                                          5: score only (output) cluster format [0]

--wrapped-scoring BOOL                 Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]

-e DOUBLE                                List matches below this E-value (range 0.0-inf) [1.000E+02]

--min-seq-id FLOAT                      List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]

--min-aln-len INT                       Minimum alignment length (range 0-INT\_MAX) [0]

--seq-id-mode INT                        0: alignment length 1: shorter, 2: longer sequence [0]

--alt-ali INT                            Show up to this many alternative alignments [0]

-c FLOAT                                 List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]

--cov-mode INT                          0: coverage of query and target

                                          1: coverage of target

                                          2: coverage of query

                                          3: target seq. length has to be at least x% of query length

                                          4: query seq. length has to be at least x% of target length

                                          5: short seq. needs to be at least x% of the other seq. length [0]

--max-rejected INT                      Maximum rejected alignments before alignment calculation for a query is stopped [2147483647]

--max-accept INT                        Maximum accepted alignments before alignment

calculation for a query is stopped [2147483647]

--score-bias FLOAT                      Score bias when computing SW alignment (in bits) [0.000]

--realign BOOL                          Compute more conservative, shorter alignments (scores and E-values not changed) [0]

--realign-score-bias FLOAT              Additional bias when computing realignment [-0.200]

--realign-max-seqs INT                  Maximum number of results to return in realignment [2147483647]

--corr-score-weight FLOAT              Weight of backtrace correlation score that is added to the alignment score [0.000]

--gap-open TWIN                        Gap open cost [aa:11,nucl:5]

--gap-extend TWIN                      Gap extension cost [aa:1,nucl:2]

--zdrop INT                            Maximal allowed difference between score values before alignment is truncated (nucleotide alignment only) [40]

--exhaustive-search-filter INT        Filter result during search: 0: do not filter, 1: filter [0]

profile:

--pca                                  Pseudo count admixture strength []

--pcb                                  Pseudo counts: Neff at half of maximum

admixture (range 0.0-inf) []

--mask-profile INT                      Mask query sequence of profile using tantan [0,1] [1]

--e-profile DOUBLE                      Include sequences matches with < E-value thr. into the profile ( $\geq 0.0$ ) [1.000E-03]

--wg BOOL                              Use global sequence weighting for profile calculation [0]

--filter-msa INT                        Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT                Only filter MSAs with more than N sequences, 0 always filters [0]

--max-seq-id FLOAT                      Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] [0.900]

--qid STR                              Reduce diversity of output MSAs using min.seq. identity with query sequences [0.0,1.0]

                                        Alternatively, can be a list of multiple thresholds:

                                        E.g.: 0.15,0.30,0.50 to defines filter buckets of ]0.15-0.30] and ]0.30-0.50] [0.0]

--qsc FLOAT                            Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT                            Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT                              Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT                  use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT              Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT                  Number of iterative profile search iterations

```

[1]
--exhaustive-search BOOL          For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL                Efficient search for LCA candidates [0]
misc:
--taxon-list STR                 Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT             prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT              Rescore diagonals with:
                                0: Hamming distance
                                1: local alignment (score only)
                                2: local alignment
                                3: global alignment
                                4: longest alignment fulfilling window quality

criterion [0]
--allow-deletion BOOL            Allow deletions in a MSA [0]
--min-length INT                 Minimum codon number in open reading frames
[30]
--max-length INT                 Maximum codon number in open reading frames
[32734]
--max-gaps INT                   Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT          Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT            Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT             Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR             Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR            Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT          1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
                                15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
                                23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
                                29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]
--translate INT                  Translate ORF to amino acid [0]
--use-all-table-starts BOOL     Use all alternatives for a start codon in the

```

genetic table, if false - only ATG (AUG) [0]

--id-offset INT                      Numeric ids in index file are offset by this value [0]

--sequence-overlap INT                Overlap between sequences [0]

--sequence-split-mode INT             Sequence split mode 0: copy data, 1: soft link data and write new index, [1]

--headers-split-mode INT              Header split mode: 0: split position, 1: original header [0]

--search-type INT                      Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment [0]

--start-sens FLOAT                     Start sensitivity [4.000]

--sens-steps INT                       Number of search steps performed from --start-sens to -s [1]

--translation-mode INT                Translation AA seq from nucleotide by 0: ORFs, 1: full reading frames [0]

--simple-best-hit BOOL                 Update the p-value by a single best hit, or by best and second best hits [1]

common:

--sub-mat TWIN                        Substitution matrix file [aa:blosom62.out,nucl:nucleotide.out]

--max-seq-len INT                      Maximum sequence length [65535]

--db-load-mode INT                     Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]

--threads INT                         Number of CPU-cores used (all by default) [10]

--compressed INT                      Write compressed output [0]

-v INT                                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

--gpu INT                              Use GPU (CUDA) if possible [0]

--gpu-server INT                       Use GPU server [0]

--gpu-server-wait-timeout INT         Wait for GPU server for 0: don't wait -1: no wait limit: >0 this many seconds [600]

--mpi-runner STR                       Use MPI on compute cluster with this MPI command (e.g. "mpirun -np 42") []

--force-reuse BOOL                     Reuse tmp filse in tmp/latest folder ignoring parameters and version changes [0]

--remove-tmp-files BOOL                Delete temporary files [0]

expert:

--filter-hits BOOL                     Filter hits by seq.id. and coverage [0]

--sort-results INT                     Sort results: 0: no sorting, 1: sort by E-value (Alignment) or seq.id. (Hamming) [0]

--create-lookup INT                    Create database lookup file (can be very large) [0]

--chain-alignments INT                Chain overlapping alignments [0]

--merge-query INT                      Combine ORFs/split sequences to a single entry [1]

--strand INT                           Strand selection only works for DNA/DNA search 0: reverse, 1: forward, 2: both [1]

references:



- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.77. mvdb

Move a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]

#### Key Options

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### Full CLI Help Snapshot

usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.78. nrtotaxmapping

Create taxonomy mapping for NR database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

### Usage

```
usage: mmseqs nrtotaxmapping <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs nrtotaxmapping` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.79. offsetalignment

Offset alignment by ORF start position.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

### Usage

```
usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB>
<i:targetOrfDB> <i:alnDB> <o:alnDB> [options]
```

### Key Options

| Option                     | Purpose                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------|
| <code>--search-type</code> | Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment |

| Option             | Purpose                                                         |
|--------------------|-----------------------------------------------------------------|
| --threads          | Number of CPU-cores used (all by default)                       |
| --compressed       | Write compressed output                                         |
| --db-load-mode     | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |
| --chain-alignments | Chain overlapping alignments                                    |
| --merge-query      | Combine ORFs/split sequences to a single entry                  |

### Full CLI Help Snapshot

```
usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB>
<i:targetOrfDB> <i:alnDB> <o:alnDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --search-type INT          Search type 0: auto 1: amino acid, 2: translated, 3:
nucleotide, 4: translated nucleotide alignment [0]
common:
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT           Write compressed output [0]
  --db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  -v INT                      Verbose level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
expert:
  --chain-alignments INT     Chain overlapping alignments [0]
  --merge-query INT          Combine ORFs/split sequences to a single entry [1]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

#### 6.2.80. orftocontig

Write ORF locations in alignment format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `sequence_manipulation`.

### Usage

```
usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB>
<o:orfsAlignedToContigDB> [options]
```

### Key Options

| Option                    | Purpose                                                       |
|---------------------------|---------------------------------------------------------------|
| <code>--threads</code>    | Number of CPU-cores used (all by default)                     |
| <code>--compressed</code> | Write compressed output                                       |
| <code>-v</code>           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB>
<o:orfsAlignedToContigDB> [options]
  By Eli Levy Karin <eli.levy.karin@gmail.com>
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
```

#### 6.2.81. `pairaln`

Pair sequences to match best protein A and B from a species.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `profiles`.

### Usage

usage: mmseqs pairaln <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs pairaln locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.82. pickconsensusrep

Select new representatives for each cluster based on consensus.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

### Usage

usage: mmseqs pickconsensusrep <inputDB(s)> <outputDB> [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs pickconsensusrep locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.83. prefilter

Double consecutive diagonal k-mer search.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

#### Usage

```
usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
```

#### Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --seed-sub-mat       | Substitution matrix file for k-mer generation                                                |
| -s                   | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                             |
| -k                   | k-mer length (0: automatically set to optimum)                                               |
| --target-search-mode | target search mode (0: regular k-mer, 1: similar k-mer)                                      |
| --k-score            | k-mer threshold for generating similar k-mer lists                                           |
| --alph-size          | Alphabet size (range 2-21)                                                                   |
| --max-seqs           | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)       |
| --split              | Split input into N equally distributed chunks. 0: set the best split automatically           |
| --split-mode         | 0: split target db; 1: split query db; 2: auto, depending on main memory                     |
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --comp-bias-corr     | Correct for locally biased amino acid composition (range 0-1)                                |

| Option                 | Purpose                                                       |
|------------------------|---------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1) |

### Full CLI Help Snapshot

```
usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Maria Hauser
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
[4.000]
  -k INT                       k-mer length (0: automatically set to optimum) [0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1: similar k-
mer) [0]
  --k-score TWIN              k-mer threshold for generating similar k-mer lists
[seq:2147483647,prof:2147483647]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT              Maximum results per query sequence allowed to pass
the prefilter (affects sensitivity) [300]
  --split INT                  Split input into N equally distributed chunks. 0:
set the best split automatically [0]
  --split-mode INT            0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
  --comp-bias-corr INT        Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --diag-score BOOL           Use ungapped diagonal scoring during prefilter [1]
  --exact-kmer-matching INT   Extract only exact k-mers for matching (range 0-1)
[0]
  --mask INT                  Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT         Repeat letters that occure > threshold in a rwo [0]
  --min-ungapped-score INT    Accept only matches with ungapped alignment score
above threshold [15]
  --add-self-matches BOOL     Artificially add entries of queries with themselves
(for clustering) [0]
  --spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
  --spaced-kmer-pattern STR   User-specified spaced k-mer pattern []
```

```

--local-tmp STR          Path where some of the temporary files will be
created []
align:
  -c FLOAT              List matches above this fraction of aligned
                        (covered) residues (see --cov-mode) [0.000]
  --cov-mode INT         0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the other
seq. length [0]
profile:
  --pca                 Pseudo count admixture strength []
  --pcb                 Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
  --taxon-list STR      Taxonomy ID, possibly multiple values separated by
', ' []
common:
  --sub-mat TWIN        Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT     Maximum sequence length [65535]
  --db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
  --threads INT         Number of CPU-cores used (all by default) [10]
  --compressed INT      Write compressed output [0]
  -v INT               Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

#### 6.2.84. prefixid

For each entry in a DB prepend the entry key to the entry itself.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.



Dependency entry: Open in map; functional module: utilities.

## Usage

usage: mmseqs prefixid <i:DB> <o:DB> [options]

## Key Options

| Option         | Purpose                                                                 |
|----------------|-------------------------------------------------------------------------|
| --prefix       | Use this prefix for all entries                                         |
| --mapping-file | Specify a file that translates the keys of a DB to new keys, TSV format |
| --tsv          | Return output in TSV format                                             |
| --threads      | Number of CPU-cores used (all by default)                               |
| --compressed   | Write compressed output                                                 |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

## Full CLI Help Snapshot

usage: mmseqs prefixid <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: misc:

--prefix STR Use this prefix for all entries []  
--mapping-file STR Specify a file that translates the keys of a DB to new  
keys, TSV format []  
--tsv BOOL Return output in TSV format [0]

common:

--threads INT Number of CPU-cores used (all by default) [10]  
--compressed INT Write compressed output [0]  
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:  
+info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.85. profile2consensus

Extract consensus sequence DB from a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

### Usage

```
usage: mmseqs profile2consensus <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2consensus` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.86. `profile2neff`

Convert a profile DB to a tab-separated list of Neff scores.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

### Usage

```
usage: mmseqs profile2neff <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2neff` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.87. **profile2pssm**

Convert a profile DB to a tab-separated PSSM file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

```
usage: mmseqs profile2pssm <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2pssm` locally for exact positional arguments in your build.

#### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

#### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.88. **profile2repseq**

Extract representative sequence DB from a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

### Usage

```
usage: mmseqs profile2repseq <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2repseq` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

## 6.2.89. `proteinaln2nucl`

Transform protein alignments to nucleotide alignments.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

### Usage

```
usage: mmseqs proteinaln2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB>
<i:aaTargetDB> <i:alnDB> <o:alnDB> [options]
```

### Key Options

| Option                    | Purpose                  |
|---------------------------|--------------------------|
| <code>--gap-open</code>   | Gap open cost            |
| <code>--gap-extend</code> | Gap extension cost       |
| <code>--sub-mat</code>    | Substitution matrix file |

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs protealn2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB>
<i:aaTargetDB> <i:alnDB> <o:alnDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: align:
  --gap-open TWIN      Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN    Gap extension cost [aa:1,nucl:2]
common:
  --sub-mat TWIN       Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --threads INT        Number of CPU-cores used (all by default) [10]
  --compressed INT     Write compressed output [0]
  -v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.90. rbh

Reciprocal best hit search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 1 upstream caller(s) and 6 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: search\_workflows.

### Usage

```
usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
```

### Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                   |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                          |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                     | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score              | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size            | Alphabet size (range 2-21)                                                             |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split                | Split input into N equally distributed chunks. 0: set the best split automatically     |
| --split-mode           | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

```
usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
  By Eli Levy Karin
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [0]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
```

```

similar k-mer) [0]
--k-score TWIN          k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN        Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT          Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT             Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT        0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL       Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT Extract only exact k-mers for matching (range
0-1) [0]
--mask INT              Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
--mask-prob FLOAT       Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT   Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT     Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT   0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR User-specified spaced k-mer pattern []
--local-tmp STR          Path where some of the temporary files will be
created []
--disk-space-limit BYTE  Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT     How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [3]
--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment

```

5: score only (output) cluster format [0]  
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during  
 the scoring process to allow wrapped diagonal scoring around end and start [0]  
 -e DOUBLE List matches below this E-value (range 0.0-inf)  
 [1.000E-03]  
 --min-seq-id FLOAT List matches above this sequence identity (for  
 clustering) (range 0.0-1.0) [0.000]  
 --min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]  
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer  
 sequence [0]  
 --alt-ali INT Show up to this many alternative alignments [0]  
 -c FLOAT List matches above this fraction of aligned  
 (covered) residues (see --cov-mode) [0.000]  
 --cov-mode INT 0: coverage of query and target  
 1: coverage of target  
 2: coverage of query  
 3: target seq. length has to be at least x% of  
 query length  
 4: query seq. length has to be at least x% of  
 target length  
 5: short seq. needs to be at least x% of the  
 other seq. length [0]  
 --max-rejected INT Maximum rejected alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --max-accept INT Maximum accepted alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --score-bias FLOAT Score bias when computing SW alignment (in  
 bits) [0.000]  
 --realign BOOL Compute more conservative, shorter alignments  
 (scores and E-values not changed) [0]  
 --realign-score-bias FLOAT Additional bias when computing realignment  
 [-0.200]  
 --realign-max-seqs INT Maximum number of results to return in  
 realignment [2147483647]  
 --corr-score-weight FLOAT Weight of backtrace correlation score that is  
 added to the alignment score [0.000]  
 --gap-open TWIN Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]  
 --zdrop INT Maximal allowed difference between score values  
 before alignment is truncated (nucleotide alignment only) [40]  
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,  
 1: filter [0]  
 profile:  
 --pca Pseudo count admixture strength []  
 --pcb Pseudo counts: Neff at half of maximum  
 admixture (range 0.0-inf) []  
 --mask-profile INT Mask query sequence of profile using tantan  
 [0,1] [1]  
 --e-profile DOUBLE Include sequences matches with < E-value thr.



into the profile ( $\geq 0.0$ ) [1.000E-03]

--wg BOOL Use global sequence weighting for profile calculation [0]

--filter-msa INT Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT Only filter MSAs with more than N sequences, 0 always filters [0]

--max-seq-id FLOAT Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] [0.900]

--qid STR Reduce diversity of output MSAs using min.seq. identity with query sequences [0.0,1.0]

Alternatively, can be a list of multiple thresholds:

E.g.: 0.15,0.30,0.50 to defines filter buckets of ]0.15-0.30] and ]0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT Number of iterative profile search iterations [1]

--exhaustive-search BOOL For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL Efficient search for LCA candidates [0]

misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT Rescore diagonals with:  
0: Hamming distance  
1: local alignment (score only)  
2: local alignment  
3: global alignment  
4: longest alignment fulfilling window quality criterion [0]

--allow-deletion BOOL Allow deletions in a MSA [0]

--min-length INT Minimum codon number in open reading frames [10]

--max-length INT Maximum codon number in open reading frames [32734]

--max-gaps INT Maximum number of codons with gaps or unknown residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT Contig start can be 0: incomplete, 1: complete,

2: both [2]  
 --contig-end-mode INT Contig end can be 0: incomplete, 1: complete,  
 2: both [2]  
 --orf-start-mode INT Orf fragment can be 0: from start to stop, 1:  
 from any to stop, 2: from last encountered start to stop (no start in the middle)  
 [1]  
 --forward-frames STR Comma-separated list of frames on the forward  
 strand to be extracted [1,2,3]  
 --reverse-frames STR Comma-separated list of frames on the reverse  
 strand to be extracted [1,2,3]  
 --translation-table INT 1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3)  
 YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE  
 9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11)  
 PROKARYOTE, 12) ALT\_YEAST, 13) ASCIDIAN\_MITOCHONDRIAL, 14)  
 ALT\_FLATWORM\_MITOCHONDRIAL  
 15) BLEPHARISMA, 16)  
 CHLOROPHYCEAN\_MITOCHONDRIAL, 21) TREMATODE\_MITOCHONDRIAL, 22)  
 SCENEDESMUS\_MITOCHONDRIAL  
 23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24)  
 PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)  
 KARYORELICT, 28) CONDYLOSTOMA  
 29) MESODINIUM, 30) PERTRICH, 31)  
 BLASTOCRITHIDIA [1]  
 --translate INT Translate ORF to amino acid [0]  
 --use-all-table-starts BOOL Use all alternatives for a start codon in the  
 genetic table, if false - only ATG (AUG) [0]  
 --id-offset INT Numeric ids in index file are offset by this  
 value [0]  
 --sequence-overlap INT Overlap between sequences [0]  
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link  
 data and write new index, [1]  
 --headers-split-mode INT Header split mode: 0: split position, 1:  
 original header [0]  
 --search-type INT Search type 0: auto 1: amino acid, 2:  
 translated, 3: nucleotide, 4: translated nucleotide alignment [0]  
 --start-sens FLOAT Start sensitivity [4.000]  
 --sens-steps INT Number of search steps performed from --start-  
 sens to -s [1]  
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs,  
 1: full reading frames [0]  
 common:  
 --sub-mat TWIN Substitution matrix file  
 [aa:blosom62.out,nucl:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:  
 mmap, 3: mmap+touch [0]  
 --threads INT Number of CPU-cores used (all by default) [10]  
 --compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:

```

+warnings, 3: +info [3]
--gpu INT                Use GPU (CUDA) if possible [0]
--gpu-server INT          Use GPU server [0]
--gpu-server-wait-timeout INT  Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL        Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT       Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT         Combine ORFs/split sequences to a single entry
[1]
--strand INT              Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.91. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

#### Usage

usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]

#### Key Options

| Option    | Purpose                                   |
|-----------|-------------------------------------------|
| --threads | Number of CPU-cores used (all by default) |

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]

By Sung-eun Jang

options: common:

--threads INT Number of CPU-cores used (all by default) [10]  
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info  
[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.92. renamedbkeys

Create a new DB with original keys renamed.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

#### Usage

usage: mmseqs renamedbkeys <i:idMapFile|stdin> <i:DB> <o:DB> [options]

#### Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --subdb-mode | Subdb mode 0: copy data 1: soft link data and write index     |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs renamedbkeys <i:idMapFile|stdin> <i:DB> <o:DB> [options]
By Milot Mirdita <milot@mirdita.de>
options: misc:
--subdb-mode INT    Subdb mode 0: copy data 1: soft link data and write index [0]
common:
--threads INT       Number of CPU-cores used (all by default) [10]
-v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.93. rescorediagonal

Compute sequence identity for diagonal.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

#### Usage

```
usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB> <o:resultDB>
[options]
```

#### Key Options

| Option             | Purpose                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|
| --add-self-matches | Artificially add entries of queries with themselves (for clustering)                                                     |
| --wrapped-scoring  | Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start |
| -e                 | List matches below this E-value (range 0.0-inf)                                                                          |
| -c                 | List matches above this fraction of aligned (covered) residues (see --cov-mode)                                          |
| -a                 | Add backtrace string (convert to alignments with mmseqs convertalis module)                                              |

| Option         | Purpose                                                                    |
|----------------|----------------------------------------------------------------------------|
| --cov-mode     | 0: coverage of query and target                                            |
| --min-seq-id   | List matches above this sequence identity (for clustering) (range 0.0-1.0) |
| --min-aln-len  | Minimum alignment length (range 0-INT_MAX)                                 |
| --seq-id-mode  | 0: alignment length 1: shorter, 2: longer sequence                         |
| --rescore-mode | Rescore diagonals with:                                                    |
| --sub-mat      | Substitution matrix file                                                   |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch            |

### Full CLI Help Snapshot

```
usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB>
<o:resultDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --add-self-matches BOOL  Artificially add entries of queries with themselves
                           (for clustering) [0]
align:
  --wrapped-scoring BOOL   Double the (nucleotide) query sequence during the
                           scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                List matches below this E-value (range 0.0-inf)
                           [1.000E-03]
  -c FLOAT                List matches above this fraction of aligned (covered)
                           residues (see --cov-mode) [0.000]
  -a BOOL                 Add backtrace string (convert to alignments with mmseqs
                           convertalis module) [0]
  --cov-mode INT           0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of query
                           length
                           4: query seq. length has to be at least x% of target
                           length
                           5: short seq. needs to be at least x% of the other seq.
                           length [0]
  --min-seq-id FLOAT       List matches above this sequence identity (for
                           clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT        Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT        0: alignment length 1: shorter, 2: longer sequence [0]
misc:
  --rescore-mode INT       Rescore diagonals with:
```

```

0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality

criterion [0]
common:
  --sub-mat TWIN          Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --db-load-mode INT      Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT           Number of CPU-cores used (all by default) [10]
  --compressed INT        Write compressed output [0]
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
expert:
  --filter-hits BOOL       Filter hits by seq.id. and coverage [0]
  --sort-results INT       Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

#### 6.2.94. result2dnamsa

Compute MSA DB with out insertions in the query for DNA sequences.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

```
usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
```

#### Key Options

| Option    | Purpose                                   |
|-----------|-------------------------------------------|
| --threads | Number of CPU-cores used (all by default) |

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --skip-query | Skip the query sequence                                       |

### Full CLI Help Snapshot

```
usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
  By Martin Steinegger (martin.steinegger@snu.ac.kr)
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
expert:
  --skip-query BOOL Skip the query sequence [0]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

#### 6.2.95. result2flat

Create flat file by adding FASTA headers to DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

```
usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB>
[options]
```

#### Key Options



| Option                          | Purpose                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>--use-fasta-header</code> | Use the id parsed from the fasta header as the index key instead of using incrementing numeric identifiers |
| <code>-v</code>                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                              |

### Full CLI Help Snapshot

```
usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --use-fasta-header BOOL  Use the id parsed from the fasta header as the index
key instead of using incrementing numeric identifiers [0]
common:
  -v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.96. result2msa

Compute MSA DB from a result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

#### Usage

```
usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]
```

#### Key Options

| Option                        | Purpose                                                       |
|-------------------------------|---------------------------------------------------------------|
| <code>--comp-bias-corr</code> | Correct for locally biased amino acid composition (range 0-1) |

| Option                 | Purpose                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                                                          |
| --gap-open             | Gap open cost                                                                                                          |
| --gap-extend           | Gap extension cost                                                                                                     |
| --filter-msa           | Filter msa: 0: do not filter, 1: filter                                                                                |
| --filter-min-enable    | Only filter MSAs with more than N sequences, 0 always filters                                                          |
| --max-seq-id           | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]                                        |
| --qid                  | Reduce diversity of output MSAs using min.seq. identity with query sequences                                           |
| --qsc                  | Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0]                |
| --cov                  | Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0]                        |
| --diff                 | Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 |
| --allow-deletion       | Allow deletions in a MSA                                                                                               |

### Full CLI Help Snapshot

```
usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
  By Martin Steinegger (martin.steinegger@snu.ac.kr) & Milot Mirdita
  <milot@mirdita.de> & Clovis Galiez
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
  (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
  (range 0-1) [1.000]
align:
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --filter-msa INT             Filter msa: 0: do not filter, 1: filter [0]
  --filter-min-enable INT      Only filter MSAs with more than N sequences, 0
  always filters [0]
```

```

--max-seq-id FLOAT          Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
--qid STR                   Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                             Alternatively, can be a list of multiple
thresholds:
                             E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
--qsc FLOAT                Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                 Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--allow-deletion BOOL      Allow deletions in a MSA [0]
--msa-format-mode INT      Format MSA as: 0: binary cA3M DB
                             1: binary ca3m w. consensus DB
                             2: aligned FASTA DB
                             3: aligned FASTA w. header summary
                             4: STOCKHOLM flat file
                             5: A3M format
                             6: A3M format w. alignment info [2]
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--summary-prefix STR       Set the cluster summary prefix [cl]
--skip-query BOOL          Skip the query sequence [0]

```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.97. result2profile

Compute profile DB from a result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

## Usage

```
usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB>
[options]
```

## Key Options

| Option                 | Purpose                                                                         |
|------------------------|---------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                   |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                   |
| -e                     | List matches below this E-value (range 0.0-inf)                                 |
| --gap-open             | Gap open cost                                                                   |
| --gap-extend           | Gap extension cost                                                              |
| --mask-profile         | Mask query sequence of profile using tantan [0,1]                               |
| --e-profile            | Include sequences matches with < E-value thr. into the profile (>=0.0)          |
| --wg                   | Use global sequence weighting for profile calculation                           |
| --filter-msa           | Filter msa: 0: do not filter, 1: filter                                         |
| --filter-min-enable    | Only filter MSAs with more than N sequences, 0 always filters                   |
| --max-seq-id           | Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] |
| --qid                  | Reduce diversity of output MSAs using min.seq. identity with query sequences    |

## Full CLI Help Snapshot

```
usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB>
[options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

```

options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
align:
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --mask-profile INT          Mask query sequence of profile using tantan [0,1]
[1]
  --e-profile DOUBLE          Include sequences matches with < E-value thr. into
the profile (>=0.0) [1.000E-03]
  --wg BOOL                   Use global sequence weighting for profile
calculation [0]
  --filter-msa INT            Filter msa: 0: do not filter, 1: filter [1]
  --filter-min-enable INT     Only filter MSAs with more than N sequences, 0
always filters [0]
  --max-seq-id FLOAT          Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
  --qid STR                   Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple
thresholds:
E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
  --qsc FLOAT                 Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
  --cov FLOAT                 Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
  --diff INT                  Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
  --pseudo-cnt-mode INT       use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
  --pca                       Pseudo count admixture strength []
  --pcb                       Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
  --profile-output-mode INT   Profile output mode: 0: binary log-odds 1: human-
readable frequencies [0]
misc:
  --allow-deletion BOOL       Allow deletions in a MSA [0]
common:
  --sub-mat TWIN              Substitution matrix file
[aa:blosom62.out,nucl:nucleotide.out]
  --db-load-mode INT          Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
  --threads INT               Number of CPU-cores used (all by default) [10]

```

```
--compressed INT      Write compressed output [0]
-v INT                Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.98. result2rbh

Filter a merged result DB to retain only reciprocal best hits.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

```
usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]
```

#### Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### Full CLI Help Snapshot

```
usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]
```

By Eli Levy Karin

options: common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT    Write compressed output [0]
-v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.99. result2repseq

Get representative sequences from result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

```
usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options]
```

#### Key Options

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| --compressed   | Write compressed output                                         |
| --threads      | Number of CPU-cores used (all by default)                       |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

#### Full CLI Help Snapshot

```
usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options]
```

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger

<martin.steinegger@snu.ac.kr>

options: common:

```
--db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
--compressed INT      Write compressed output [0]
--threads INT          Number of CPU-cores used (all by default) [10]
-v INT                Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.100. result2stats

Compute statistics for each entry in a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

```
usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB>
[options]
```

#### Key Options

| Option       | Purpose                                                                  |
|--------------|--------------------------------------------------------------------------|
| --stat       | One of: linecount, mean, min, max, doolittle, charges, seqlen, firstline |
| --tsv        | Return output in TSV format                                              |
| --compressed | Write compressed output                                                  |
| --threads    | Number of CPU-cores used (all by default)                                |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info            |

#### Full CLI Help Snapshot

```
usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB>
[options]
  By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --stat STR          One of: linecount, mean, min, max, doolittle, charges,
seqlen, firstline []
  --tsv BOOL          Return output in TSV format [0]
common:
  --compressed INT    Write compressed output [0]
  --threads INT        Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching



for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.101. reverseseq

Reverse (without complement) sequences.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

#### Usage

```
usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]
```

#### Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### Full CLI Help Snapshot

```
usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]
```

By Eli Levy Karin <eli.levy.karin@gmail.com>

options: common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT   Write compressed output [0]
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.102. rmdb

Remove a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 19 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

### Usage

```
usage: mmseqs rmdb <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs rmdb` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.103. search

Sensitive homology search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 8 upstream caller(s) and 22 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: `search_workflows`.

### Usage

```
usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
```

### Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering)                   |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                                          |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                                       |
| -k                     | k-mer length (0: automatically set to optimum)                                         |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)                                |
| --k-score              | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size            | Alphabet size (range 2-21)                                                             |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split                | Split input into N equally distributed chunks. 0: set the best split automatically     |
| --split-mode           | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

```
usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [5.700]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
```

```

similar k-mer) [0]
--k-score TWIN          k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN        Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT          Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT             Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT        0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL       Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT Extract only exact k-mers for matching (range
0-1) [0]
--mask INT              Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT       Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT   Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT     Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT  0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR User-specified spaced k-mer pattern []
--local-tmp STR         Path where some of the temporary files will be
created []
--disk-space-limit BYTE Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT    How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [2]
--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment

```

5: score only (output) cluster format [0]  
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during  
 the scoring process to allow wrapped diagonal scoring around end and start [0]  
 -e DOUBLE List matches below this E-value (range 0.0-inf)  
 [1.000E-03]  
 --min-seq-id FLOAT List matches above this sequence identity (for  
 clustering) (range 0.0-1.0) [0.000]  
 --min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]  
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer  
 sequence [0]  
 --alt-ali INT Show up to this many alternative alignments [0]  
 -c FLOAT List matches above this fraction of aligned  
 (covered) residues (see --cov-mode) [0.000]  
 --cov-mode INT 0: coverage of query and target  
 1: coverage of target  
 2: coverage of query  
 3: target seq. length has to be at least x% of  
 query length  
 4: query seq. length has to be at least x% of  
 target length  
 5: short seq. needs to be at least x% of the  
 other seq. length [0]  
 --max-rejected INT Maximum rejected alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --max-accept INT Maximum accepted alignments before alignment  
 calculation for a query is stopped [2147483647]  
 --score-bias FLOAT Score bias when computing SW alignment (in  
 bits) [0.000]  
 --realign BOOL Compute more conservative, shorter alignments  
 (scores and E-values not changed) [0]  
 --realign-score-bias FLOAT Additional bias when computing realignment  
 [-0.200]  
 --realign-max-seqs INT Maximum number of results to return in  
 realignment [2147483647]  
 --corr-score-weight FLOAT Weight of backtrace correlation score that is  
 added to the alignment score [0.000]  
 --gap-open TWIN Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]  
 --zdrop INT Maximal allowed difference between score values  
 before alignment is truncated (nucleotide alignment only) [40]  
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,  
 1: filter [0]  
 profile:  
 --pca Pseudo count admixture strength []  
 --pcb Pseudo counts: Neff at half of maximum  
 admixture (range 0.0-inf) []  
 --mask-profile INT Mask query sequence of profile using tantan  
 [0,1] [1]  
 --e-profile DOUBLE Include sequences matches with < E-value thr.

into the profile ( $\geq 0.0$ ) [1.000E-01]

--wg BOOL Use global sequence weighting for profile calculation [0]

--filter-msa INT Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT Only filter MSAs with more than N sequences, 0 always filters [0]

--max-seq-id FLOAT Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] [0.900]

--qid STR Reduce diversity of output MSAs using min.seq. identity with query sequences [0.0,1.0]

Alternatively, can be a list of multiple thresholds:

E.g.: 0.15,0.30,0.50 to defines filter buckets of ]0.15-0.30] and ]0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT Number of iterative profile search iterations [1]

--exhaustive-search BOOL For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL Efficient search for LCA candidates [0]

misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT Rescore diagonals with:  
0: Hamming distance  
1: local alignment (score only)  
2: local alignment  
3: global alignment  
4: longest alignment fulfilling window quality criterion [0]

--allow-deletion BOOL Allow deletions in a MSA [0]

--min-length INT Minimum codon number in open reading frames [30]

--max-length INT Maximum codon number in open reading frames [32734]

--max-gaps INT Maximum number of codons with gaps or unknown residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT Contig start can be 0: incomplete, 1: complete,

2: both [2]  
 --contig-end-mode INT Contig end can be 0: incomplete, 1: complete,  
 2: both [2]  
 --orf-start-mode INT Orf fragment can be 0: from start to stop, 1:  
 from any to stop, 2: from last encountered start to stop (no start in the middle)  
 [1]  
 --forward-frames STR Comma-separated list of frames on the forward  
 strand to be extracted [1,2,3]  
 --reverse-frames STR Comma-separated list of frames on the reverse  
 strand to be extracted [1,2,3]  
 --translation-table INT 1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3)  
 YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE  
 9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11)  
 PROKARYOTE, 12) ALT\_YEAST, 13) ASCIDIAN\_MITOCHONDRIAL, 14)  
 ALT\_FLATWORM\_MITOCHONDRIAL  
 15) BLEPHARISMA, 16)  
 CHLOROPHYCEAN\_MITOCHONDRIAL, 21) TREMATODE\_MITOCHONDRIAL, 22)  
 SCENEDESMUS\_MITOCHONDRIAL  
 23) THRAUSTOCHYTRIUM\_MITOCHONDRIAL, 24)  
 PTEROBRANCHIA\_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)  
 KARYORELICT, 28) CONDYLOSTOMA  
 29) MESODINIUM, 30) PERTRICH, 31)  
 BLASTOCRITHIDIA [1]  
 --translate INT Translate ORF to amino acid [0]  
 --use-all-table-starts BOOL Use all alternatives for a start codon in the  
 genetic table, if false - only ATG (AUG) [0]  
 --id-offset INT Numeric ids in index file are offset by this  
 value [0]  
 --sequence-overlap INT Overlap between sequences [0]  
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link  
 data and write new index, [1]  
 --headers-split-mode INT Header split mode: 0: split position, 1:  
 original header [0]  
 --search-type INT Search type 0: auto 1: amino acid, 2:  
 translated, 3: nucleotide, 4: translated nucleotide alignment [0]  
 --start-sens FLOAT Start sensitivity [4.000]  
 --sens-steps INT Number of search steps performed from --start-  
 sens to -s [1]  
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs,  
 1: full reading frames [0]  
 common:  
 --sub-mat TWIN Substitution matrix file  
 [aa:blosum62.out,nucl:nucleotide.out]  
 --max-seq-len INT Maximum sequence length [65535]  
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:  
 mmap, 3: mmap+touch [0]  
 --threads INT Number of CPU-cores used (all by default) [10]  
 --compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:

```

+warnings, 3: +info [3]
--gpu INT                Use GPU (CUDA) if possible [0]
--gpu-server INT          Use GPU server [0]
--gpu-server-wait-timeout INT  Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL        Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT       Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT         Combine ORFs/split sequences to a single entry
[1]
--strand INT              Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

#### examples:

```

# Search multiple FASTA against FASTA (like BLASTP, TBLASTN, BLASTX, BLASTN --
search-type 3, TBLASTX --search-type 2)
mmseqs search queryDB targetDB resultDB tmp
mmseqs convertalis queryDB targetDB resultDB result.m8

```

```

# Iterative profile search (like PSI-BLAST)
mmseqs search queryDB targetDB resultDB tmp --num-iterations 2

```

```

# Profile search against small databases (e.g. PFAM, eggNOG)
mmseqs databases PFAM pfam_db tmp
mmseqs search queryDB pfam_db resultDB tmp

```

```

# Exhaustive search against sequences or profiles (works for large DBs)
mmseqs search queryDB targetDB resultDB tmp --exhaustive-search

```

```

# Increasing sensitivity search (from 2 to 7 in 3 steps)
mmseqs search queryDB targetDB resultDB --start-sens 2 -s 7 --sens-steps 3

```

#### references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M,
Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv,
2024.11.13.623350 (2024)

```



#### 6.2.104. **sequence2profile**

Turn sequence into profile by adding context specific pseudo counts.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

##### **Usage**

```
usage: mmseqs sequence2profile <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs sequence2profile` locally for exact positional arguments in your build.

##### **Key Options**

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

##### **Full CLI Help Snapshot**

###### **Note**

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.105. **setextendeddbtype**

Write an extended DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

##### **Usage**

```
usage: mmseqs setextendeddbtype <i:DB> [options]
```

## Key Options

| Option                         | Purpose                                                                 |
|--------------------------------|-------------------------------------------------------------------------|
| <code>--extended-dbtype</code> | Set extended dbtype 1: compressed, 2: need src, 4: context pseudoe cnts |
| <code>-v</code>                | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

## Full CLI Help Snapshot

```
usage: mmseqs setextendeddbtype <i:DB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --extended-dbtype INT    Set extended dbtype 1: compressed, 2: need src, 4:
context pseudoe cnts [0]
common:
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

### examples:

```
# Print entries with keys 1, 2 and 3 from a sequence DB to stdout
mmseqs setextendeddbtype db --extended-dbtype 2
```

### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

## 6.2.106. sortresult

Sort a result DB in the same order as the prefilter or align module.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

## Usage

```
usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]
```

## Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --compressed | Write compressed output                                       |
| --threads    | Number of CPU-cores used (all by default)                     |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: common:
  --compressed INT    Write compressed output [0]
  --threads INT       Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.107. splitdb

Split DB into subsets.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

### Usage

```
usage: mmseqs splitdb <i:DB> <o:DB> [options]
```

### Key Options

| Option       | Purpose                                                |
|--------------|--------------------------------------------------------|
| --split      | Split input into N equally distributed chunks          |
| --split-aa   | Try to find the best split boundaries by entry lengths |
| --compressed | Write compressed output                                |

| Option | Purpose                                                       |
|--------|---------------------------------------------------------------|
| -v     | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

usage: mmseqs splitdb <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| --split INT      | Split input into N equally distributed chunks [0]                 |
| --split-aa BOOL  | Try to find the best split boundaries by entry lengths [0]        |
| --compressed INT | Write compressed output [0]                                       |
| -v INT           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3] |

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.108. splitsequence

Split sequences by length.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

### Usage

usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]

### Key Options

| Option                | Purpose                                                                  |
|-----------------------|--------------------------------------------------------------------------|
| --sequence-overlap    | Overlap between sequences                                                |
| --sequence-split-mode | Sequence split mode 0: copy data, 1: soft link data and write new index, |
| --headers-split-mode  | Header split mode: 0: split position, 1: original header                 |

| Option          | Purpose                                                       |
|-----------------|---------------------------------------------------------------|
| --max-seq-len   | Maximum sequence length                                       |
| --threads       | Number of CPU-cores used (all by default)                     |
| --compressed    | Write compressed output                                       |
| -v              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |
| --create-lookup | Create database lookup file (can be very large)               |

### Full CLI Help Snapshot

```
usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --sequence-overlap INT      Overlap between sequences [300]
  --sequence-split-mode INT  Sequence split mode 0: copy data, 1: soft link data
and write new index, [1]
  --headers-split-mode INT   Header split mode: 0: split position, 1: original
header [0]
common:
  --max-seq-len INT          Maximum sequence length [10000]
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT           Write compressed output [0]
  -v INT                     Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
  --create-lookup INT        Create database lookup file (can be very large) [0]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

#### 6.2.109. subtractdbs

Remove all entries from first DB occurring in second DB by key.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB> [options]

## Key Options

| Option       | Purpose                                                                |
|--------------|------------------------------------------------------------------------|
| -e           | List matches below this E-value (range 0.0-inf)                        |
| --e-profile  | Include sequences matches with < E-value thr. into the profile (>=0.0) |
| --threads    | Number of CPU-cores used (all by default)                              |
| --compressed | Write compressed output                                                |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info          |

## Full CLI Help Snapshot

```
usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: align:
  -e DOUBLE          List matches below this E-value (range 0.0-inf) [1.000E-03]
profile:
  --e-profile DOUBLE Include sequences matches with < E-value thr. into the
profile (>=0.0) [1.000E-03]
common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

### 6.2.110. suffixid

For each entry in a DB append the entry key to the entry itself.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

### Usage

```
usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]
```

### Key Options

| Option         | Purpose                                                                 |
|----------------|-------------------------------------------------------------------------|
| --prefix       | Use this prefix for all entries                                         |
| --mapping-file | Specify a file that translates the keys of a DB to new keys, TSV format |
| --tsv          | Return output in TSV format                                             |
| --threads      | Number of CPU-cores used (all by default)                               |
| --compressed   | Write compressed output                                                 |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info           |

### Full CLI Help Snapshot

```
usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: misc:

```
--prefix STR      Use this prefix for all entries []
--mapping-file STR Specify a file that translates the keys of a DB to new
keys, TSV format []
--tsv BOOL        Return output in TSV format [0]
```

common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT   Write compressed output [0]
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.111. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

## Usage

```
usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]
```

## Key Options

| Option                    | Purpose                                                       |
|---------------------------|---------------------------------------------------------------|
| <code>--threads</code>    | Number of CPU-cores used (all by default)                     |
| <code>--compressed</code> | Write compressed output                                       |
| <code>-v</code>           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

## Full CLI Help Snapshot

```
usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT   Write compressed output [0]
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

- Mirdita M, von den Driesch L, Galiez C, Martin M, Soding J, Steinegger M: Uniclust databases of clustered and deeply annotated protein sequences and alignments. *Nucleic Acids Research* 45(D1), D170-D176 (2017)

### 6.2.112. summarizeheaders

Summarize FASTA headers of result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.



Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

### Usage

```
usage: mmseqs summarizeheaders <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs summarizeheaders` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.113. `summarizeresult`

Extract annotations from alignment DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

### Usage

```
usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]
```

### Key Options

| Option | Purpose                                                                                      |
|--------|----------------------------------------------------------------------------------------------|
| -a     | Add backtrace string (convert to alignments with mmseqs convertalis module)                  |
| -c     | List matches above this fraction of aligned (covered) residues (see <code>-cov-mode</code> ) |

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --overlap    | Maximum overlap of covered regions                            |
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
  <martin.steinegger@snu.ac.kr>
options: align:
  -a BOOL          Add backtrace string (convert to alignments with mmseqs
convertalis module) [0]
  -c FLOAT          List matches above this fraction of aligned (covered)
residues (see --cov-mode) [0.000]
misc:
  --overlap FLOAT   Maximum overlap of covered regions [0.000]
common:
  --threads INT     Number of CPU-cores used (all by default) [10]
  --compressed INT  Write compressed output [0]
  -v INT            Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
  - Mirdita M, von den Driesch L, Galiez C, Martin M, Soding J, Steinegger M:
Uniclust databases of clustered and deeply annotated protein sequences and
alignments. Nucleic Acids Research 45(D1), D170-D176 (2017)
```

#### 6.2.114. summarizetabs

Extract annotations from HHblits BLAST-tab-formatted results.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

usage: mmseqs summarizetabs <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs summarizetabs locally for exact positional arguments in your build.

## Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

## Full CLI Help Snapshot

### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

### 6.2.115. swapdb

Transpose DB with integer values in first column.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]

## Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| --threads            | Number of CPU-cores used (all by default)                                                    |
| --compressed         | Write compressed output                                                                      |
| -v                   | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                |

## Full CLI Help Snapshot

usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]  
 By Martin Steinegger <martin.steinegger@snu.ac.kr>, Clovis Galiez & Eli Levy  
 Karin  
 options: prefilter:  
 --split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G.  
 Default (0) to all available system memory [0]  
 common:  
 --threads INT Number of CPU-cores used (all by default) [10]  
 --compressed INT Write compressed output [0]  
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings,  
 3: +info [3]  
 references:  
 - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching  
 for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028  
 (2017)

### 6.2.116. swapresults

Transpose prefilter/alignment DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result\_handling.

#### Usage

usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>  
 [options]

#### Key Options

| Option               | Purpose                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| --split-memory-limit | Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory |
| -e                   | List matches below this E-value (range 0.0-inf)                                              |
| --gap-open           | Gap open cost                                                                                |
| --gap-extend         | Gap extension cost                                                                           |
| --sub-mat            | Substitution matrix file                                                                     |
| --threads            | Number of CPU-cores used (all by default)                                                    |

| Option         | Purpose                                                         |
|----------------|-----------------------------------------------------------------|
| --compressed   | Write compressed output                                         |
| --db-load-mode | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch |
| -v             | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info   |

### Full CLI Help Snapshot

```
usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>, Clovis Galiez & Eli Levy
  Karin
options: prefilter:
  --split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G.
  Default (0) to all available system memory [0]
align:
  -e DOUBLE                List matches below this E-value (range 0.0-inf)
  [1.000E-03]
  --gap-open TWIN           Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN        Gap extension cost [aa:1,nucl:2]
common:
  --sub-mat TWIN            Substitution matrix file
  [aa:blosum62.out,nucl:nucleotide.out]
  --threads INT             Number of CPU-cores used (all by default) [10]
  --compressed INT          Write compressed output [0]
  --db-load-mode INT        Database preload mode 0: auto, 1: fread, 2: mmap, 3:
  mmap+touch [0]
  -v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
  3: +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
```

#### 6.2.117. tar2db

Convert content of tar archives to any DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

## Usage

```
usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]
```

## Key Options

| Option                       | Purpose                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--output-dbtype</code> | Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| <code>--tar-include</code>   | Include file names based on this regex                                                                                                                                                                                                                                                                                                       |
| <code>--tar-exclude</code>   | Exclude file names based on this regex                                                                                                                                                                                                                                                                                                       |
| <code>--compressed</code>    | Write compressed output                                                                                                                                                                                                                                                                                                                      |
| <code>--threads</code>       | Number of CPU-cores used (all by default)                                                                                                                                                                                                                                                                                                    |
| <code>-v</code>              | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                                                                                                                                                                                                                                |

## Full CLI Help Snapshot

```
usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: misc:

```
--output-dbtype INT  Set database type for resulting database: Amino acid
sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6,
Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA
or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional
prefiltering result 14, Offsetted headers 15 [12]
```

```
--tar-include STR    Include file names based on this regex [.*]
```

```
--tar-exclude STR    Exclude file names based on this regex [^$]
```

common:

```
--compressed INT    Write compressed output [0]
```

```
--threads INT        Number of CPU-cores used (all by default) [10]
```

```
-v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
```

```
+info [3]
```

examples:

```
# Assuming tar archive containing three aligned FASTA files:
# * folder/msa1.fa.gz * folder/msa2.fa * folder/msa3.fa
# Create a msaDB with three DB entries each containing a separate MSA
mmseqs tar2db archive.tar.gz msaDB --output-dbtype 11
```

#### references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.118. taxonomy

Taxonomic classification.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 2 upstream caller(s) and 15 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### Usage

usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]

#### Key Options

| Option                 | Purpose                                                              |
|------------------------|----------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)        |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)        |
| --add-self-matches     | Artificially add entries of queries with themselves (for clustering) |
| --seed-sub-mat         | Substitution matrix file for k-mer generation                        |
| -s                     | Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive                     |
| -k                     | k-mer length (0: automatically set to optimum)                       |
| --target-search-mode   | target search mode (0: regular k-mer, 1: similar k-mer)              |

| Option       | Purpose                                                                                |
|--------------|----------------------------------------------------------------------------------------|
| --k-score    | k-mer threshold for generating similar k-mer lists                                     |
| --alph-size  | Alphabet size (range 2-21)                                                             |
| --max-seqs   | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| --split      | Split input into N equally distributed chunks.<br>0: set the best split automatically  |
| --split-mode | 0: split target db; 1: split query db; 2: auto, depending on main memory               |

### Full CLI Help Snapshot

```
usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr> & Eli Levy Karin <eli.levy.karin@gmail.com>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [2.000]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT               Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
  --split INT                  Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --diag-score BOOL            Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT     Extract only exact k-mers for matching (range
0-1) [0]
```



--mask INT Mask sequences in prefilter stage with tantan:  
 0: w/o low complexity masking, 1: with low complexity masking [1]  
 --mask-prob FLOAT Mask sequences is probablity is above threshold  
 [0.900]  
 --mask-lower-case INT Lowercase letters will be excluded from k-mer  
 search 0: include region, 1: exclude region [0]  
 --mask-n-repeat INT Repeat letters that occure > threshold in a rwo  
 [0]  
 --min-ungapped-score INT Accept only matches with ungapped alignment  
 score above threshold [15]  
 --spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use  
 spaced k-mers [1]  
 --spaced-kmer-pattern STR User-specified spaced k-mer pattern []  
 --local-tmp STR Path where some of the temporary files will be  
 created []  
 --disk-space-limit BYTE Set max disk space to use for reverse profile  
 searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the  
 temp folder [0]  
 align:  
 -a B00L Add backtrace string (convert to alignments  
 with mmseqs convertalis module) [0]  
 --alignment-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment [1]  
 --alignment-output-mode INT How to compute the alignment:  
 0: automatic  
 1: only score and end\_pos  
 2: also start\_pos and cov  
 3: also seq.id  
 4: only ungapped alignment  
 5: score only (output) cluster format [0]  
 --wrapped-scoring B00L Double the (nucleotide) query sequence during  
 the scoring process to allow wrapped diagonal scoring around end and start [0]  
 -e DOUBLE List matches below this E-value (range 0.0-inf)  
 [1.000E+00]  
 --min-seq-id FLOAT List matches above this sequence identity (for  
 clustering) (range 0.0-1.0) [0.000]  
 --min-aln-len INT Minimum alignment length (range 0-INT\_MAX) [0]  
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer  
 sequence [0]  
 --alt-ali INT Show up to this many alternative alignments [0]  
 -c FLOAT List matches above this fraction of aligned  
 (covered) residues (see --cov-mode) [0.000]  
 --cov-mode INT 0: coverage of query and target  
 1: coverage of target  
 2: coverage of query

query length 3: target seq. length has to be at least x% of  
 target length 4: query seq. length has to be at least x% of  
 other seq. length [0] 5: short seq. needs to be at least x% of the  
 --max-rejected INT Maximum rejected alignments before alignment  
 calculation for a query is stopped [5]  
 --max-accept INT Maximum accepted alignments before alignment  
 calculation for a query is stopped [30]  
 --score-bias FLOAT Score bias when computing SW alignment (in  
 bits) [0.000]  
 --realign BOOL Compute more conservative, shorter alignments  
 (scores and E-values not changed) [0]  
 --realign-score-bias FLOAT Additional bias when computing realignment  
 [-0.200]  
 --realign-max-seqs INT Maximum number of results to return in  
 realignment [2147483647]  
 --corr-score-weight FLOAT Weight of backtrace correlation score that is  
 added to the alignment score [0.000]  
 --gap-open TWIN Gap open cost [aa:11,nucl:5]  
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]  
 --zdrop INT Maximal allowed difference between score values  
 before alignment is truncated (nucleotide alignment only) [40]  
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,  
 1: filter [0]  
 profile:  
 --pca Pseudo count admixture strength []  
 --pcb Pseudo counts: Neff at half of maximum  
 admixture (range 0.0-inf) []  
 --mask-profile INT Mask query sequence of profile using tantan  
 [0,1] [1]  
 --e-profile DOUBLE Include sequences matches with < E-value thr.  
 into the profile (>=0.0) [1.000E-03]  
 --wg BOOL Use global sequence weighting for profile  
 calculation [0]  
 --filter-msa INT Filter msa: 0: do not filter, 1: filter [1]  
 --filter-min-enable INT Only filter MSAs with more than N sequences, 0  
 always filters [0]  
 --max-seq-id FLOAT Reduce redundancy of output MSA using max.  
 pairwise sequence identity [0.0,1.0] [0.900]  
 --qid STR Reduce diversity of output MSAs using min.seq.  
 identity with query sequences [0.0,1.0]  
 Alternatively, can be a list of multiple  
 thresholds:  
 E.g.: 0.15,0.30,0.50 to defines filter buckets  
 of ]0.15-0.30] and ]0.30-0.50] [0.0]  
 --qsc FLOAT Reduce diversity of output MSAs using min.  
 score per aligned residue with query sequences [-50.0,100.0] [-20.000]

```

--cov FLOAT                Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                 Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT      use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT   Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--exhaustive-search BOOL    For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL           Efficient search for LCA candidates [0]
misc:
--orf-filter INT            Prefilter query ORFs with non-selective search
Only used during nucleotide-vs-protein

classification
NOTE: Consider disabling when classifying short
reads [1]
--orf-filter-e DOUBLE       E-value threshold used for query ORF
prefiltering [1.000E+02]
--orf-filter-s FLOAT        Sensitivity used for query ORF prefiltering
[2.000]
--lca-mode INT              LCA Mode 1: single search LCA , 2/3:
approximate 2bLCA, 4: top hit [3]
--tax-output-mode INT       0: output LCA, 1: output alignment 2: output
both [0]
--majority FLOAT            minimal fraction of agreement among
taxonomically assigned sequences of a set [0.500]
--vote-mode INT             Mode of assigning weights to compute majority.
0: uniform, 1: minus log E-value, 2: score [1]
--lca-ranks STR             Add column with specified ranks (',' separated)
[]
--tax-lineage INT           0: don't show, 1: add all lineage names, 2: add
all lineage taxids [0]
--blacklist STR             Comma separated list of ignored taxa in LCA
computation [12908:unclassified sequences,28384:other sequences]
--taxon-list STR            Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT        prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT          Rescore diagonals with:
0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality

criterion [0]
--allow-deletion BOOL       Allow deletions in a MSA [0]
--min-length INT            Minimum codon number in open reading frames
[30]

```

```

--max-length INT                Maximum codon number in open reading frames
[32734]
--max-gaps INT                  Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT         Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR           Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR           Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT         1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)
BLASTOCRITHIDIA [1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT          Overlap between sequences [0]
--sequence-split-mode INT       Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT        Header split mode: 0: split position, 1:
original header [0]
--search-type INT              Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--translation-mode INT          Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
common:
--compressed INT               Write compressed output [0]
--threads INT                  Number of CPU-cores used (all by default) [10]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--sub-mat TWIN                  Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]

```

```

--max-seq-len INT           Maximum sequence length [65535]
--db-load-mode INT          Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--gpu INT                   Use GPU (CUDA) if possible [0]
--gpu-server INT            Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR            Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL          Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL     Delete temporary files [0]
expert:
--filter-hits BOOL          Filter hits by seq.id. and coverage [0]
--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT         Create database lookup file (can be very large)
[0]
--chain-alignments INT      Chain overlapping alignments [0]
--merge-query INT           Combine ORFs/split sequences to a single entry
[1]
--strand INT                Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

#### examples:

```

# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Assign taxonomy based on 2bLCA
mmseqs taxonomy queryDB swissprotDB result tmp

# Assign taxonomy based on top hit
mmseqs taxonomy queryDB swissprotDB result tmp --lca-mode 4

# Assign taxonomy without ORF prefilter
# Classifies higher percentage for short nucleotide input (e.g. short reads) at
the cost of speed
mmseqs taxonomy queryNuclDB swissprotDB result tmp --orf-filter 0

# Create a Krona report
mmseqs taxonomyreport swissprotDB result report.html --report-mode 1

```

#### references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)

```

### 6.2.119. taxonomyreport

Create a taxonomy report in Kraken or Krona format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

#### Usage

```
usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB>
<o:taxonomyReport> [options]
```

#### Key Options

| Option        | Purpose                                                       |
|---------------|---------------------------------------------------------------|
| --report-mode | Taxonomy report mode 0: Kraken 1: Krona                       |
| --threads     | Number of CPU-cores used (all by default)                     |
| -v            | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

#### Full CLI Help Snapshot

```
usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB>
<o:taxonomyReport> [options]
  By Milot Mirdita <milot@mirdita.de> & Florian Breitwieser <florian.bw@gmail.com>
options: misc:
  --report-mode INT    Taxonomy report mode 0: Kraken 1: Krona [0]
common:
  --threads INT        Number of CPU-cores used (all by default) [10]
  -v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)
```

### 6.2.120. touchdb

Preload DB into memory (page cache).

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

### Usage

```
usage: mmseqs touchdb <i:DB> [options]
```

### Key Options

| Option    | Purpose                                                       |
|-----------|---------------------------------------------------------------|
| --threads | Number of CPU-cores used (all by default)                     |
| -v        | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

```
usage: mmseqs touchdb <i:DB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

```
--threads INT    Number of CPU-cores used (all by default) [10]
-v INT           Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

### 6.2.121. transitivealign

Transfer alignments via transitivity.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

### Usage

```
usage: mmseqs transitivealign <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs transitivealign` locally for exact positional arguments in your build.

### Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

### Full CLI Help Snapshot

#### Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

#### 6.2.122. `translateaa`

Translate proteins to lexicographically lowest codons.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `sequence_manipulation`.

### Usage

usage: `mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]`

### Key Options

| Option                    | Purpose                                                       |
|---------------------------|---------------------------------------------------------------|
| <code>--threads</code>    | Number of CPU-cores used (all by default)                     |
| <code>--compressed</code> | Write compressed output                                       |
| <code>-v</code>           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

usage: `mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]`

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

|                               |                                                               |
|-------------------------------|---------------------------------------------------------------|
| <code>--threads INT</code>    | Number of CPU-cores used (all by default) [10]                |
| <code>--compressed INT</code> | Write compressed output [0]                                   |
| <code>-v INT</code>           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |



[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

### 6.2.123. translatenucs

Translate nucleotides to proteins.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence\_manipulation.

### Usage

usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]

### Key Options

| Option              | Purpose                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| --translation-table | 1)<br>CANONICAL, 2) VERT_MITOCHONDRIAL,<br>3) YEAST_MITOCHONDRIAL, 4)<br>MOLD_MITOCHONDRIAL, 5)<br>INVERT_MITOCHONDRIAL, 6) CILIATE |
| --add-orf-stop      | Add stop codon “*” at complete start and end                                                                                        |
| -v                  | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                                       |
| --compressed        | Write compressed output                                                                                                             |
| --threads           | Number of CPU-cores used (all by default)                                                                                           |

### Full CLI Help Snapshot

usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: misc:

--translation-table INT 1) CANONICAL, 2) VERT\_MITOCHONDRIAL, 3)  
YEAST\_MITOCHONDRIAL, 4) MOLD\_MITOCHONDRIAL, 5) INVERT\_MITOCHONDRIAL, 6) CILIATE  
9) FLATWORM\_MITOCHONDRIAL, 10) EUPLOTID, 11)



| Option       | Purpose                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------|
|              | MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15 |
| --compressed | Write compressed output                                                                                        |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info                                                  |

### Full CLI Help Snapshot

```
usage: mmseqs tsv2db <i:tsvFile> <o:resultDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --add-self-matches BOOL  Artificially add entries of queries with themselves
                           (for clustering) [0]
misc:
  --output-dbtype INT      Set database type for resulting database: Amino acid
                           sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6,
                           Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA
                           or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional
                           prefiltering result 14, Offsetted headers 15 [12]
common:
  --compressed INT        Write compressed output [0]
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
                           +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
    for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
    (2017)
```

#### 6.2.125. tsv2exprofiledb

Create a expandable profile db from TSV files.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 5 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

#### Usage

usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]

### Key Options

| Option       | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| --gpu        | Use GPU (CUDA) if possible                                    |
| --threads    | Number of CPU-cores used (all by default)                     |
| --compressed | Write compressed output                                       |
| -v           | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info |

### Full CLI Help Snapshot

usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--gpu INT            Use GPU (CUDA) if possible [0]  
--threads INT        Number of CPU-cores used (all by default) [10]  
--compressed INT    Write compressed output [1]  
-v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info  
[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

#### 6.2.126. ungappedprefilter

Optimal diagonal score search.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

### Usage

usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]

### Key Options

| Option                 | Purpose                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| --comp-bias-corr       | Correct for locally biased amino acid composition (range 0-1)                          |
| --comp-bias-corr-scale | Correct for locally biased amino acid composition (range 0-1)                          |
| --min-ungapped-score   | Accept only matches with ungapped alignment score above threshold                      |
| --max-seqs             | Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) |
| -c                     | List matches above this fraction of aligned (covered) residues (see --cov-mode)        |
| -e                     | List matches below this E-value (range 0.0-inf)                                        |
| --cov-mode             | 0: coverage of query and target                                                        |
| --taxon-list           | Taxonomy ID, possibly multiple values separated by ‘,’                                 |
| --prefilter-mode       | prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped          |
| --sub-mat              | Substitution matrix file                                                               |
| --db-load-mode         | Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch                        |
| --gpu                  | Use GPU (CUDA) if possible                                                             |

### Full CLI Help Snapshot

usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB>

[options]

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger

<martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition (range 0-1) [1]  
--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition (range 0-1) [1.000]  
--min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]  
--max-seqs INT Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]

align:

-c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]  
-e DOUBLE List matches below this E-value (range 0.0-inf)

```

[1.000E-03]
--cov-mode INT          0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
misc:
--taxon-list STR        Taxonomy ID, possibly multiple values separated
by ',' [0]
--prefilter-mode INT    prefilter mode: 0: kmer/ungapped 1: ungapped, 2:
nofilter, 3: ungapped&gapped [0]
common:
--sub-mat TWIN          Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT      Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--gpu INT               Use GPU (CUDA) if possible [0]
--gpu-server INT        Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--threads INT           Number of CPU-cores used (all by default) [10]
--compressed INT        Write compressed output [0]
-v INT                  Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M,
Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv,
2024.11.13.623350 (2024)

```

### 6.2.127. unpackdb

Unpack a DB into separate files.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

```
usage: mmseqs unpackdb <i:DB> <o:outDir> [options]
```

## Key Options

| Option             | Purpose                                                          |
|--------------------|------------------------------------------------------------------|
| --unpack-name-mode | Name unpacked files by 0: DB key, 1: accession (through .lookup) |
| --unpack-suffix    | File suffix for unpacked files.                                  |
| --threads          | Number of CPU-cores used (all by default)                        |
| -v                 | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info    |

## Full CLI Help Snapshot

```
usage: mmseqs unpackdb <i:DB> <o:outDir> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: misc:
```

```
--unpack-name-mode INT    Name unpacked files by 0: DB key, 1: accession  
(through .lookup) [1]
```

```
--unpack-suffix STR       File suffix for unpacked files.  
                           Add .gz suffix to write compressed files. []
```

```
common:
```

```
--threads INT             Number of CPU-cores used (all by default) [10]
```

```
-v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:  
+info [3]
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching  
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028  
(2017)
```

### 6.2.128. view

Print DB entries given in `-id-list` to stdout.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

## Usage

usage: mmseqs view <i:DB> [options]

## Key Options

| Option           | Purpose                                                                     |
|------------------|-----------------------------------------------------------------------------|
| --id-list        | Entries to be printed separated by ‘;                                       |
| --id-mode        | Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) |
| --idx-entry-type | 0: sequence, 1: src sequence, 2: header, 3: src header                      |
| -v               | Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info               |

## Full CLI Help Snapshot

usage: mmseqs view <i:DB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--id-list STR Entries to be printed separated by ',' []

--id-mode INT Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup) [0]

--idx-entry-type INT 0: sequence, 1: src sequence, 2: header, 3: src header [0]

common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

examples:

# Print entries with keys 1, 2 and 3 from a sequence DB to stdout

mmseqs view sequencedb --id-list 1,2,3

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

## 7. MMseqs2 Dependency Map

This file is generated from MMseqs2/src/MMseqsBase.cpp and workflow scripts.

| Metric                 | Value |
|------------------------|-------|
| Total visible commands | 128   |

n/a in connection fields means no direct edge was resolved by static extraction.



## 7.1. Easy Workflows

### 7.1.1. **easy-cluster**

Slower, sensitive clustering.

| Aspect                  | Value                                                                           |
|-------------------------|---------------------------------------------------------------------------------|
| Layer                   | workflow                                                                        |
| Category flags          | COMMAND_EASY                                                                    |
| Calls                   | cluster, createdb, createseqfiledb, createtsv, result2flat, result2repseq, rmdb |
| Called by               | n/a                                                                             |
| Workflow scripts        | n/a                                                                             |
| Command reference       | <a href="#">Open page</a>                                                       |
| Functional module entry | <a href="#">Open module page</a>                                                |

### 7.1.2. **easy-linclust**

Fast linear time cluster, less sensitive clustering.

| Aspect                  | Value                                                                            |
|-------------------------|----------------------------------------------------------------------------------|
| Layer                   | workflow                                                                         |
| Category flags          | COMMAND_EASY                                                                     |
| Calls                   | createdb, createseqfiledb, createtsv, linclust, result2flat, result2repseq, rmdb |
| Called by               | n/a                                                                              |
| Workflow scripts        | n/a                                                                              |
| Command reference       | <a href="#">Open page</a>                                                        |
| Functional module entry | <a href="#">Open module page</a>                                                 |

### 7.1.3. **easy-linsearch**

Fast, less sensitive homology search.

| Aspect         | Value                                                                           |
|----------------|---------------------------------------------------------------------------------|
| Layer          | workflow                                                                        |
| Category flags | COMMAND_EASY   COMMAND_EXPERT                                                   |
| Calls          | convertalis, createdb, createlinindex, linsearch, rmdb, search, summarizeresult |

| Aspect                  | Value            |
|-------------------------|------------------|
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.1.4. **easy-rbh**

Find reciprocal best hit.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | workflow                         |
| Category flags          | COMMAND_EASY                     |
| Calls                   | convertalis, createdb, rbh, rmdb |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | Open page                        |
| Functional module entry | Open module page                 |

#### 7.1.5. **easy-search**

Sensitive homology search.

| Aspect                  | Value                                                                           |
|-------------------------|---------------------------------------------------------------------------------|
| Layer                   | workflow                                                                        |
| Category flags          | COMMAND_EASY                                                                    |
| Calls                   | convertalis, createdb, createlinindex, linsearch, rmdb, search, summarizeresult |
| Called by               | n/a                                                                             |
| Workflow scripts        | n/a                                                                             |
| Command reference       | Open page                                                                       |
| Functional module entry | Open module page                                                                |

#### 7.1.6. **easy-taxonomy**

Taxonomic classification.

| Aspect | Value    |
|--------|----------|
| Layer  | workflow |

| Aspect                  | Value                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Category flags          | COMMAND_EASY                                                                                                             |
| Calls                   | addtaxonomy, convertalis, createdb, createtsv, filterdb, lca, rmdb, summarizealis, swapresults, taxonomy, taxonomyreport |
| Called by               | n/a                                                                                                                      |
| Workflow scripts        | n/a                                                                                                                      |
| Command reference       | Open page                                                                                                                |
| Functional module entry | Open module page                                                                                                         |

## 7.2. Search Workflows

### 7.2.1. linsearch

Fast, less sensitive homology search.

| Aspect                  | Value                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| Layer                   | high_level_api                                                                                           |
| Category flags          | COMMAND_MAIN COMMAND_EXPERT                                                                              |
| Calls                   | align, concatdbs, extractorfs, filterdb, kmersearch, offsetalignment, rescorediagonal, rmdb, swapresults |
| Called by               | easy-linsearch, easy-search                                                                              |
| Workflow scripts        | n/a                                                                                                      |
| Command reference       | Open page                                                                                                |
| Functional module entry | Open module page                                                                                         |

### 7.2.2. map

Map nearly identical sequences.

| Aspect           | Value          |
|------------------|----------------|
| Layer            | high_level_api |
| Category flags   | COMMAND_MAIN   |
| Calls            | search         |
| Called by        | n/a            |
| Workflow scripts | n/a            |

| Aspect                  | Value            |
|-------------------------|------------------|
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.2.3. rbh

Reciprocal best hit search.

| Aspect                  | Value                                                     |
|-------------------------|-----------------------------------------------------------|
| Layer                   | high_level_api                                            |
| Category flags          | COMMAND_MAIN                                              |
| Calls                   | filterdb, mergedbs, result2rbh, rmdb, search, swapresults |
| Called by               | easy-rbh                                                  |
| Workflow scripts        | easyrbh.sh                                                |
| Command reference       | Open page                                                 |
| Functional module entry | Open module page                                          |

### 7.2.4. search

Sensitive homology search.

| Aspect           | Value                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer            | high_level_api                                                                                                                                                                                                                                                                           |
| Category flags   | COMMAND_MAIN                                                                                                                                                                                                                                                                             |
| Calls            | align, createsubdb, expand2profile, expandaln, extractframes, extractorfs, filterresult, lcaalign, mergedbs, mvdb, offsetalignment, prefilter, profile2consensus, rescorediagonal, result2profile, result2stats, rmdb, search, splitsequence, subtractdb, swapresults, ungappedprefilter |
| Called by        | clusterupdate, easy-linsearch, easy-search, map, multihitsearch, rbh, search, taxonomy                                                                                                                                                                                                   |
| Workflow scripts | enrich.sh, iterativepp.sh, map.sh, multihitsearch.sh, rbh.sh, taxonomy.sh, update_clustering.sh                                                                                                                                                                                          |

| Aspect                  | Value            |
|-------------------------|------------------|
| Command reference       | Open page        |
| Functional module entry | Open module page |

## 7.3. Clustering

### 7.3.1. clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

| Aspect                  | Value                                                                        |
|-------------------------|------------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                                |
| Category flags          | COMMAND_CLUSTER                                                              |
| Calls                   | n/a                                                                          |
| Called by               | cluster, linclust                                                            |
| Workflow scripts        | cascaded_clustering.sh, clustering.sh, linclust.sh, nucleotide_clustering.sh |
| Command reference       | Open page                                                                    |
| Functional module entry | Open module page                                                             |

### 7.3.2. cluster

Slower, sensitive clustering.

| Aspect                  | Value                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | high_level_api                                                                                                                                                                                  |
| Category flags          | COMMAND_MAIN                                                                                                                                                                                    |
| Calls                   | align, clust, clusthash, concatdb, createsubdb, extractframes, filterdb, linclust, mergeclusters, mergedbs, mvdb, offsetalignment, prefilter, rescorediagonal, rmdb, subtractdb, swapdb, tsv2db |
| Called by               | clusterupdate, easy-cluster                                                                                                                                                                     |
| Workflow scripts        | update_clustering.sh                                                                                                                                                                            |
| Command reference       | Open page                                                                                                                                                                                       |
| Functional module entry | Open module page                                                                                                                                                                                |

### 7.3.3. clusterupdate

Update previous clustering with new sequences.

| Aspect                  | Value                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | high_level_api                                                                                                                     |
| Category flags          | COMMAND_MAIN                                                                                                                       |
| Calls                   | cluster, concatdbs, createsubdb, diffseqdbs, filterdb, mergedbs, mvdb, prefixid, renamedbkeys, result2repseq, rmdb, search, swapdb |
| Called by               | n/a                                                                                                                                |
| Workflow scripts        | n/a                                                                                                                                |
| Command reference       | <a href="#">Open page</a>                                                                                                          |
| Functional module entry | <a href="#">Open module page</a>                                                                                                   |

### 7.3.4. clusthash

Hash-based clustering of equal length sequences.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | mid_level_api                    |
| Category flags          | COMMAND_CLUSTER                  |
| Calls                   | n/a                              |
| Called by               | cluster                          |
| Workflow scripts        | clustering.sh                    |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.3.5. linclust

Fast, less sensitive clustering.

| Aspect         | Value                                                                                  |
|----------------|----------------------------------------------------------------------------------------|
| Layer          | high_level_api                                                                         |
| Category flags | COMMAND_MAIN                                                                           |
| Calls          | align, clust, createsubdb, filterdb, kmermatcher, mergeclusters, rescorediagonal, rmdb |

| Aspect                  | Value                                               |
|-------------------------|-----------------------------------------------------|
| Called by               | cluster, easy-linclud                               |
| Workflow scripts        | cascaded_clustering.sh,<br>nucleotide_clustering.sh |
| Command reference       | Open page                                           |
| Functional module entry | Open module page                                    |

### 7.3.6. mergeclusters

Merge multiple cascaded clustering steps.

| Aspect                  | Value                                                                          |
|-------------------------|--------------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                                  |
| Category flags          | COMMAND_CLUSTER                                                                |
| Calls                   | n/a                                                                            |
| Called by               | cluster, linclud                                                               |
| Workflow scripts        | cascaded_clustering.sh, clustering.sh,<br>linclud.sh, nucleotide_clustering.sh |
| Command reference       | Open page                                                                      |
| Functional module entry | Open module page                                                               |

### 7.3.7. pickconsensusrep

Select new representatives for each cluster based on consensus.

| Aspect                  | Value                                                                   |
|-------------------------|-------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                           |
| Category flags          | COMMAND_CLUSTER                                                         |
| Calls                   | align, msa2profile, prefixid,<br>renamedbkeys, result2msa, rmdb, tsv2db |
| Called by               | n/a                                                                     |
| Workflow scripts        | n/a                                                                     |
| Command reference       | Open page                                                               |
| Functional module entry | Open module page                                                        |

## 7.4. Prefiltering

### 7.4.1. countkmer

Count k-mers.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_SPECIAL                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.4.2. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | mid_level_api                    |
| Category flags          | COMMAND_PREFILTER                |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.4.3. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

| Aspect            | Value                     |
|-------------------|---------------------------|
| Layer             | mid_level_api             |
| Category flags    | COMMAND_PREFILTER         |
| Calls             | n/a                       |
| Called by         | linclust                  |
| Workflow scripts  | linclust.sh               |
| Command reference | <a href="#">Open page</a> |



| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Functional module entry | <a href="#">Open module page</a> |

#### 7.4.4. **kmersearch**

Find bottom-m-hashed k-mer matches between target and query DB.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | mid_level_api                    |
| Category flags          | COMMAND_PREFILTER                |
| Calls                   | n/a                              |
| Called by               | linsearch                        |
| Workflow scripts        | linsearch.sh                     |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.4.5. **prefilter**

Double consecutive diagonal k-mer search.

| Aspect                  | Value                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                                                                                                                                                                                     |
| Category flags          | COMMAND_PREFILTER                                                                                                                                                                                                                 |
| Calls                   | n/a                                                                                                                                                                                                                               |
| Called by               | cluster, search, taxonomy                                                                                                                                                                                                         |
| Workflow scripts        | blastp.sh,<br>blastpgp.sh, cascaded_clustering.sh,<br>clustering.sh, enrich.sh, iterativepp.sh,<br>nucleotide_clustering.sh,<br>searchslicedtargetprofile.sh,<br>searchtargetprofile.sh, taxpercontig.sh,<br>translated_search.sh |
| Command reference       | <a href="#">Open page</a>                                                                                                                                                                                                         |
| Functional module entry | <a href="#">Open module page</a>                                                                                                                                                                                                  |

#### 7.4.6. **ungappedprefilter**

Optimal diagonal score search.

| Aspect                  | Value                  |
|-------------------------|------------------------|
| Layer                   | mid_level_api          |
| Category flags          | COMMAND_PREFILTER      |
| Calls                   | n/a                    |
| Called by               | search                 |
| Workflow scripts        | blastp.sh, blastpgp.sh |
| Command reference       | Open page              |
| Functional module entry | Open module page       |

## 7.5. Alignment

### 7.5.1. align

Optimal gapped local alignment.

| Aspect                  | Value                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                                                        |
| Category flags          | COMMAND_ALIGNMENT                                                                                    |
| Calls                   | n/a                                                                                                  |
| Called by               | cluster, linclust, linsearch,<br>pickconsensusrep, search                                            |
| Workflow scripts        | iterativepp.sh,<br>nucleotide_clustering.sh,<br>pickconsensusrep.sh,<br>searchslicedtargetprofile.sh |
| Command reference       | Open page                                                                                            |
| Functional module entry | Open module page                                                                                     |

### 7.5.2. alignall

Within-result all-vs-all gapped local alignment.

| Aspect           | Value             |
|------------------|-------------------|
| Layer            | mid_level_api     |
| Category flags   | COMMAND_ALIGNMENT |
| Calls            | n/a               |
| Called by        | n/a               |
| Workflow scripts | n/a               |

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.5.3. alignbykmer

Heuristic gapped local k-mer based alignment.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | mid_level_api                    |
| Category flags          | COMMAND_ALIGNMENT                |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.5.4. expandaln

Expand an alignment result based on another.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | mid_level_api                    |
| Category flags          | COMMAND_PROFILE_PROFILE          |
| Calls                   | n/a                              |
| Called by               | search                           |
| Workflow scripts        | enrich.sh, iterativepp.sh        |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.5.5. fwbw

Forward Backward Alignment.

| Aspect         | Value             |
|----------------|-------------------|
| Layer          | mid_level_api     |
| Category flags | COMMAND_ALIGNMENT |
| Calls          | n/a               |

| Aspect                  | Value            |
|-------------------------|------------------|
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.5.6. **offsetalignment**

Offset alignment by ORF start position.

| Aspect                  | Value                                                                         |
|-------------------------|-------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                 |
| Category flags          | COMMAND_RESULT                                                                |
| Calls                   | n/a                                                                           |
| Called by               | cluster, linsearch, search                                                    |
| Workflow scripts        | blastn.sh,<br>linsearch.sh, nucleotide_clustering.sh,<br>translated_search.sh |
| Command reference       | Open page                                                                     |
| Functional module entry | Open module page                                                              |

#### 7.5.7. **proteinaln2nucl**

Transform protein alignments to nucleotide alignments.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_RESULT   |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.5.8. **rescorediagonal**

Compute sequence identity for diagonal.

| Aspect                  | Value                                                                      |
|-------------------------|----------------------------------------------------------------------------|
| Layer                   | mid_level_api                                                              |
| Category flags          | COMMAND_ALIGNMENT                                                          |
| Calls                   | n/a                                                                        |
| Called by               | cluster, linclust, linsearch, search, taxonomy                             |
| Workflow scripts        | linclust.sh,<br>linsearch.sh, nucleotide_clustering.sh,<br>taxpercontig.sh |
| Command reference       | Open page                                                                  |
| Functional module entry | Open module page                                                           |

### 7.5.9. **transitivealign**

Transfer alignments via transitivity.

| Aspect                  | Value             |
|-------------------------|-------------------|
| Layer                   | mid_level_api     |
| Category flags          | COMMAND_ALIGNMENT |
| Calls                   | n/a               |
| Called by               | n/a               |
| Workflow scripts        | n/a               |
| Command reference       | Open page         |
| Functional module entry | Open module page  |

## 7.6. Profiles

### 7.6.1. **convertca3m**

Convert a cA3M DB to a result DB.

| Aspect           | Value                   |
|------------------|-------------------------|
| Layer            | mid_level_api           |
| Category flags   | COMMAND_PROFILE_PROFILE |
| Calls            | n/a                     |
| Called by        | n/a                     |
| Workflow scripts | n/a                     |

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.6.2. **convertmsa**

Convert Stockholm/PFAM MSA file to a MSA DB.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_DATABASE_CREATION        |
| Calls                   | n/a                              |
| Called by               | databases                        |
| Workflow scripts        | databases.sh                     |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.6.3. **convertprofiledb**

Convert a HH-suite HHM DB to a profile DB.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_PROFILE                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.6.4. **expand2profile**

Expand an alignment result based on another and create a profile.

| Aspect         | Value                   |
|----------------|-------------------------|
| Layer          | mid_level_api           |
| Category flags | COMMAND_PROFILE_PROFILE |
| Calls          | n/a                     |

| Aspect                  | Value            |
|-------------------------|------------------|
| Called by               | search           |
| Workflow scripts        | iterativepp.sh   |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.5. msa2profile

Convert a MSA DB to a profile DB.

| Aspect                  | Value                                          |
|-------------------------|------------------------------------------------|
| Layer                   | low_level_api                                  |
| Category flags          | COMMAND_PROFILE<br>  COMMAND_DATABASE_CREATION |
| Calls                   | n/a                                            |
| Called by               | databases, pickconsensusrep                    |
| Workflow scripts        | databases.sh, pickconsensusrep.sh              |
| Command reference       | Open page                                      |
| Functional module entry | Open module page                               |

#### 7.6.6. msa2result

Convert a MSA DB to a profile DB.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_PROFILE   COMMAND_EXPERT |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | Open page                        |
| Functional module entry | Open module page                 |

#### 7.6.7. pairaln

Pair sequences to match best protein A and B from a species.

| Aspect | Value         |
|--------|---------------|
| Layer  | low_level_api |

| Aspect                  | Value            |
|-------------------------|------------------|
| Category flags          | COMMAND_EXPERT   |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.8. **profile2consensus**

Extract consensus sequence DB from a profile DB.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_PROFILE  |
| Calls                   | n/a              |
| Called by               | search           |
| Workflow scripts        | iterativepp.sh   |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.9. **profile2neff**

Convert a profile DB to a tab-separated list of Neff scores.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_PROFILE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.10. **profile2pssm**

Convert a profile DB to a tab-separated PSSM file.



| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_PROFILE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.11. **profile2repseq**

Extract representative sequence DB from a profile DB.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_PROFILE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.12. **result2profile**

Compute profile DB from a result DB.

| Aspect                  | Value                  |
|-------------------------|------------------------|
| Layer                   | low_level_api          |
| Category flags          | COMMAND_PROFILE        |
| Calls                   | n/a                    |
| Called by               | search                 |
| Workflow scripts        | blastpgp.sh, enrich.sh |
| Command reference       | Open page              |
| Functional module entry | Open module page       |

#### 7.6.13. **sequence2profile**

Turn sequence into profile by adding context specific pseudo counts.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_PROFILE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.6.14. tsv2exprofiledb

Create a expandable profile db from TSV files.

| Aspect                  | Value                                 |
|-------------------------|---------------------------------------|
| Layer                   | mid_level_api                         |
| Category flags          | COMMAND_PROFILE_PROFILE               |
| Calls                   | aliasdb, compress, mvdb, rmdb, tsv2db |
| Called by               | n/a                                   |
| Workflow scripts        | n/a                                   |
| Command reference       | Open page                             |
| Functional module entry | Open module page                      |

## 7.7. Database

### 7.7.1. aliasdb

Create relative symlink of DB to another name in the same folder.

| Aspect                  | Value              |
|-------------------------|--------------------|
| Layer                   | low_level_api      |
| Category flags          | COMMAND_STORAGE    |
| Calls                   | n/a                |
| Called by               | tsv2exprofiledb    |
| Workflow scripts        | tsv2exprofiledb.sh |
| Command reference       | Open page          |
| Functional module entry | Open module page   |

### 7.7.2. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

| Aspect                  | Value                                                        |
|-------------------------|--------------------------------------------------------------|
| Layer                   | low_level_api                                                |
| Category flags          | COMMAND_SET                                                  |
| Calls                   | n/a                                                          |
| Called by               | cluster, clusterupdate, linsearch                            |
| Workflow scripts        | linsearch.sh, nucleotide_clustering.sh, update_clustering.sh |
| Command reference       | Open page                                                    |
| Functional module entry | Open module page                                             |

### 7.7.3. cpdb

Copy a DB.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_STORAGE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.7.4. createdb

Convert FASTA/Q file(s) to a sequence DB.

| Aspect         | Value                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Layer          | low_level_api                                                                                             |
| Category flags | COMMAND_DATABASE_CREATION                                                                                 |
| Calls          | n/a                                                                                                       |
| Called by      | databases, easy-cluster, easy-lincludst, easy-linsearch, easy-rbh, easy-search, easy-taxonomy, multihitdb |

| Aspect                  | Value                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------|
| Workflow scripts        | databases.sh, easycluster.sh, easyrbh.sh, easysearch.sh, easytaxonomy.sh, multihitdb.sh |
| Command reference       | Open page                                                                               |
| Functional module entry | Open module page                                                                        |

#### 7.7.5. createindex

Store precomputed index on disk to reduce search overhead.

| Aspect                  | Value                                           |
|-------------------------|-------------------------------------------------|
| Layer                   | low_level_api                                   |
| Category flags          | COMMAND_DATABASE_CREATION                       |
| Calls                   | extractframes, extractorfs, rmdb, splitsequence |
| Called by               | n/a                                             |
| Workflow scripts        | n/a                                             |
| Command reference       | Open page                                       |
| Functional module entry | Open module page                                |

#### 7.7.6. createlinindex

Create linsearch index.

| Aspect                  | Value                                           |
|-------------------------|-------------------------------------------------|
| Layer                   | low_level_api                                   |
| Category flags          | COMMAND_DATABASE_CREATION<br>  COMMAND_EXPERT   |
| Calls                   | extractframes, extractorfs, rmdb, splitsequence |
| Called by               | easy-linsearch, easy-search                     |
| Workflow scripts        | easysearch.sh                                   |
| Command reference       | Open page                                       |
| Functional module entry | Open module page                                |

#### 7.7.7. createsubdb

Create a subset of a DB from list of DB keys.

| Aspect                  | Value                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                                                                                    |
| Category flags          | COMMAND_SET                                                                                                                                                      |
| Calls                   | n/a                                                                                                                                                              |
| Called by               | cluster, clusterupdate, linclust, search, taxonomy                                                                                                               |
| Workflow scripts        | blastp.sh,<br>cascaded_clustering.sh, clustering.sh,<br>linclust.sh, nucleotide_clustering.sh,<br>taxpercontig.sh, translated_search.sh,<br>update_clustering.sh |
| Command reference       | Open page                                                                                                                                                        |
| Functional module entry | Open module page                                                                                                                                                 |

#### 7.7.8. databases

List and download databases.

| Aspect                  | Value                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                |
| Category flags          | COMMAND_DATABASE_CREATION                                                                    |
| Calls                   | convertmsa, createdb, createtaxdb,<br>msa2profile, nrtotaxmapping, prefixid,<br>rmdb, tar2db |
| Called by               | n/a                                                                                          |
| Workflow scripts        | n/a                                                                                          |
| Command reference       | Open page                                                                                    |
| Functional module entry | Open module page                                                                             |

#### 7.7.9. db2tar

Archive contents of a DB to a tar archive.

| Aspect         | Value                                      |
|----------------|--------------------------------------------|
| Layer          | low_level_api                              |
| Category flags | COMMAND_DATABASE_CREATION   COMMAND_EXPERT |
| Calls          | n/a                                        |
| Called by      | n/a                                        |

| Aspect                  | Value            |
|-------------------------|------------------|
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.7.10. **lndb**

Symlink a DB.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_STORAGE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.7.11. **mergedbs**

Merge entries from multiple DBs.

| Aspect                  | Value                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                                                                                                    |
| Category flags          | COMMAND_SET                                                                                                                                                                      |
| Calls                   | n/a                                                                                                                                                                              |
| Called by               | cluster, clusterupdate, rbh, search                                                                                                                                              |
| Workflow scripts        | blastp.sh,<br>blastpgp.sh,      cascaded_clustering.sh,<br>enrich.sh,                      iterativepp.sh,<br>rbh.sh,      searchslicedtargetprofile.sh,<br>update_clustering.sh |
| Command reference       | Open page                                                                                                                                                                        |
| Functional module entry | Open module page                                                                                                                                                                 |

#### 7.7.12. **mvdb**

Move a DB.

| Aspect                  | Value                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                                          |
| Category flags          | COMMAND_STORAGE                                                                                                        |
| Calls                   | n/a                                                                                                                    |
| Called by               | cluster, clusterupdate, search, taxonomy, tsv2exprofiledb                                                              |
| Workflow scripts        | blastp.sh, cascaded_clustering.sh, searchslicedtargetprofile.sh, taxonomy.sh, tsv2exprofiledb.sh, update_clustering.sh |
| Command reference       | Open page                                                                                                              |
| Functional module entry | Open module page                                                                                                       |

#### 7.7.13. renamedbkeys

Create a new DB with original keys renamed.

| Aspect                  | Value                           |
|-------------------------|---------------------------------|
| Layer                   | low_level_api                   |
| Category flags          | COMMAND_DB                      |
| Calls                   | n/a                             |
| Called by               | clusterupdate, pickconsensusrep |
| Workflow scripts        | update_clustering.sh            |
| Command reference       | Open page                       |
| Functional module entry | Open module page                |

#### 7.7.14. rmdb

Remove a DB.

| Aspect         | Value                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer          | low_level_api                                                                                                                                     |
| Category flags | COMMAND_STORAGE                                                                                                                                   |
| Calls          | n/a                                                                                                                                               |
| Called by      | cluster, clusterupdate, createindex, createlinindex, databases, easy-cluster, easy-linclud, easy-linsearch, easy-rbh, easy-search, easy-taxonomy, |

| Aspect                  | Value                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | linclust, linsearch, multihitsearch, pickconsensusrep, rbh, search, taxonomy, tsv2exprofiledb                                                                                                                                                                                                                                                                                                                                                |
| Workflow scripts        | blastn.sh, blastp.sh, blastpgp.sh, cascaded_clustering.sh, clustering.sh, createindex.sh, databases.sh, easycluster.sh, easyrbh.sh, easysearch.sh, easytaxonomy.sh, iterativepp.sh, linclust.sh, linsearch.sh, multihitsearch.sh, nucleotide_clustering.sh, pickconsensusrep.sh, rbh.sh, searchsllicedtargetprofile.sh, searchtargetprofile.sh, taxonomy.sh, taxpercontig.sh, translated_search.sh, tsv2exprofiledb.sh, update_clustering.sh |
| Command reference       | <a href="#">Open page</a>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Functional module entry | <a href="#">Open module page</a>                                                                                                                                                                                                                                                                                                                                                                                                             |

#### 7.7.15. **splitdb**

Split DB into subsets.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_SET                      |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.7.16. **splitsequence**

Split sequences by length.

| Aspect | Value         |
|--------|---------------|
| Layer  | low_level_api |



| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Category flags          | COMMAND_SEQUENCE                  |
| Calls                   | n/a                               |
| Called by               | createindex, createlindex, search |
| Workflow scripts        | blastn.sh, createindex.sh         |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

#### 7.7.17. **subtractdbs**

Remove all entries from first DB occurring in second DB by key.

| Aspect                  | Value                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                  |
| Category flags          | COMMAND_SET                                                                                    |
| Calls                   | n/a                                                                                            |
| Called by               | cluster, search                                                                                |
| Workflow scripts        | blastpgp.sh, cascaded_clustering.sh,<br>enrich.sh, iterativepp.sh,<br>nucleotide_clustering.sh |
| Command reference       | Open page                                                                                      |
| Functional module entry | Open module page                                                                               |

#### 7.7.18. **swapdb**

Transpose DB with integer values in first column.

| Aspect                  | Value                                                                           |
|-------------------------|---------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                   |
| Category flags          | COMMAND_DB                                                                      |
| Calls                   | n/a                                                                             |
| Called by               | cluster, clusterupdate, multihitdb,<br>taxonomy                                 |
| Workflow scripts        | cascaded_clustering.sh, multihitdb.sh,<br>taxpercontig.sh, update_clustering.sh |
| Command reference       | Open page                                                                       |
| Functional module entry | Open module page                                                                |

### 7.7.19. tar2db

Convert content of tar archives to any DB.

| Aspect                  | Value                                      |
|-------------------------|--------------------------------------------|
| Layer                   | low_level_api                              |
| Category flags          | COMMAND_DATABASE_CREATION   COMMAND_EXPERT |
| Calls                   | n/a                                        |
| Called by               | databases                                  |
| Workflow scripts        | databases.sh                               |
| Command reference       | Open page                                  |
| Functional module entry | Open module page                           |

### 7.7.20. tsv2db

Convert a TSV file to any DB.

| Aspect                  | Value                                                                             |
|-------------------------|-----------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                     |
| Category flags          | COMMAND_DATABASE_CREATION<br>  COMMAND_EXPERT                                     |
| Calls                   | n/a                                                                               |
| Called by               | cluster, multihitdb, pickconsensusrep,<br>tsv2exprofiledb                         |
| Workflow scripts        | cascaded_clustering.sh, multihitdb.sh,<br>pickconsensusrep.sh, tsv2exprofiledb.sh |
| Command reference       | Open page                                                                         |
| Functional module entry | Open module page                                                                  |

## 7.8. Result Handling

### 7.8.1. convert2fasta

Convert sequence DB to FASTA format.

| Aspect         | Value                     |
|----------------|---------------------------|
| Layer          | low_level_api             |
| Category flags | COMMAND_FORMAT_CONVERSION |
| Calls          | n/a                       |
| Called by      | n/a                       |

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.8.2. **convertalis**

Convert alignment DB to BLAST-tab, SAM or custom format.

| Aspect                  | Value                                                |
|-------------------------|------------------------------------------------------|
| Layer                   | low_level_api                                        |
| Category flags          | COMMAND_FORMAT_CONVERSION                            |
| Calls                   | n/a                                                  |
| Called by               | easy-linsearch, easy-rbh, easy-search, easy-taxonomy |
| Workflow scripts        | easyrbh.sh,<br>easysearch.sh, easytaxonomy.sh        |
| Command reference       | <a href="#">Open page</a>                            |
| Functional module entry | <a href="#">Open module page</a>                     |

### 7.8.3. **createseqfiledb**

Create a DB of unaligned FASTA entries.

| Aspect                  | Value                                         |
|-------------------------|-----------------------------------------------|
| Layer                   | low_level_api                                 |
| Category flags          | COMMAND_FORMAT_CONVERSION<br>  COMMAND_EXPERT |
| Calls                   | n/a                                           |
| Called by               | easy-cluster, easy-lincludst                  |
| Workflow scripts        | easycluster.sh                                |
| Command reference       | <a href="#">Open page</a>                     |
| Functional module entry | <a href="#">Open module page</a>              |

### 7.8.4. **createtsv**

Convert result DB to tab-separated flat file.

| Aspect                  | Value                                     |
|-------------------------|-------------------------------------------|
| Layer                   | low_level_api                             |
| Category flags          | COMMAND_FORMAT_CONVERSION                 |
| Calls                   | n/a                                       |
| Called by               | easy-cluster, easy-linclud, easy-taxonomy |
| Workflow scripts        | easycluster.sh, easytaxonomy.sh           |
| Command reference       | Open page                                 |
| Functional module entry | Open module page                          |

#### 7.8.5. extractdomains

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SPECIAL  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.8.6. filterresult

Pairwise alignment result filter.

| Aspect                  | Value                        |
|-------------------------|------------------------------|
| Layer                   | low_level_api                |
| Category flags          | COMMAND_RESULT               |
| Calls                   | n/a                          |
| Called by               | search                       |
| Workflow scripts        | searchslicedtargetprofile.sh |
| Command reference       | Open page                    |
| Functional module entry | Open module page             |

### 7.8.7. **result2dnamsa**

Compute MSA DB with out insertions in the query for DNA sequences.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_RESULT   |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.8.8. **result2flat**

Create flat file by adding FASTA headers to DB entries.

| Aspect                  | Value                                         |
|-------------------------|-----------------------------------------------|
| Layer                   | low_level_api                                 |
| Category flags          | COMMAND_FORMAT_CONVERSION<br>  COMMAND_EXPERT |
| Calls                   | n/a                                           |
| Called by               | easy-cluster, easy-linclud                    |
| Workflow scripts        | easycluster.sh                                |
| Command reference       | Open page                                     |
| Functional module entry | Open module page                              |

### 7.8.9. **result2msa**

Compute MSA DB from a result DB.

| Aspect            | Value               |
|-------------------|---------------------|
| Layer             | low_level_api       |
| Category flags    | COMMAND_RESULT      |
| Calls             | n/a                 |
| Called by         | pickconsensusrep    |
| Workflow scripts  | pickconsensusrep.sh |
| Command reference | Open page           |

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Functional module entry | <a href="#">Open module page</a> |

#### 7.8.10. **result2rbh**

Filter a merged result DB to retain only reciprocal best hits.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_RESULT                   |
| Calls                   | n/a                              |
| Called by               | rbh                              |
| Workflow scripts        | rbh.sh                           |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.8.11. **result2repseq**

Get representative sequences from result DB.

| Aspect                  | Value                                     |
|-------------------------|-------------------------------------------|
| Layer                   | low_level_api                             |
| Category flags          | COMMAND_RESULT                            |
| Calls                   | n/a                                       |
| Called by               | clusterupdate, easy-cluster, easy-linclud |
| Workflow scripts        | easycluster.sh, update_clustering.sh      |
| Command reference       | <a href="#">Open page</a>                 |
| Functional module entry | <a href="#">Open module page</a>          |

#### 7.8.12. **result2stats**

Compute statistics for each entry in a DB.

| Aspect         | Value              |
|----------------|--------------------|
| Layer          | low_level_api      |
| Category flags | COMMAND_RESULT     |
| Calls          | n/a                |
| Called by      | multihitdb, search |

| Aspect                  | Value                                          |
|-------------------------|------------------------------------------------|
| Workflow scripts        | multihitdb.sh,<br>searchslicedtargetprofile.sh |
| Command reference       | Open page                                      |
| Functional module entry | Open module page                               |

#### 7.8.13. sortresult

Sort a result DB in the same order as the prefilter or align module.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_RESULT   |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.8.14. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_RESULT   |
| Calls                   | n/a              |
| Called by               | easy-taxonomy    |
| Workflow scripts        | easytaxonomy.sh  |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.8.15. summarizeheaders

Summarize FASTA headers of result DB.

| Aspect         | Value           |
|----------------|-----------------|
| Layer          | low_level_api   |
| Category flags | COMMAND_SPECIAL |

| Aspect                  | Value            |
|-------------------------|------------------|
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.8.16. **summarizerresult**

Extract annotations from alignment DB.

| Aspect                  | Value                       |
|-------------------------|-----------------------------|
| Layer                   | low_level_api               |
| Category flags          | COMMAND_RESULT              |
| Calls                   | n/a                         |
| Called by               | easy-linsearch, easy-search |
| Workflow scripts        | easysearch.sh               |
| Command reference       | Open page                   |
| Functional module entry | Open module page            |

#### 7.8.17. **swapresults**

Transpose prefilter/alignment DB.

| Aspect                  | Value                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                               |
| Category flags          | COMMAND_RESULT                                                                              |
| Calls                   | n/a                                                                                         |
| Called by               | easy-taxonomy, linsearch, rbh, search                                                       |
| Workflow scripts        | easytaxonomy.sh, linsearch.sh, rbh.sh, searchslicedtargetprofile.sh, searchtargetprofile.sh |
| Command reference       | Open page                                                                                   |
| Functional module entry | Open module page                                                                            |



## 7.9. Sequence Manipulation

### 7.9.1. **extractalignedregion**

Extract aligned sequence region from query.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.9.2. **extractframes**

Extract frames from a nucleotide sequence DB.

| Aspect                  | Value                                                                     |
|-------------------------|---------------------------------------------------------------------------|
| Layer                   | low_level_api                                                             |
| Category flags          | COMMAND_SEQUENCE                                                          |
| Calls                   | n/a                                                                       |
| Called by               | cluster, createindex, createlinindex, search                              |
| Workflow scripts        | blastn.sh, createindex.sh, nucleotide_clustering.sh, translated_search.sh |
| Command reference       | Open page                                                                 |
| Functional module entry | Open module page                                                          |

### 7.9.3. **extractorfs**

Six-frame extraction of open reading frames.

| Aspect         | Value            |
|----------------|------------------|
| Layer          | low_level_api    |
| Category flags | COMMAND_SEQUENCE |
| Calls          | n/a              |

| Aspect                  | Value                                                                |
|-------------------------|----------------------------------------------------------------------|
| Called by               | createindex, createlinindex, linsearch, multihitdb, search, taxonomy |
| Workflow scripts        | createindex.sh, multihitdb.sh, taxpercontig.sh, translated_search.sh |
| Command reference       | Open page                                                            |
| Functional module entry | Open module page                                                     |

#### 7.9.4. masksequence

Soft mask sequence DB using tantan.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.9.5. orftocontig

Write ORF locations in alignment format.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | multihitdb       |
| Workflow scripts        | multihitdb.sh    |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.9.6. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_EXPERT   |
| Calls                   | n/a              |
| Called by               | taxonomy         |
| Workflow scripts        | taxpercontig.sh  |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.9.7. **reverseseq**

Reverse (without complement) sequences.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.9.8. **translateaa**

Translate proteins to lexicographically lowest codons.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.9.9. **translatenucs**

Translate nucleotides to proteins.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SEQUENCE |
| Calls                   | n/a              |
| Called by               | multihitdb       |
| Workflow scripts        | multihitdb.sh    |
| Command reference       | Open page        |
| Functional module entry | Open module page |

## 7.10. Taxonomy

### 7.10.1. addtaxonomy

Add taxonomic labels to result DB.

| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Layer                   | low_level_api                     |
| Category flags          | COMMAND_TAXONOMY   COMMAND_EXPERT |
| Calls                   | n/a                               |
| Called by               | easy-taxonomy                     |
| Workflow scripts        | easytaxonomy.sh                   |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

### 7.10.2. aggregatetax

Aggregate multiple taxon labels to a single label.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_TAXONOMY |
| Calls                   | n/a              |
| Called by               | taxonomy         |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.10.3. **aggregatetaxweights**

Aggregate multiple taxon labels to a single label.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_TAXONOMY |
| Calls                   | n/a              |
| Called by               | taxonomy         |
| Workflow scripts        | taxpercontig.sh  |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.10.4. **createbintaxmapping**

Create binary taxonomy mapping from tabular taxonomy mapping.

| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Layer                   | low_level_api                     |
| Category flags          | COMMAND_TAXONOMY   COMMAND_EXPERT |
| Calls                   | n/a                               |
| Called by               | n/a                               |
| Workflow scripts        | n/a                               |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

### 7.10.5. **createbintaxonomy**

Create binary taxonomy from NCBI input.

| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Layer                   | low_level_api                     |
| Category flags          | COMMAND_TAXONOMY   COMMAND_EXPERT |
| Calls                   | n/a                               |
| Called by               | createtaxdb                       |
| Workflow scripts        | createtaxdb.sh                    |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

#### 7.10.6. **createdmptaxonomy**

Create dmp files from binary taxonomy.

| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Layer                   | low_level_api                     |
| Category flags          | COMMAND_TAXONOMY   COMMAND_EXPERT |
| Calls                   | n/a                               |
| Called by               | n/a                               |
| Workflow scripts        | n/a                               |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

#### 7.10.7. **createtaxdb**

Add taxonomic labels to sequence DB.

| Aspect                  | Value             |
|-------------------------|-------------------|
| Layer                   | low_level_api     |
| Category flags          | COMMAND_TAXONOMY  |
| Calls                   | createbintaxonomy |
| Called by               | databases         |
| Workflow scripts        | databases.sh      |
| Command reference       | Open page         |
| Functional module entry | Open module page  |

#### 7.10.8. **filtertaxdb**

Filter taxonomy result database.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_TAXONOMY |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.10.9. **filtertaxseqdb**

Filter taxonomy sequence database.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_TAXONOMY |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.10.10. **lca**

Compute the lowest common ancestor.

| Aspect                  | Value                   |
|-------------------------|-------------------------|
| Layer                   | low_level_api           |
| Category flags          | COMMAND_TAXONOMY        |
| Calls                   | n/a                     |
| Called by               | easy-taxonomy, taxonomy |
| Workflow scripts        | taxonomy.sh             |
| Command reference       | Open page               |
| Functional module entry | Open module page        |

#### 7.10.11. **lcaalign**

Efficient gapped alignment for lca computation.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_TAXONOMY |
| Calls                   | n/a              |
| Called by               | search           |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.10.12. majoritylca

Compute the lowest common ancestor using majority voting.

| Aspect                  | Value                             |
|-------------------------|-----------------------------------|
| Layer                   | low_level_api                     |
| Category flags          | COMMAND_TAXONOMY   COMMAND_EXPERT |
| Calls                   | n/a                               |
| Called by               | n/a                               |
| Workflow scripts        | n/a                               |
| Command reference       | Open page                         |
| Functional module entry | Open module page                  |

#### 7.10.13. nrtotaxmapping

Create taxonomy mapping for NR database.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SPECIAL  |
| Calls                   | n/a              |
| Called by               | databases        |
| Workflow scripts        | databases.sh     |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.10.14. taxonomy

Taxonomic classification.

| Aspect         | Value                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer          | high_level_api                                                                                                                                                                     |
| Category flags | COMMAND_MAIN                                                                                                                                                                       |
| Calls          | aggregatetax, aggregatetaxweights, createsubdb, extractorfs, filterdb, lca, mergeresultsbyset, mvdb, prefilter, recoverlongestorf, rescorediagonal, rmdb, search, swapdb, taxonomy |
| Called by      | easy-taxonomy, taxonomy                                                                                                                                                            |



| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Workflow scripts        | easytaxonomy.sh, taxpercontig.sh |
| Command reference       | Open page                        |
| Functional module entry | Open module page                 |

#### 7.10.15. taxonomyreport

Create a taxonomy report in Kraken or Krona format.

| Aspect                  | Value                                        |
|-------------------------|----------------------------------------------|
| Layer                   | low_level_api                                |
| Category flags          | COMMAND_TAXONOMY   COMMAND_FORMAT_CONVERSION |
| Calls                   | n/a                                          |
| Called by               | easy-taxonomy                                |
| Workflow scripts        | easytaxonomy.sh                              |
| Command reference       | Open page                                    |
| Functional module entry | Open module page                             |

### 7.11. Multi Hit

#### 7.11.1. besthitperset

For each set of sequences compute the best element and update p-value.

| Aspect                  | Value             |
|-------------------------|-------------------|
| Layer                   | high_level_api    |
| Category flags          | COMMAND_MULTIHIT  |
| Calls                   | n/a               |
| Called by               | multihitsearch    |
| Workflow scripts        | multihitsearch.sh |
| Command reference       | Open page         |
| Functional module entry | Open module page  |

#### 7.11.2. combinepvalperset

For each set compute the combined p-value.

| Aspect | Value          |
|--------|----------------|
| Layer  | high_level_api |

| Aspect                  | Value            |
|-------------------------|------------------|
| Category flags          | COMMAND_MULTIHIT |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.11.3. **mergeresultsbyset**

Merge results from multiple ORFs back to their respective contig.

| Aspect                  | Value                              |
|-------------------------|------------------------------------|
| Layer                   | high_level_api                     |
| Category flags          | COMMAND_MULTIHIT                   |
| Calls                   | n/a                                |
| Called by               | multihitsearch, taxonomy           |
| Workflow scripts        | multihitsearch.sh, taxpercontig.sh |
| Command reference       | Open page                          |
| Functional module entry | Open module page                   |

### 7.11.4. **multihitdb**

Create sequence DB for multi hit searches.

| Aspect                  | Value                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------|
| Layer                   | high_level_api                                                                            |
| Category flags          | COMMAND_MULTIHIT                                                                          |
| Calls                   | createdb, extractorfs, filterdb, orftocontig, result2stats, swapdb, translatenucs, tsv2db |
| Called by               | n/a                                                                                       |
| Workflow scripts        | n/a                                                                                       |
| Command reference       | Open page                                                                                 |
| Functional module entry | Open module page                                                                          |

### 7.11.5. multihitsearch

Search with a grouped set of sequences against another grouped set.

| Aspect                  | Value                                          |
|-------------------------|------------------------------------------------|
| Layer                   | high_level_api                                 |
| Category flags          | COMMAND_MULTIHIT                               |
| Calls                   | besthitperset, mergeresultsbyset, rmdb, search |
| Called by               | n/a                                            |
| Workflow scripts        | n/a                                            |
| Command reference       | Open page                                      |
| Functional module entry | Open module page                               |

## 7.12. Utilities

### 7.12.1. apply

Execute given program on each DB entry.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_DB       |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.12.2. compress

Compress DB entries.

| Aspect           | Value              |
|------------------|--------------------|
| Layer            | low_level_api      |
| Category flags   | COMMAND_STORAGE    |
| Calls            | n/a                |
| Called by        | tsv2exprofiledb    |
| Workflow scripts | tsv2exprofiledb.sh |

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.12.3. **convertkb**

Convert UniProtKB data to a DB.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_SPECIAL                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.12.4. **decompress**

Decompress DB entries.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_STORAGE                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

#### 7.12.5. **diffseqdbs**

Compute diff of two sequence DBs.

| Aspect         | Value           |
|----------------|-----------------|
| Layer          | low_level_api   |
| Category flags | COMMAND_SPECIAL |
| Calls          | n/a             |

| Aspect                  | Value                |
|-------------------------|----------------------|
| Called by               | clusterupdate        |
| Workflow scripts        | update_clustering.sh |
| Command reference       | Open page            |
| Functional module entry | Open module page     |

#### 7.12.6. filterdb

DB filtering by given conditions.

| Aspect                  | Value                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Layer                   | low_level_api                                                                                                                                 |
| Category flags          | COMMAND_DB                                                                                                                                    |
| Calls                   | n/a                                                                                                                                           |
| Called by               | cluster, clusterupdate, easy-taxonomy, linclust, linsearch, multihitdb, rbh, taxonomy                                                         |
| Workflow scripts        | cascaded_clustering.sh, easytaxonomy.sh, linclust.sh, linsearch.sh, multihitdb.sh, rbh.sh, taxonomy.sh, taxpercontig.sh, update_clustering.sh |
| Command reference       | Open page                                                                                                                                     |
| Functional module entry | Open module page                                                                                                                              |

#### 7.12.7. gff2db

Extract regions from a sequence database based on a GFF3 file.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_SPECIAL  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

### 7.12.8. gpuserver

Start a GPU server.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_STORAGE                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.12.9. maskbygff

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Layer                   | low_level_api                    |
| Category flags          | COMMAND_SPECIAL                  |
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |

### 7.12.10. prefixid

For each entry in a DB prepend the entry key to the entry itself.

| Aspect            | Value                                                      |
|-------------------|------------------------------------------------------------|
| Layer             | low_level_api                                              |
| Category flags    | COMMAND_DB                                                 |
| Calls             | n/a                                                        |
| Called by         | clusterupdate,<br>databases, pickconsensusrep              |
| Workflow scripts  | databases.sh, pickconsensusrep.sh,<br>update_clustering.sh |
| Command reference | <a href="#">Open page</a>                                  |

| Aspect                  | Value            |
|-------------------------|------------------|
| Functional module entry | Open module page |

#### 7.12.11. **setextendeddbtype**

Write an extended DB.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_DB       |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.12.12. **suffixid**

For each entry in a DB append the entry key to the entry itself.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_DB       |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.12.13. **summarizetabs**

Extract annotations from HHblits BLAST-tab-formatted results.

| Aspect         | Value           |
|----------------|-----------------|
| Layer          | low_level_api   |
| Category flags | COMMAND_SPECIAL |
| Calls          | n/a             |
| Called by      | n/a             |

| Aspect                  | Value            |
|-------------------------|------------------|
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.12.14. **touchdb**

Preload DB into memory (page cache).

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_STORAGE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.12.15. **unpackdb**

Unpack a DB into separate files.

| Aspect                  | Value            |
|-------------------------|------------------|
| Layer                   | low_level_api    |
| Category flags          | COMMAND_STORAGE  |
| Calls                   | n/a              |
| Called by               | n/a              |
| Workflow scripts        | n/a              |
| Command reference       | Open page        |
| Functional module entry | Open module page |

#### 7.12.16. **view**

Print DB entries given in -id-list to stdout.

| Aspect         | Value         |
|----------------|---------------|
| Layer          | low_level_api |
| Category flags | COMMAND_DB    |



| Aspect                  | Value                            |
|-------------------------|----------------------------------|
| Calls                   | n/a                              |
| Called by               | n/a                              |
| Workflow scripts        | n/a                              |
| Command reference       | <a href="#">Open page</a>        |
| Functional module entry | <a href="#">Open module page</a> |